Seq2seq and Attention

Ignat Romanov

(based on materials from Lena Voita)

Lecture-blog and lots of additional materials are here:
https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

Sequence to Sequence Task

- Translation between natural languages
- More generally, translation between any sequences

What is going to happen:

- Seq2seq Basics
- Attention
- Transformer
- Subword Segmentation: BPE
-  Analysis and Interpretability

What is going to happen:

- Seq2seq Basics
  - Machine Translation Task
  - Encoder-Decoder Framework
  - Conditional LMs
  - The Simplest RNN Model
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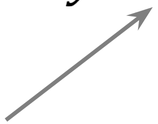
Translation

Human Translation

$$y^* = \arg\max_y p(y|x)$$



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model parameters

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- **learning**

How to find θ?

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- **search**

  How to find the argmax?

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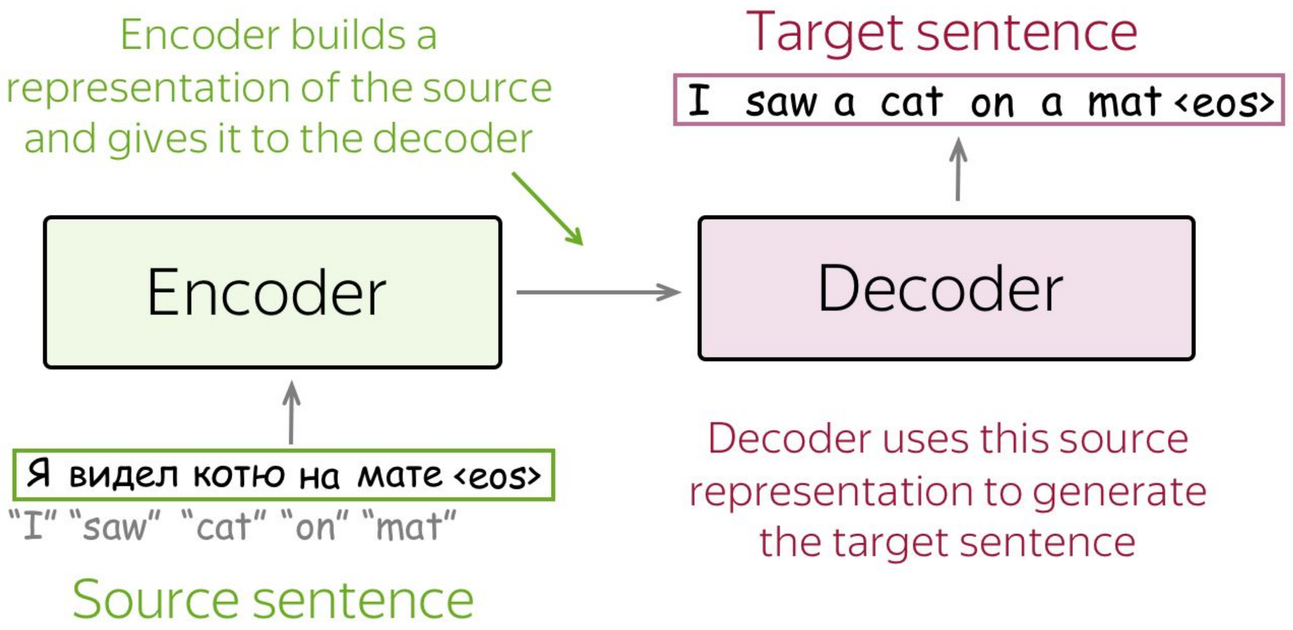
Encoder-Decoder Framework

The standard modeling paradigm:

- **Encoder** – reads the source sentence and produces its representation



Encoder-Decoder Framework

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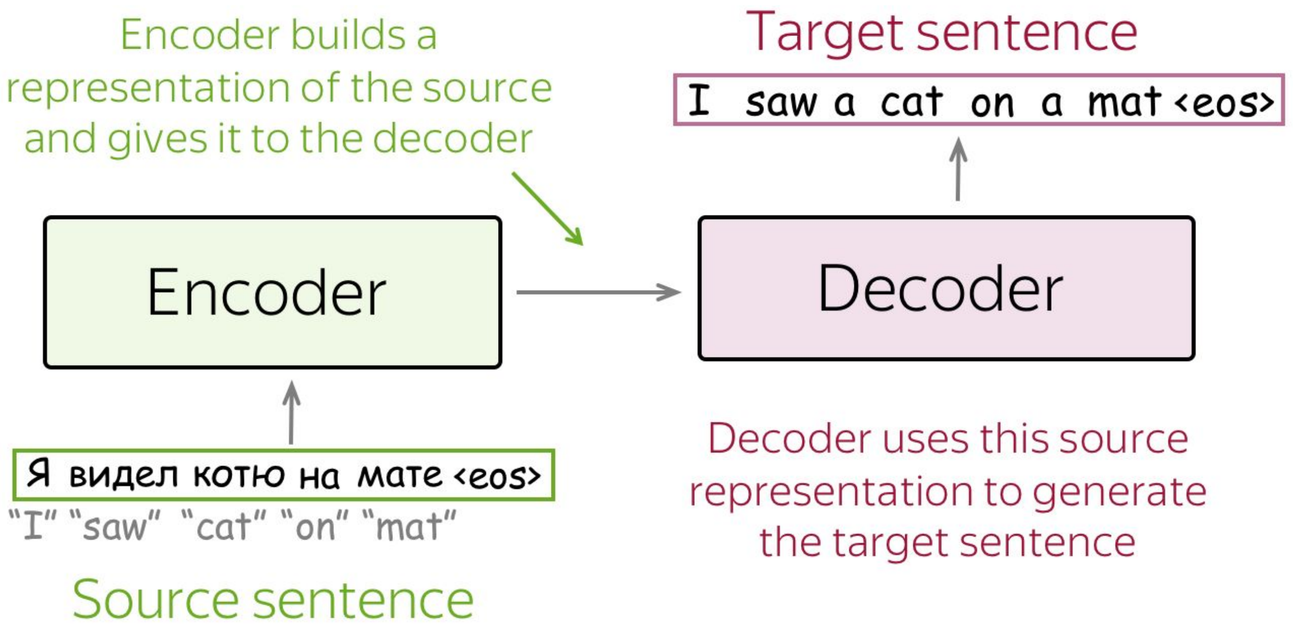
- **Encoder** – reads the source sentence and produces its representation
- **Decoder** - uses source representation from the encoder to generate the target sequence.



Conditional Language Models

Language Models: $P(y_1, y_2, \ldots, y_n) = \prod_{t=1}^{n} p(y_t | y_{<t})$

(left-to-right)

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condition on source x

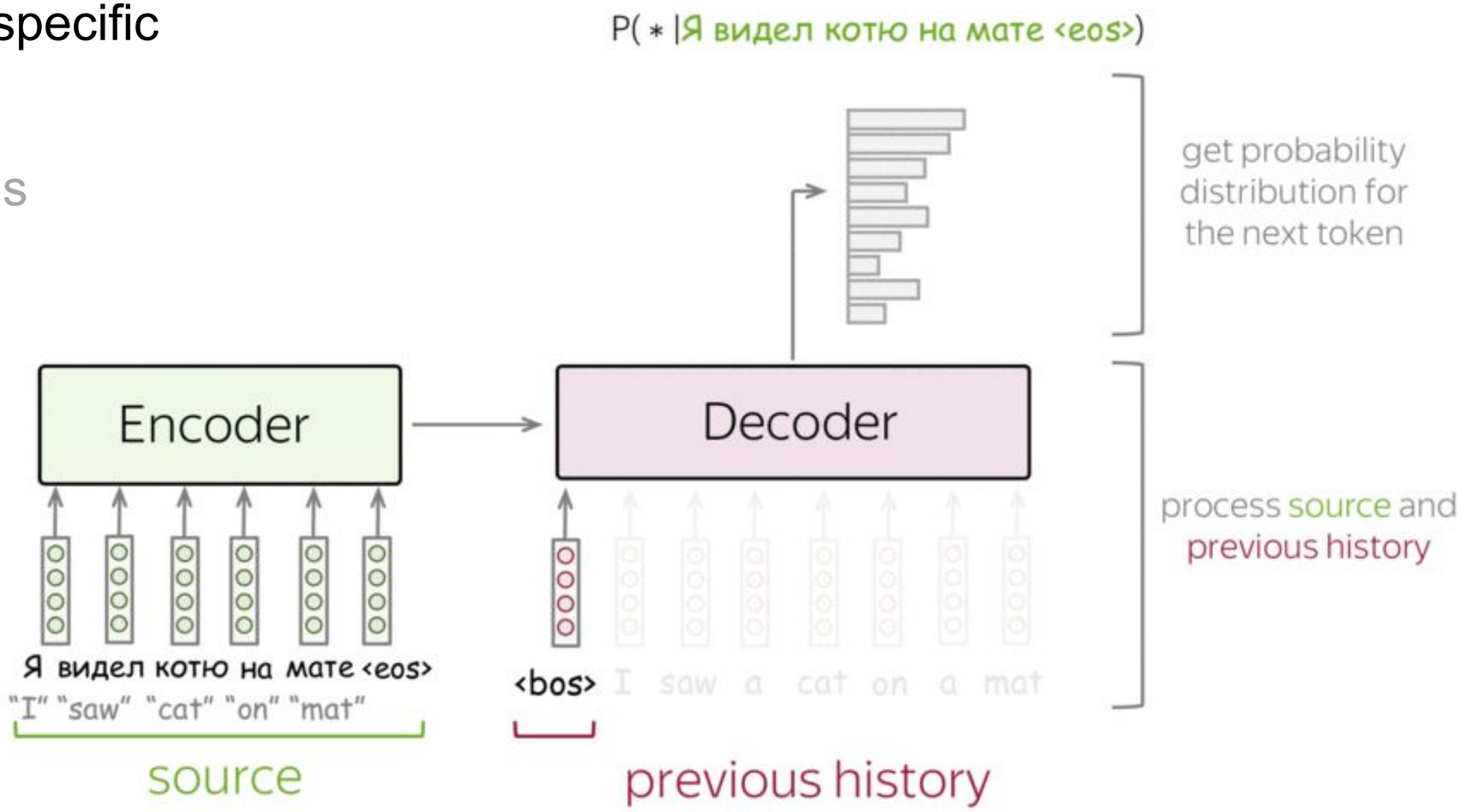
General View

- process context – model-specific

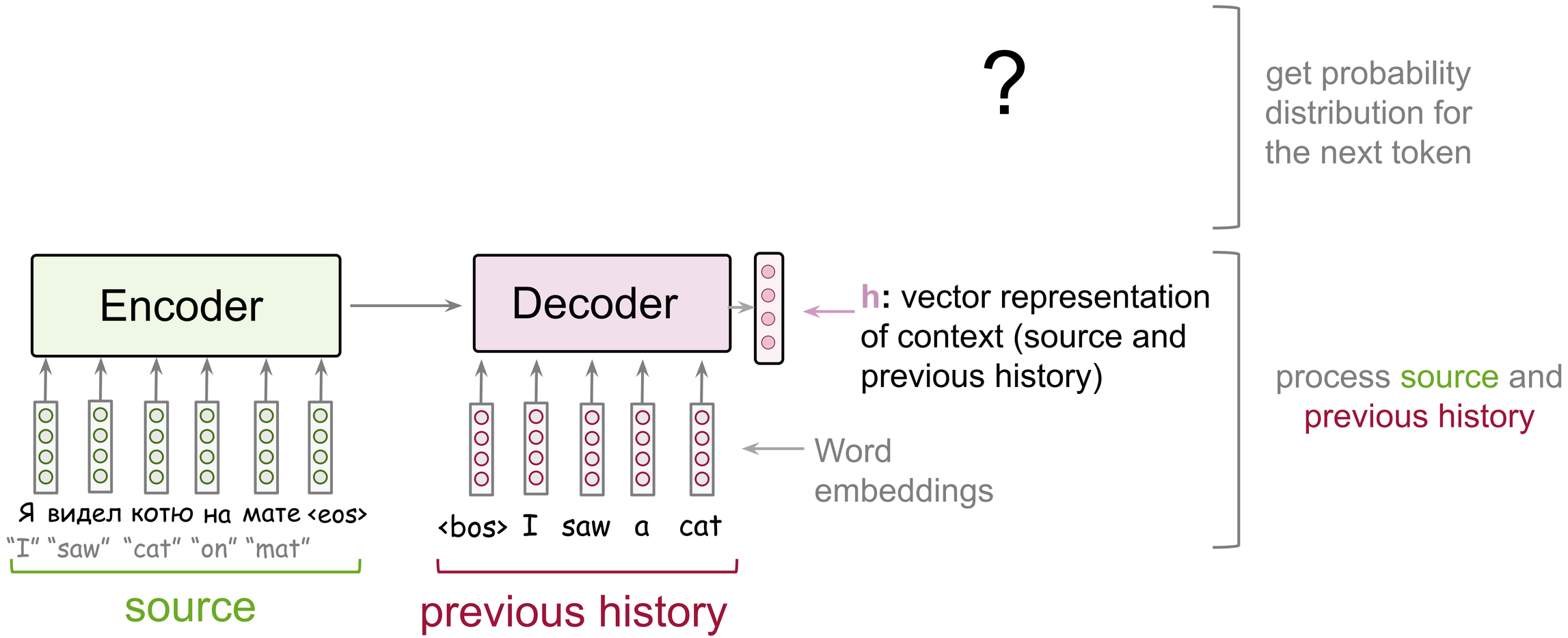
  Get vector representation of the source and previous target tokens

- evaluate probabilities – model-agnostic

  Predict probability distribution for the next target token



High-Level Pipeline



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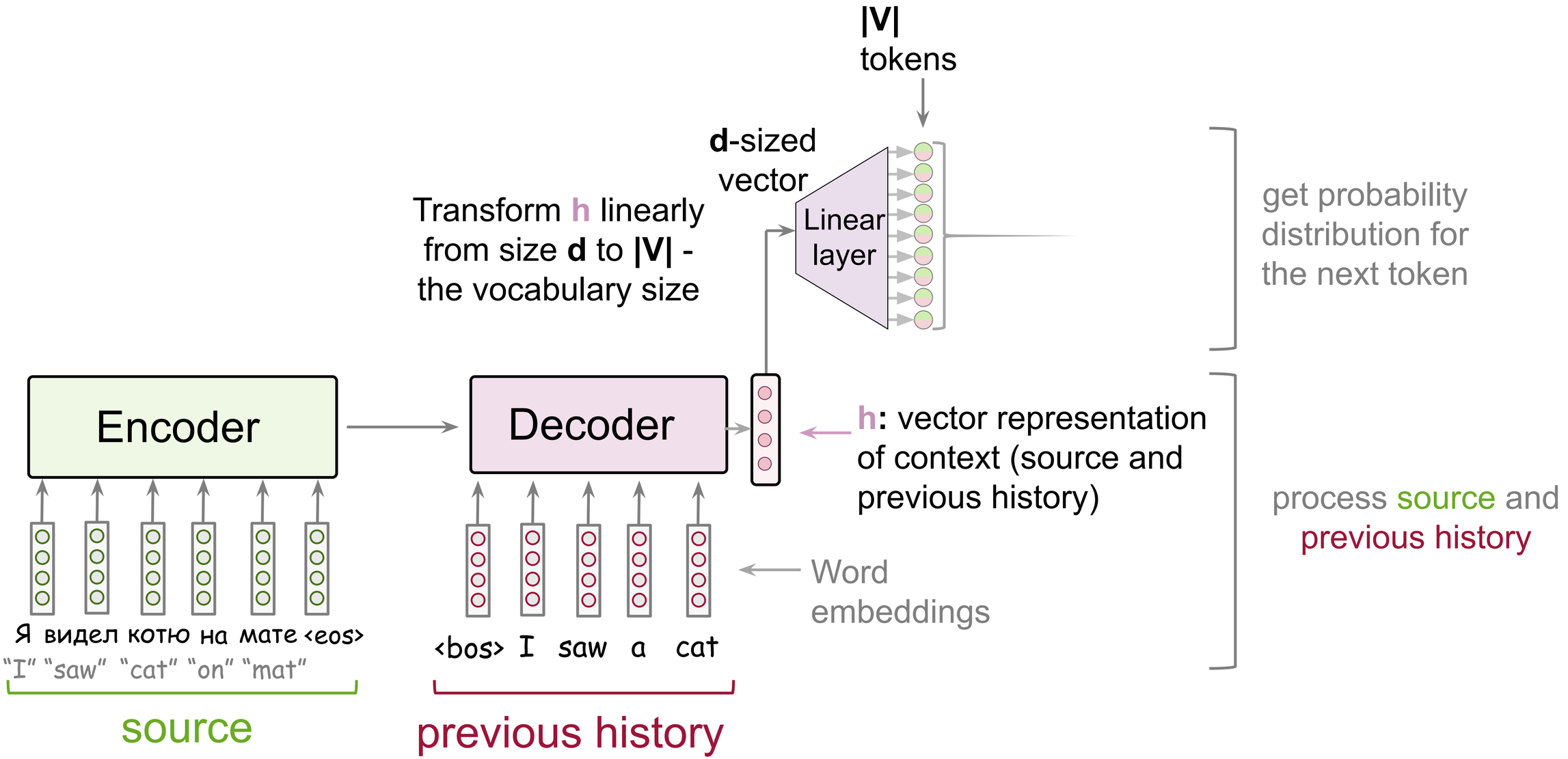

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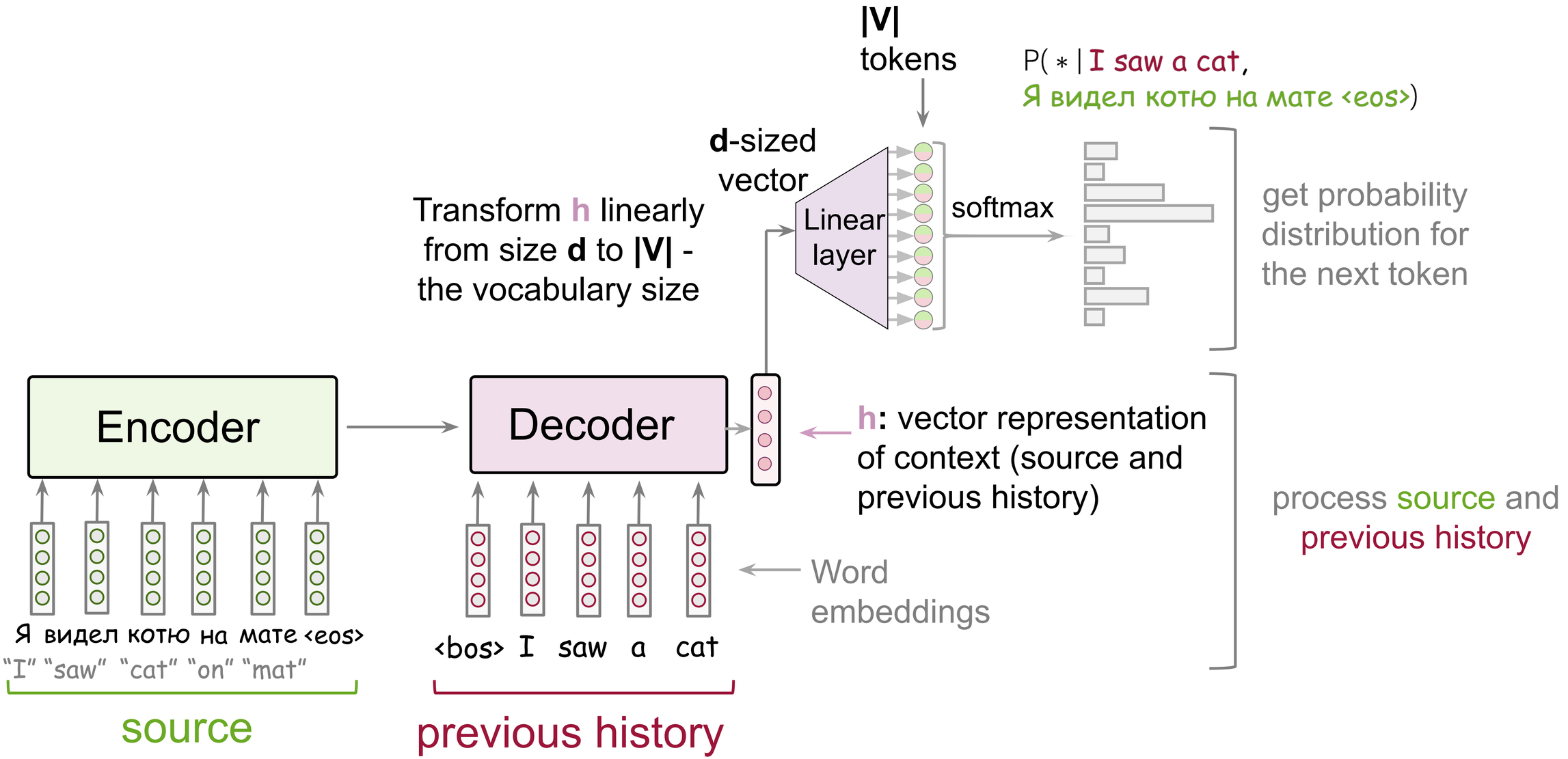
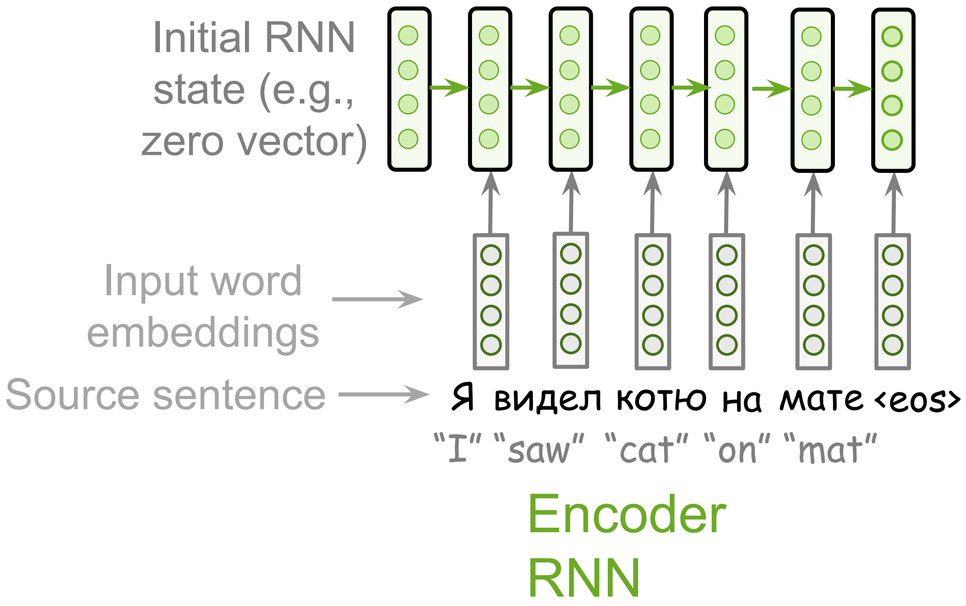


The Simplest Model: RNN Encoder and Decoder



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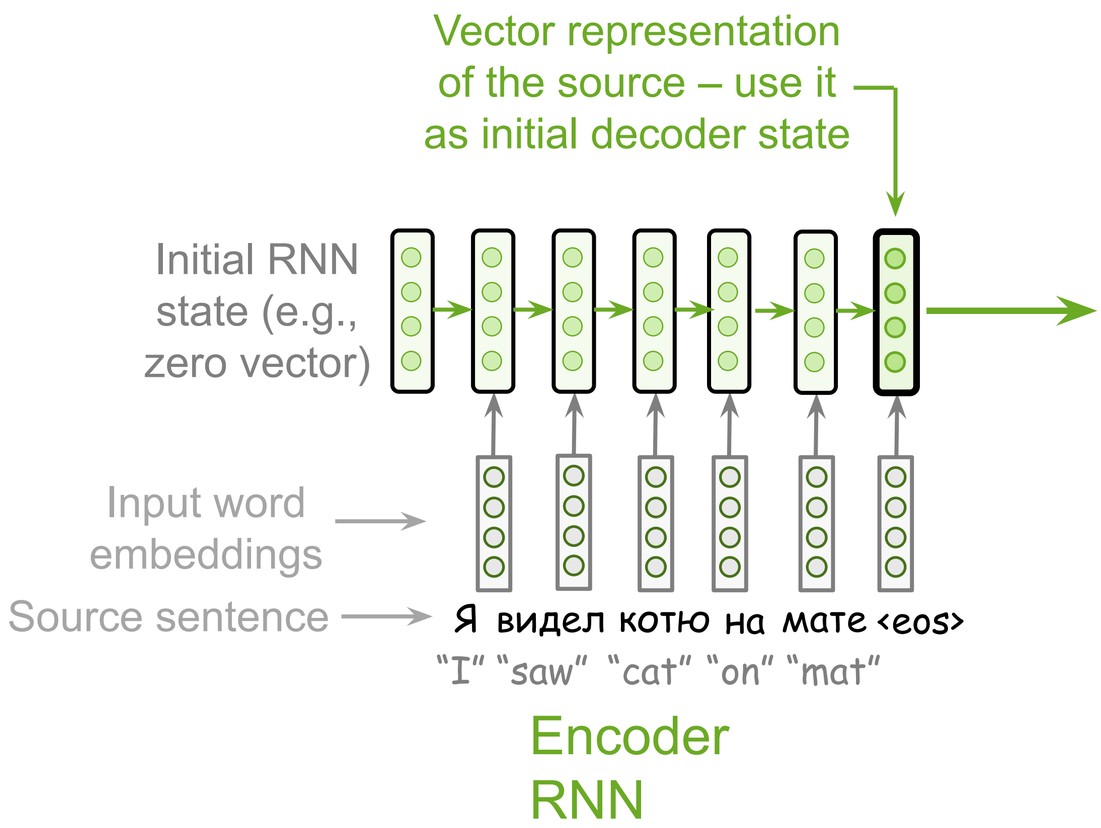

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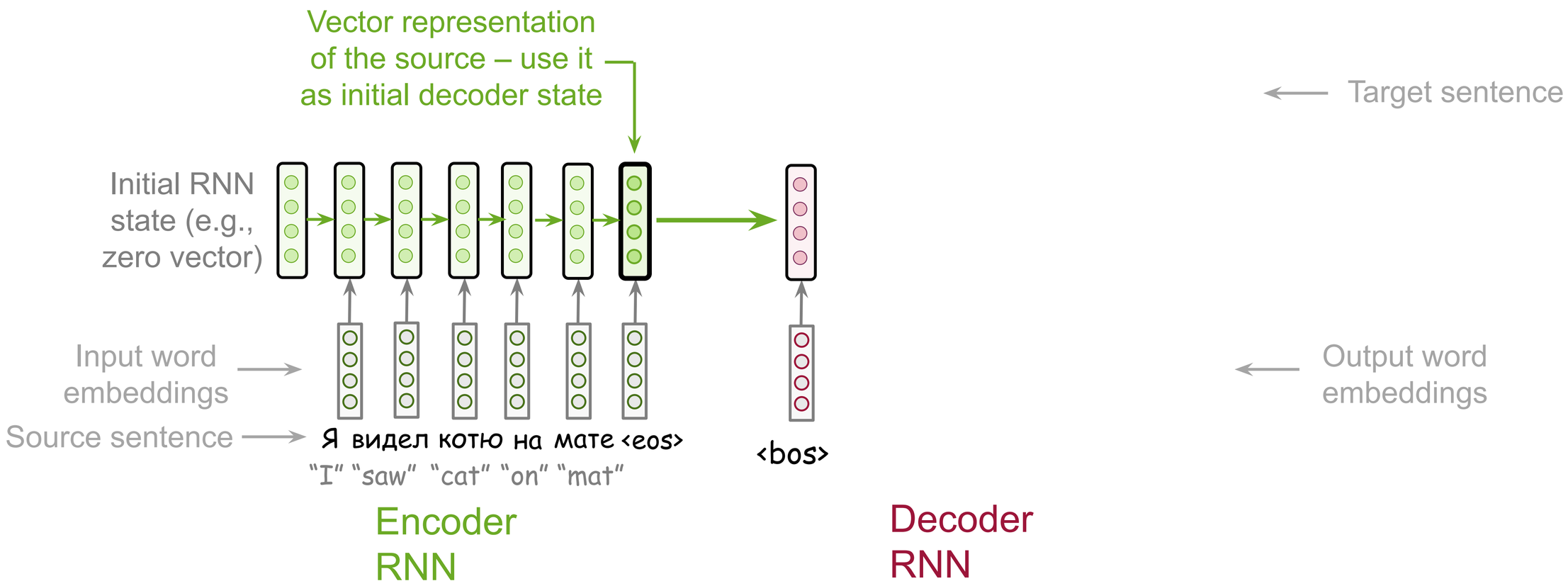

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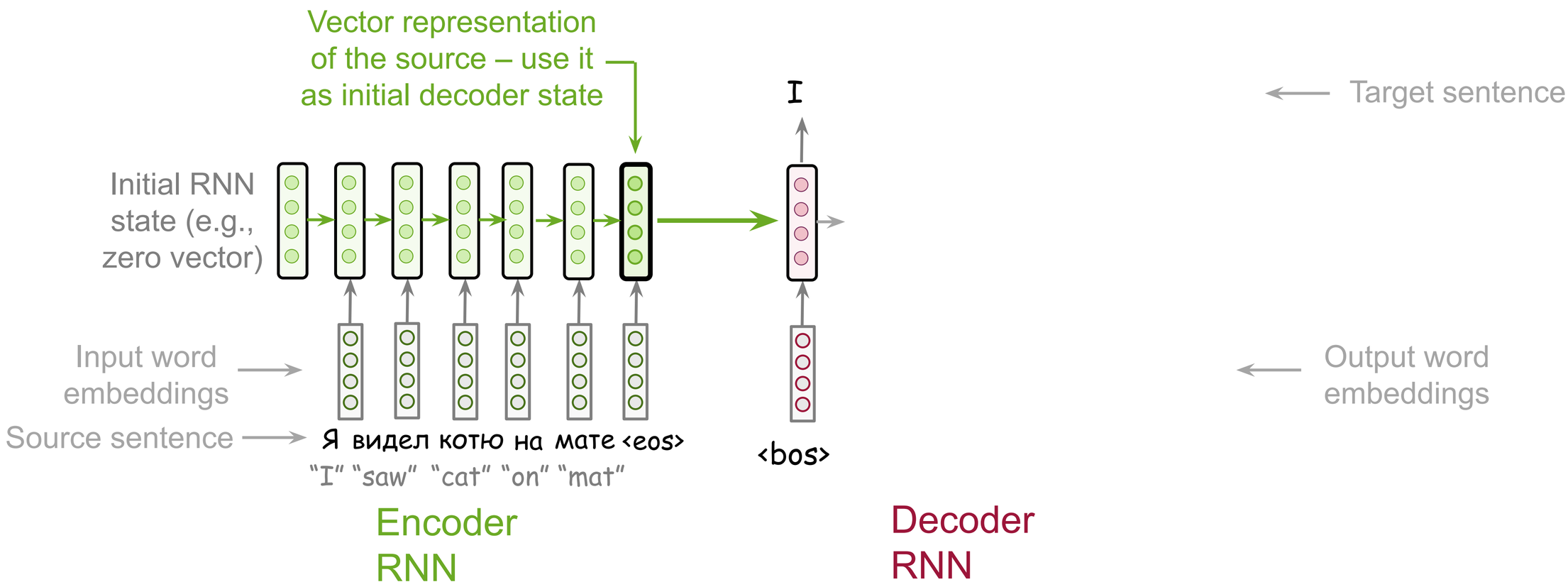

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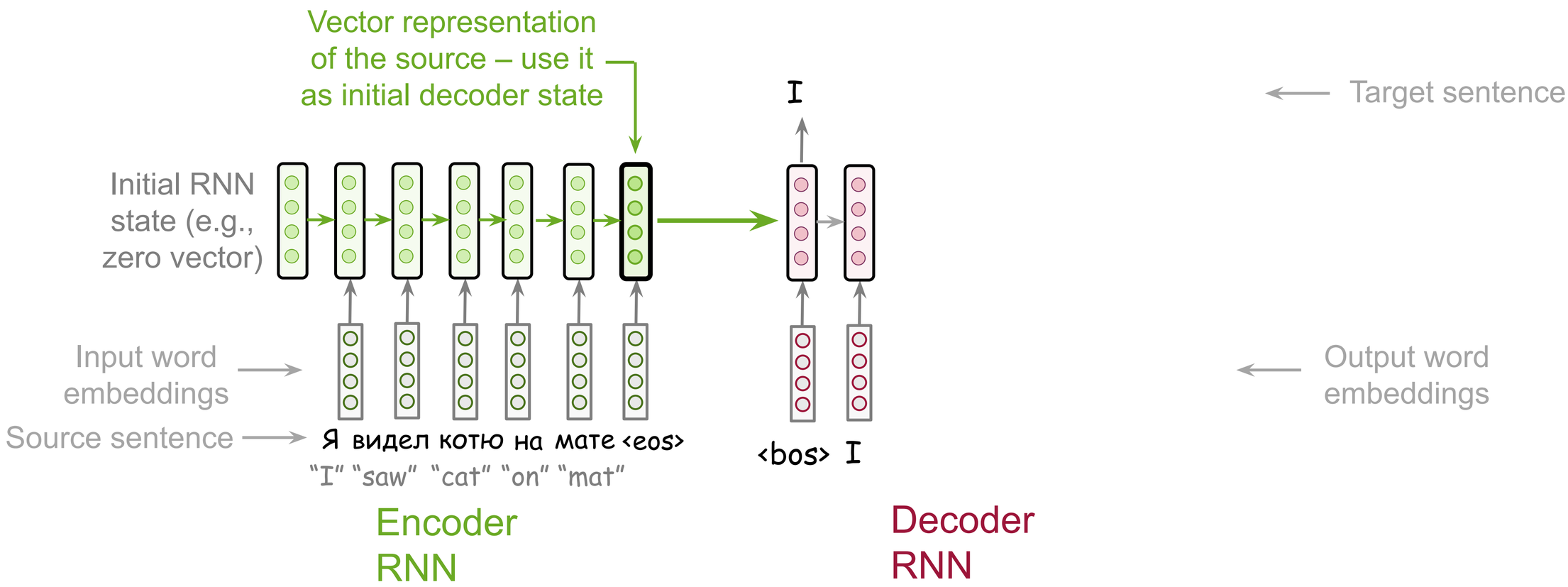

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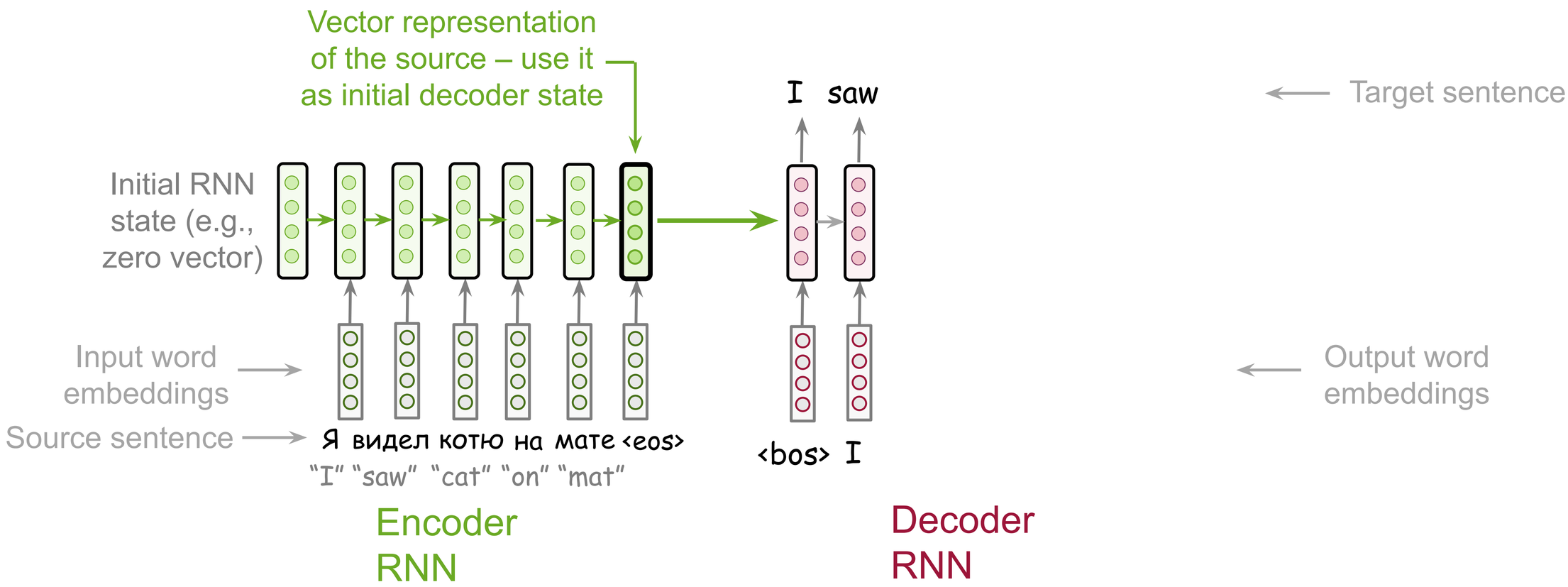

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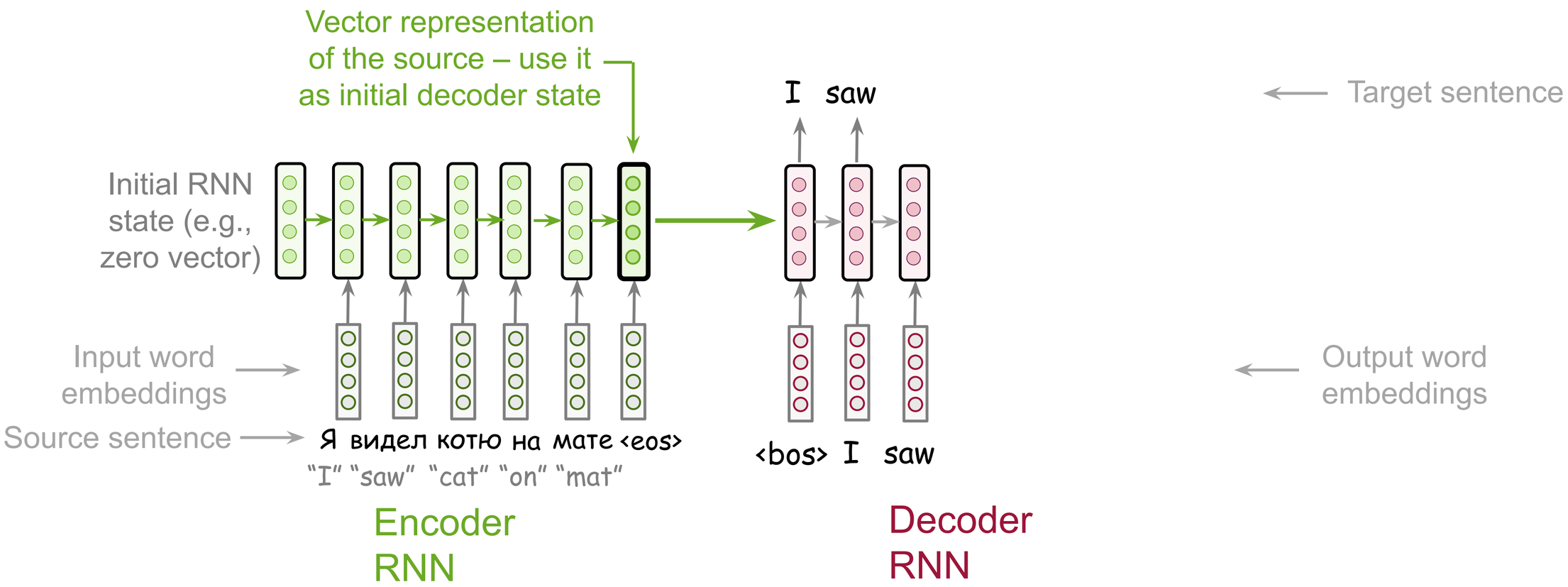

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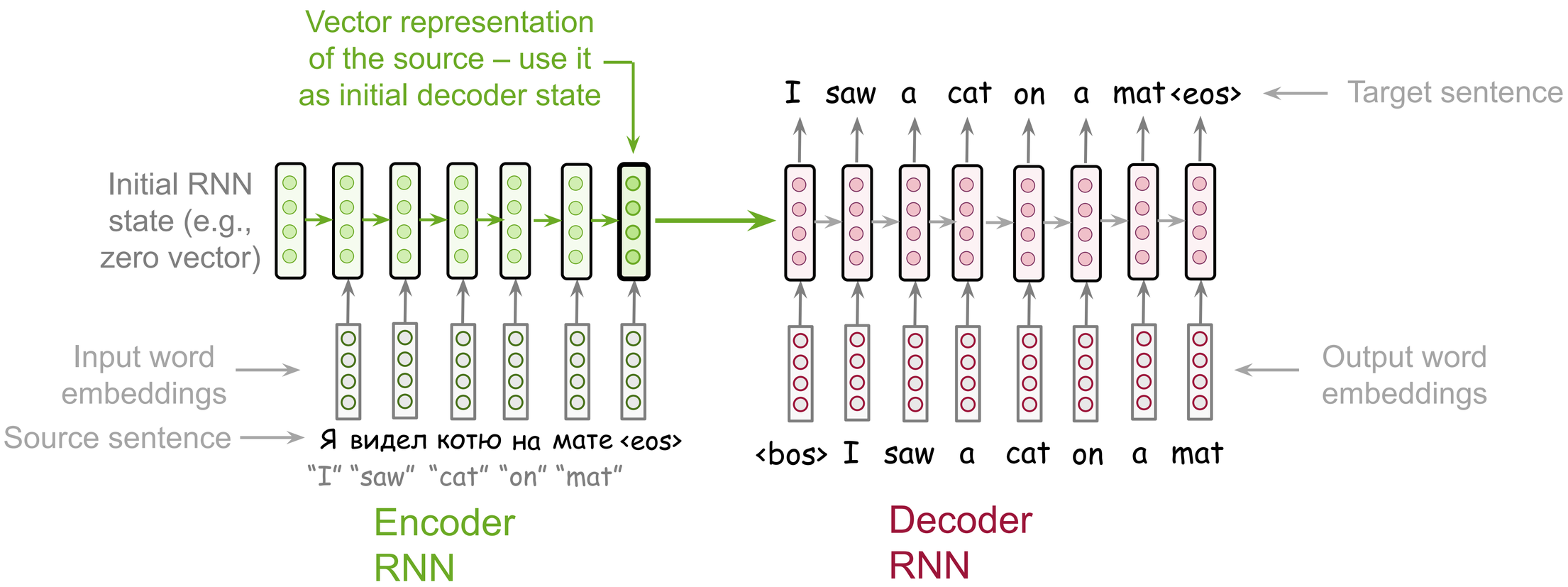
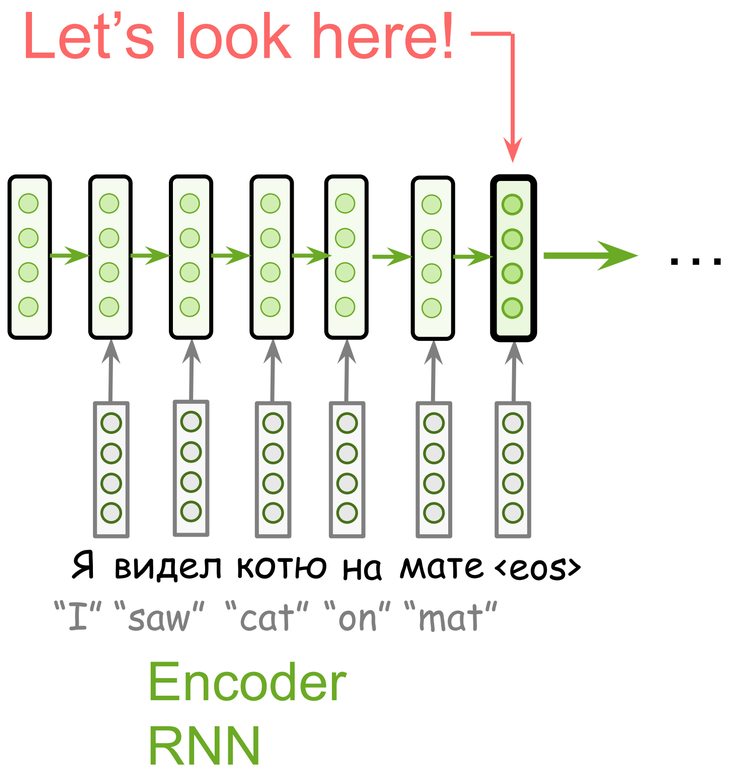
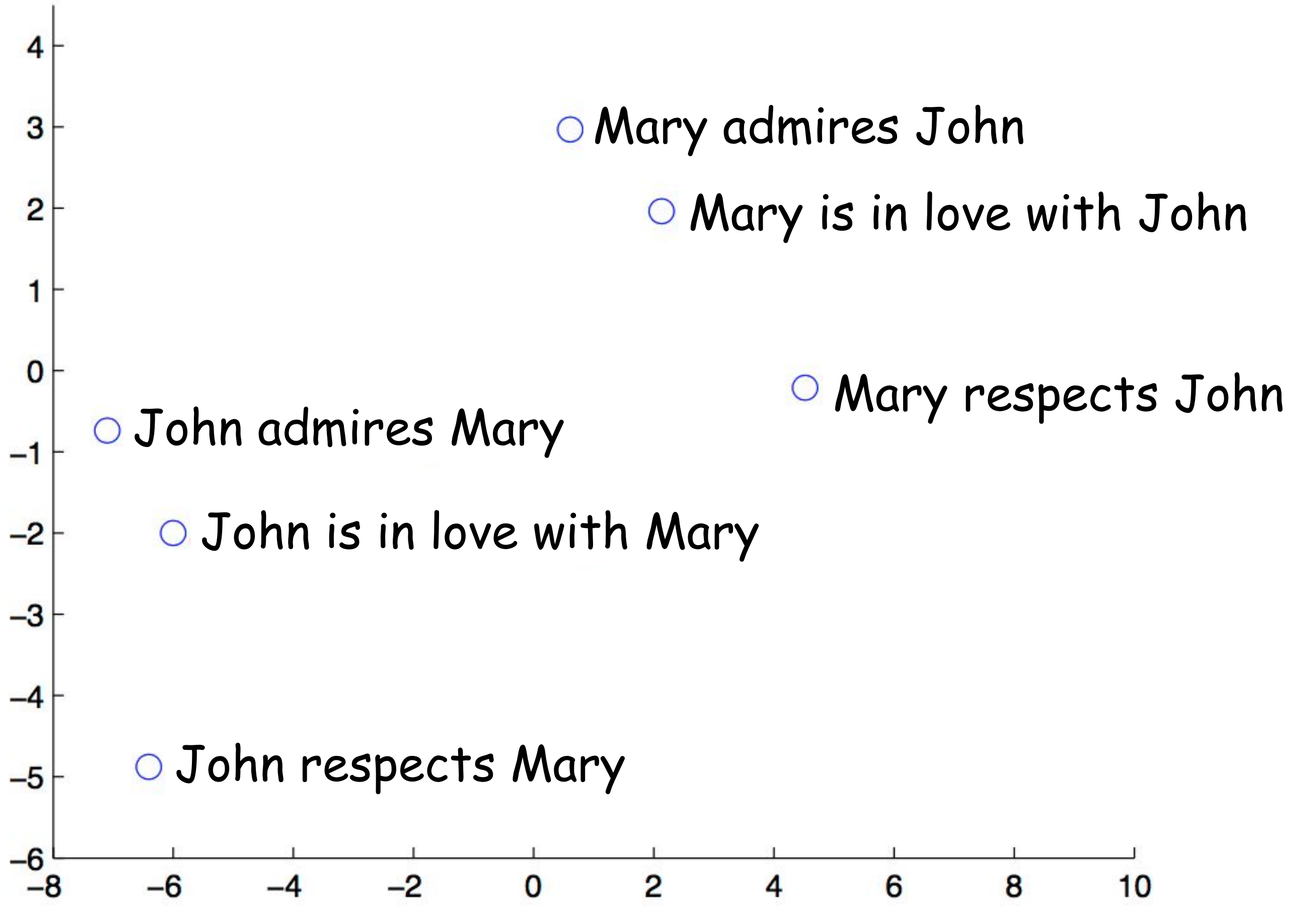


What does final encoder state represent?





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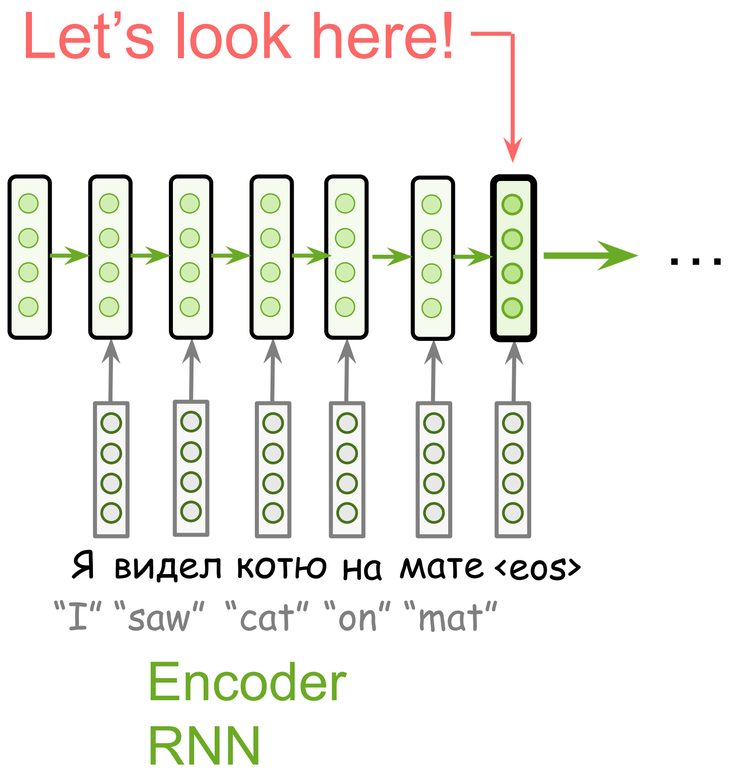

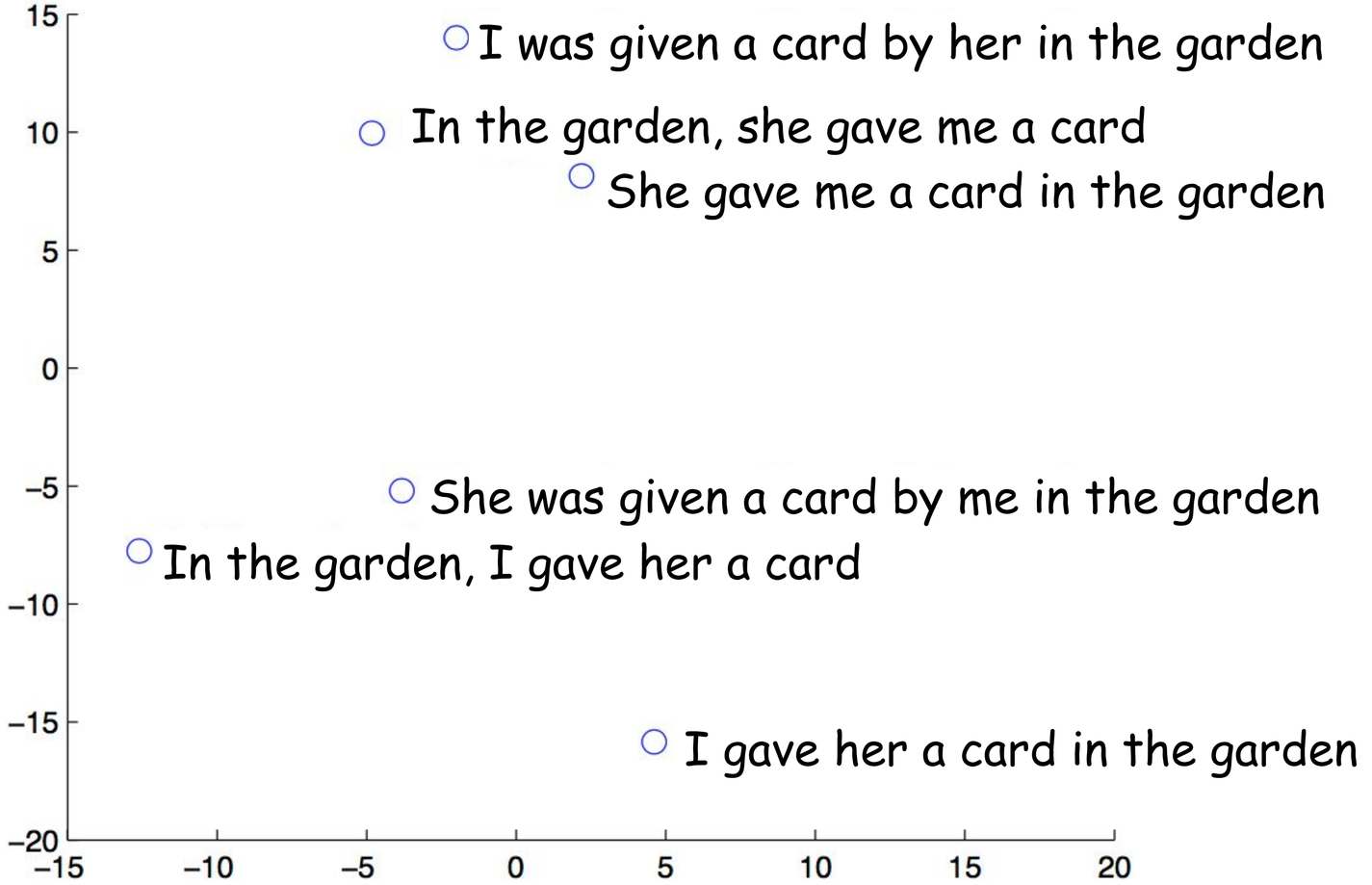

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Cross-Entropy – again!

Source sequence:

Я видел котю на мате <eos>

"I" "saw" "cat" "on" "mat"

Target sequence:

I saw a cat on a mat <eos>

one training example

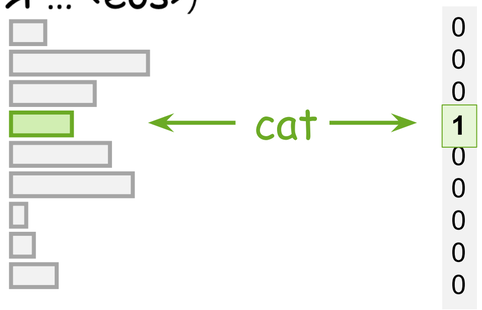
one step for this example

previous tokens

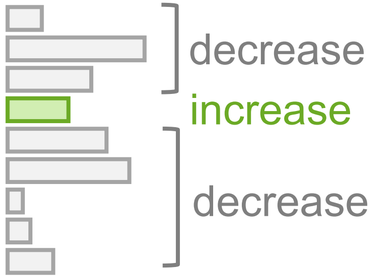
we want the model to predict this

Model prediction: p(* |I saw a, Я ... <eos>)

Target



Loss = -log (p(cat)) → min



Cross-Entropy – again!

Encoder: read source

we are here

Source: Я видел котю на мате <eos>

"I" "saw" "cat" "on" "mat"

Target: I saw a cat on a mat <eos>

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How To Generate a Translation?

$$y' = \arg\max_{y} p(y|x) = \arg\max_{y} \prod_{t=1}^{n} p(y_t|y_{<t}, x)$$

Greedy Decoding

$$y' = \arg\max_{y} p(y|x) = \arg\max_{y} \prod_{t=1}^{n} p(y_t|y_{<t}, x)$$

Straightforward:

- **greedy** - at each step, pick token with the highest probability

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Wait a minute…

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$$\max_y \prod_{t=1}^{n} p(y_t|y_{<t}, x) \neq \prod_{t=1}^{n} \max_{y_t} p(y_t|y_{<t}, x)$$

- this is bad!

Beam Search

- At each step, keep several best hypotheses

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<bos>

Start with the begin of sentence token or with an empty sequence

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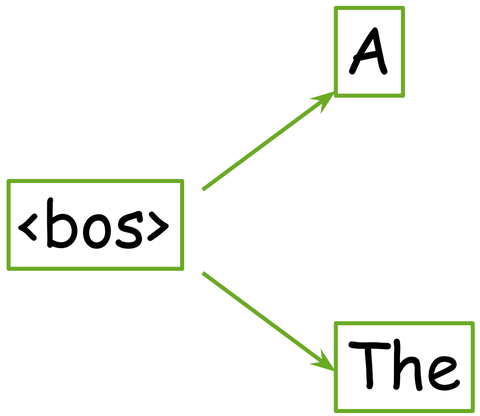

Generate: `beam_size` most probable tokens

Beam Search

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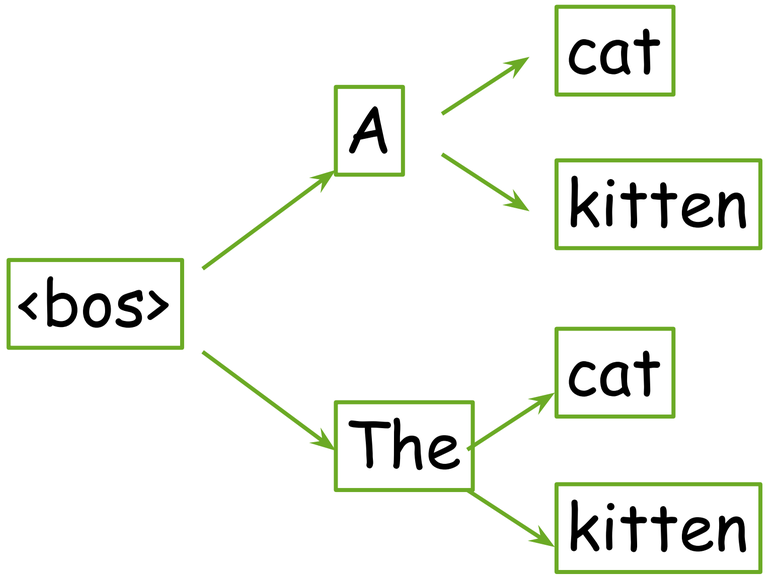

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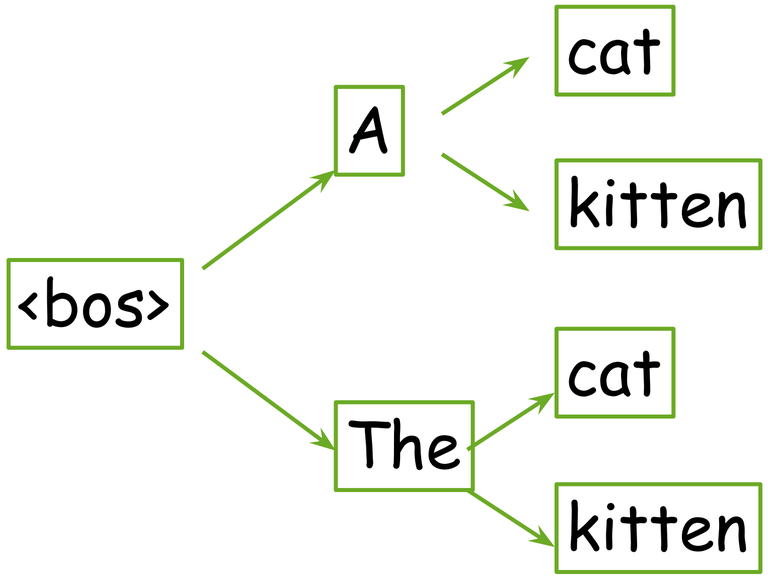

Look at probabilities: P(The cat) > P(A kitten) > P(A cat) > P(The kitten)

Top 2 hypos

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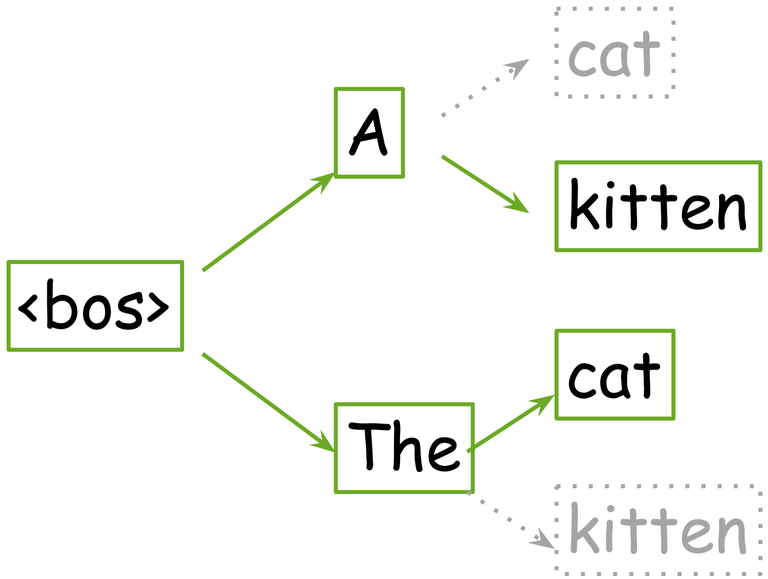

Pick top `beam_size` hypos, terminate the rest

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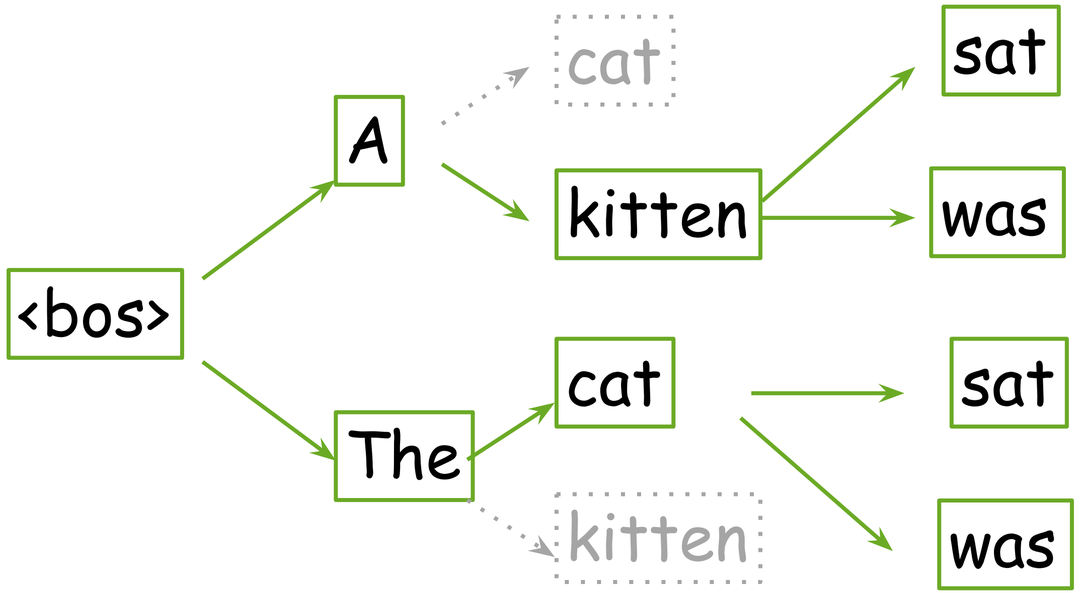

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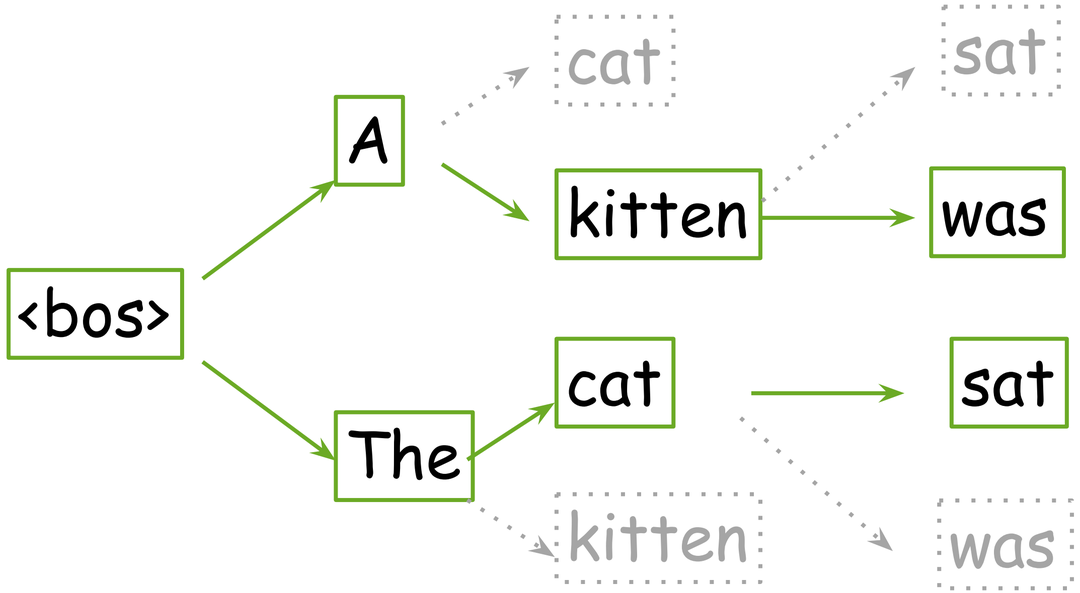

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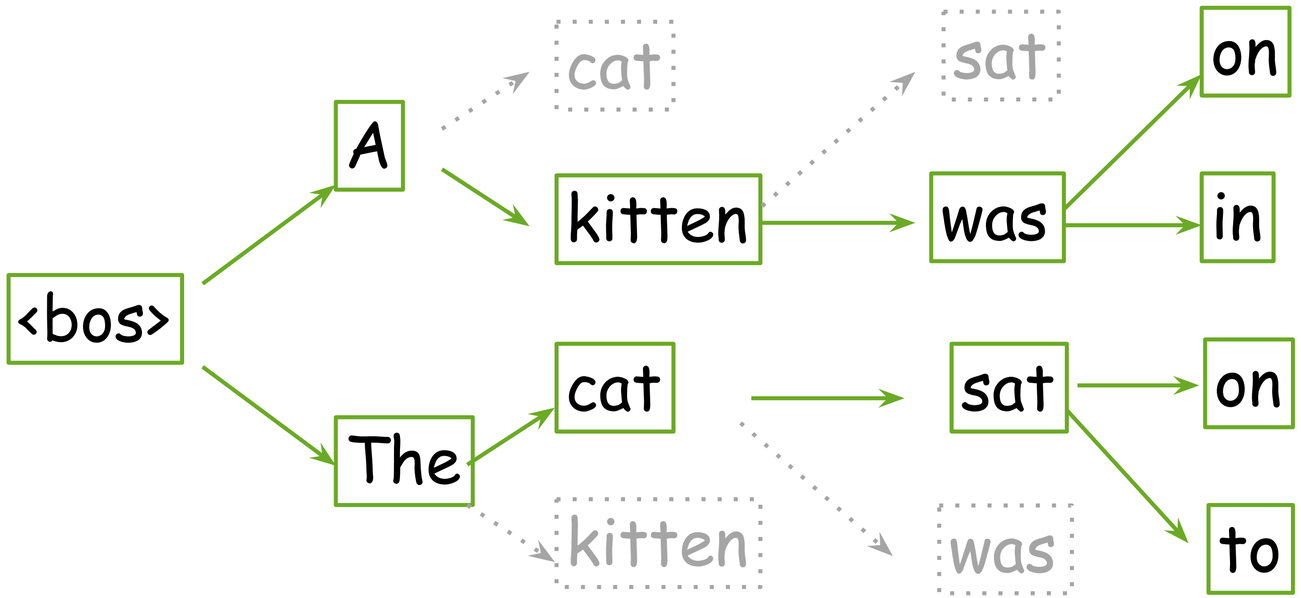

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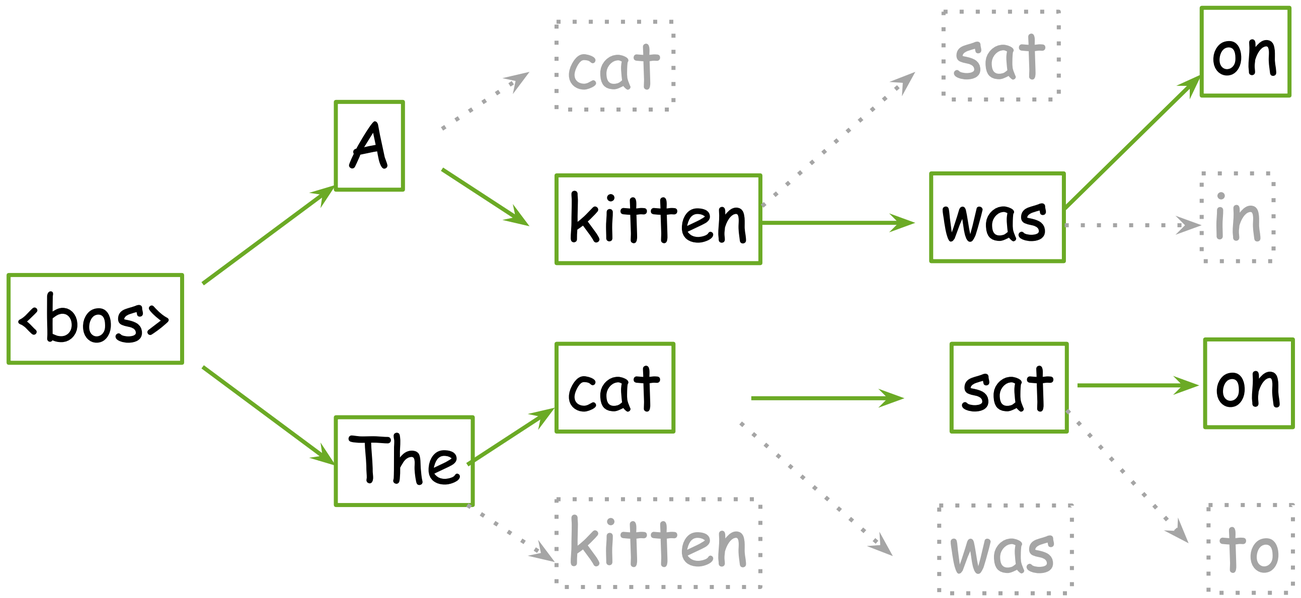

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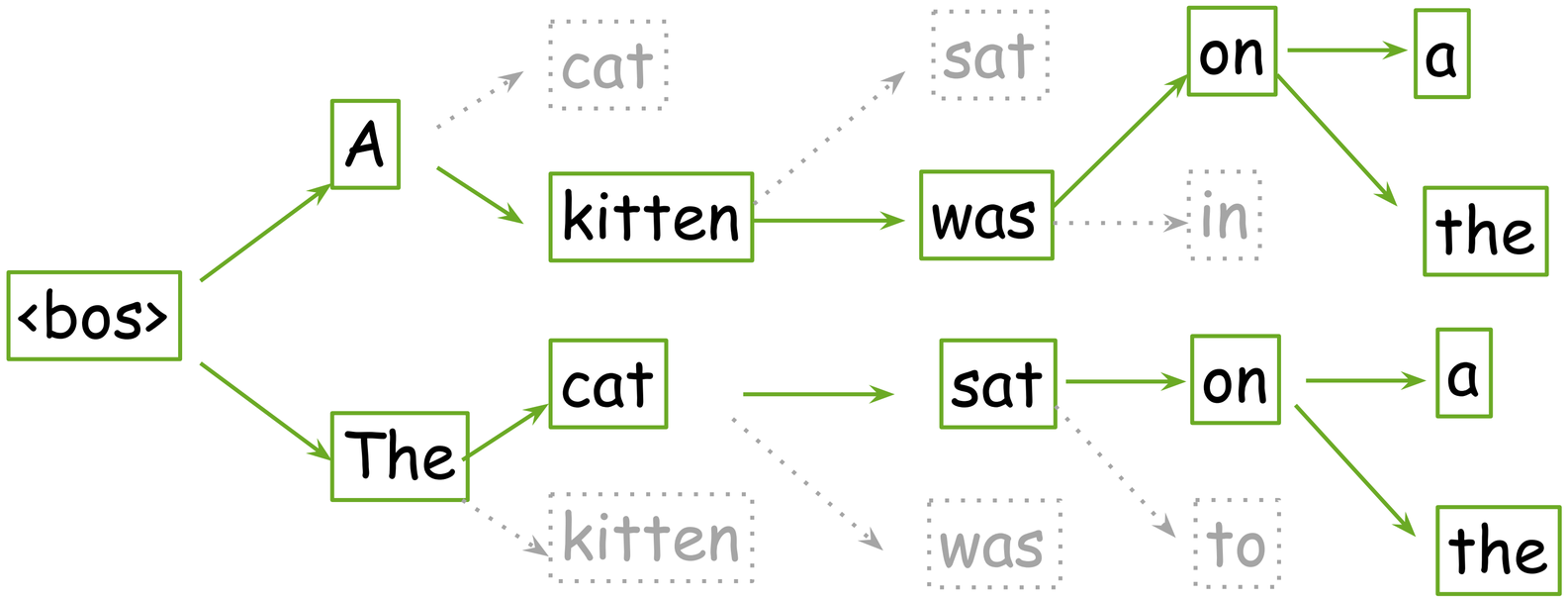

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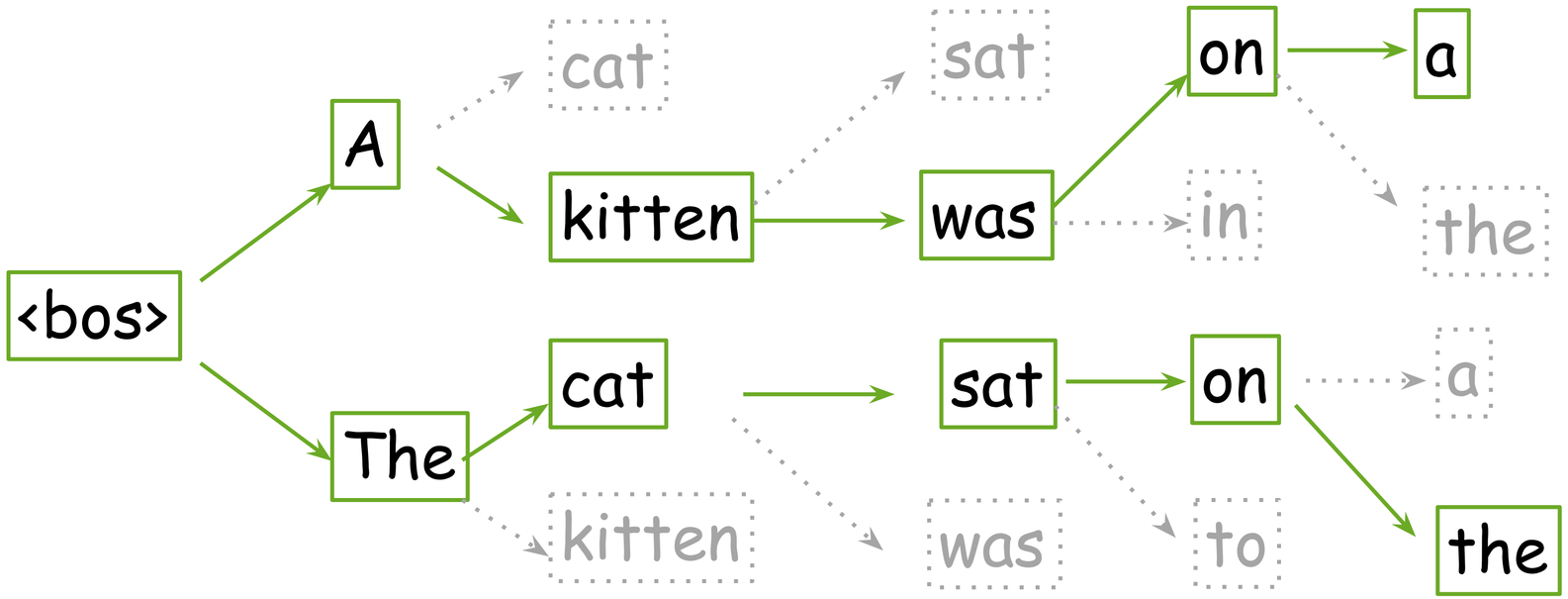

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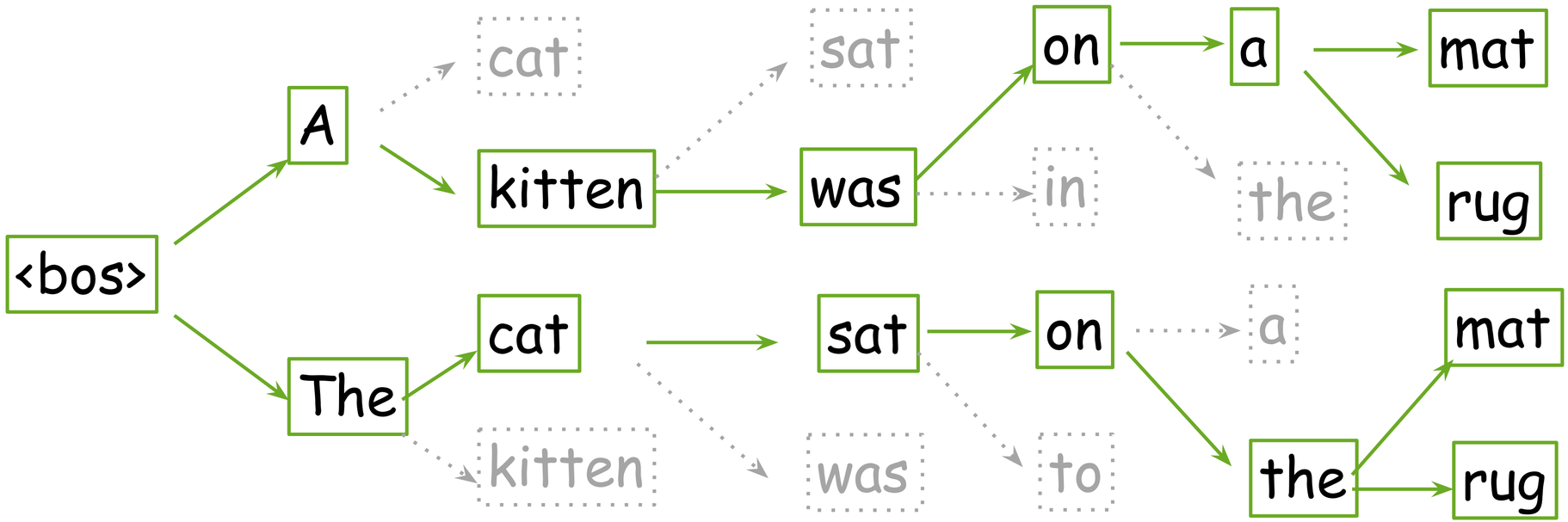

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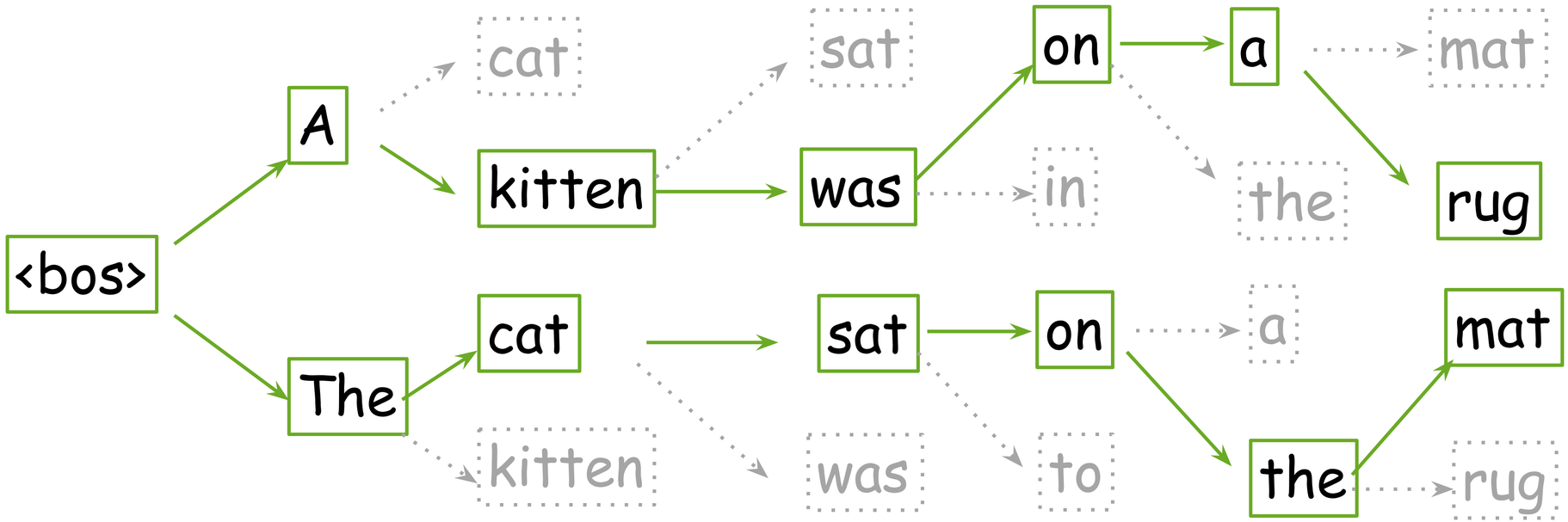

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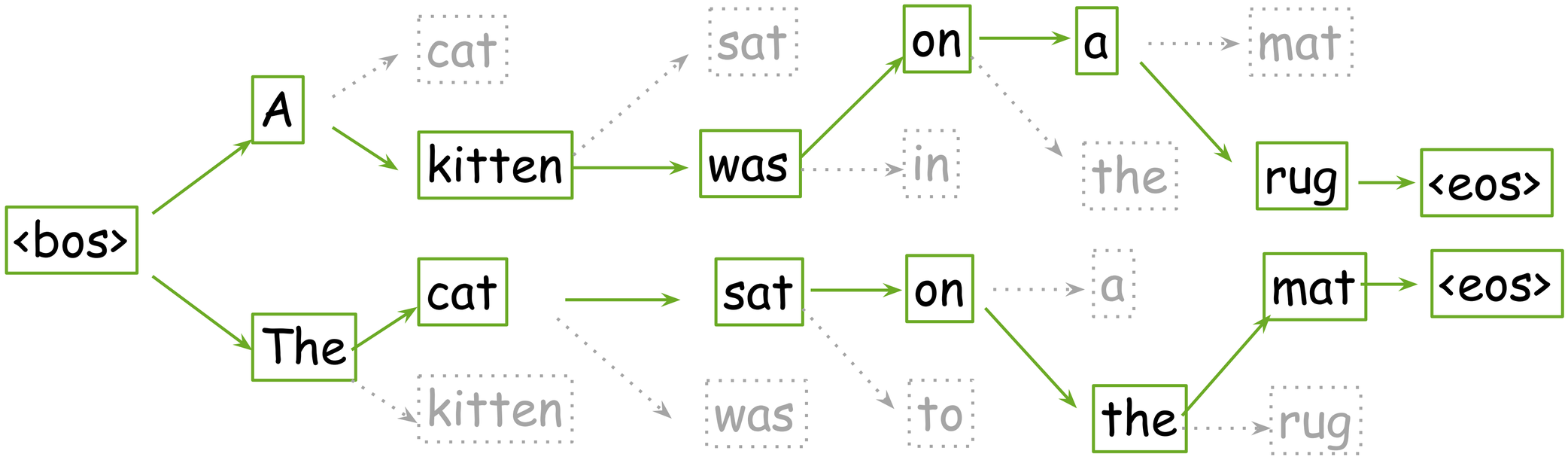
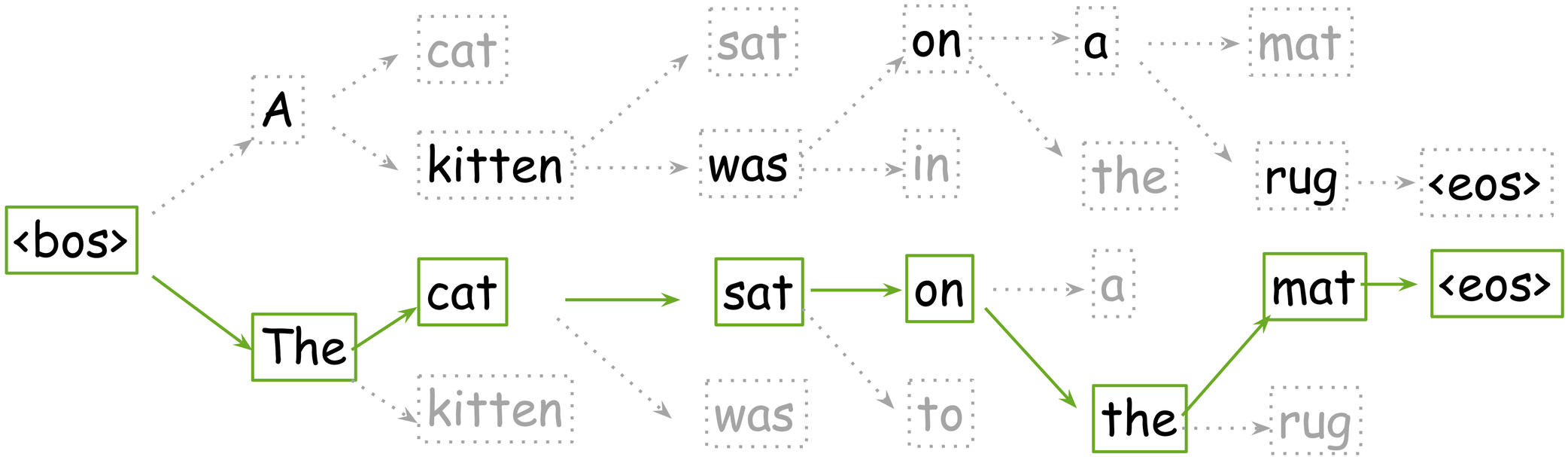


All hypotheses are complete - generation ended

Beam Search

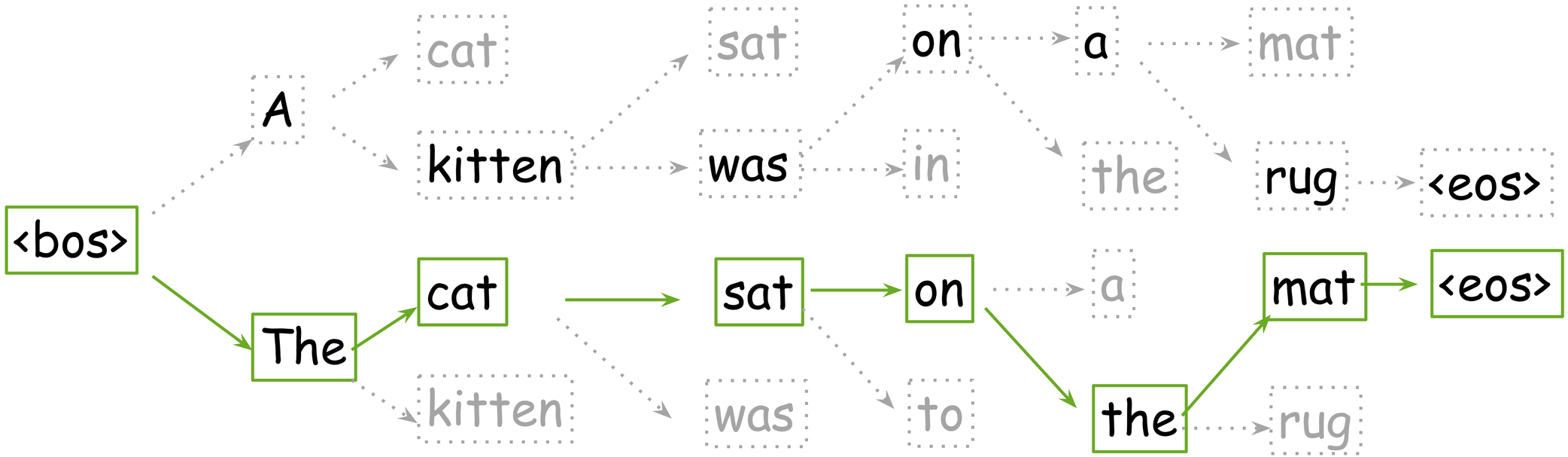
- At each step, keep several best hypotheses



Pick the hypothesis with the highest probability

Beam Search

- At each step, keep several best hypotheses



Result: The cat sat on the mat

What is going to happen:

- Seq2seq Basics
- Attention
- Transformer
- Subword Segmentation: BPE
-  Analysis and Interpretability

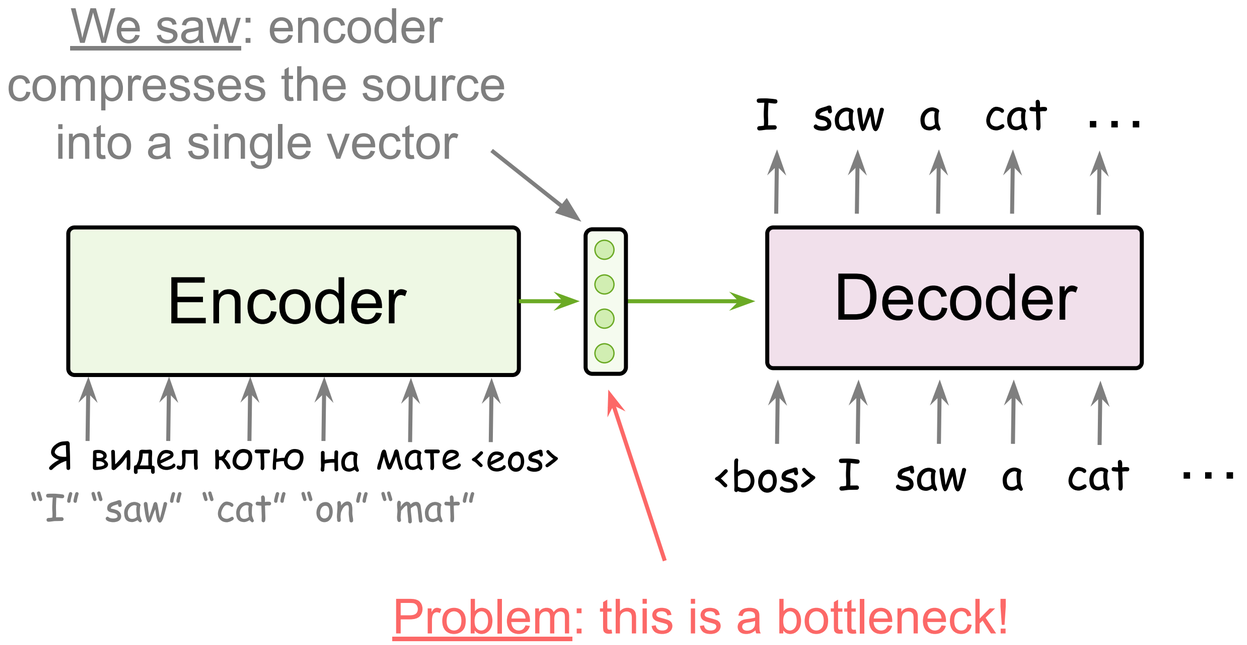
What is going to happen:

- Seq2seq Basics
- Attention
  - Why do we need it?
  - Attention: High-Level
  - Attention Score Functions
  - Models: Bahdanau vs Luong
- Transformer
- Subword Segmentation: BPE
-  Analysis and Interpretability

The Problem of Fixed Encoder Representation

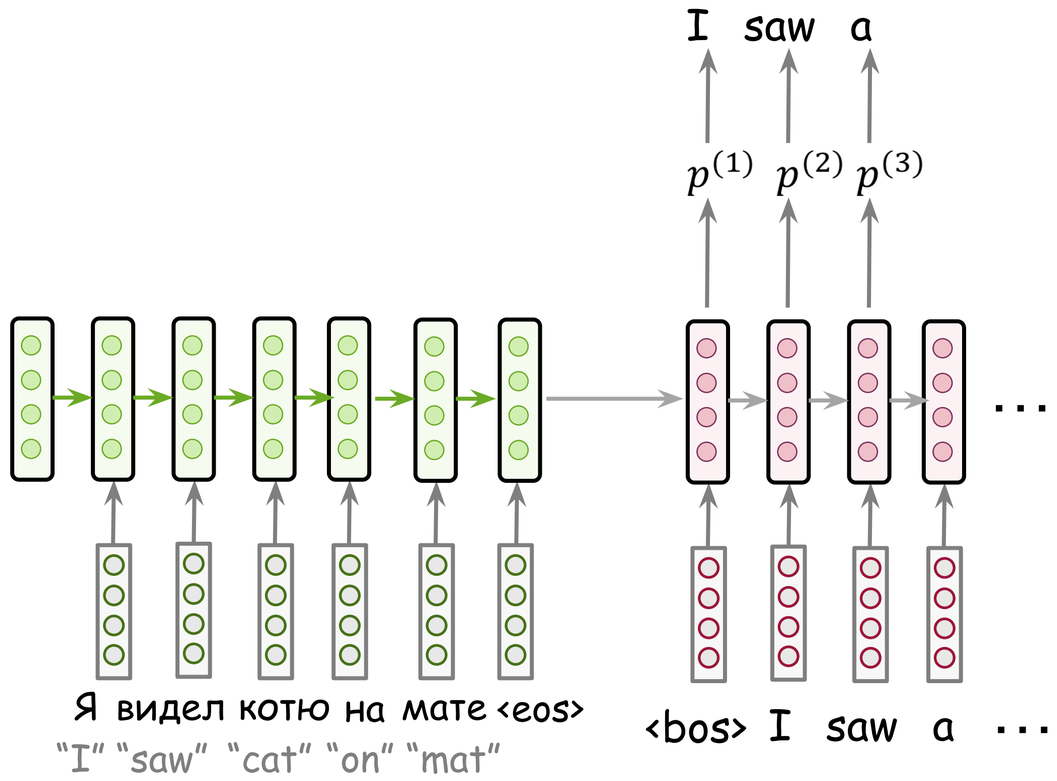
Fixed source representation is bad:

- for **encoder**, it is hard to compress a sentence
- for **decoder**, different information may be needed at different steps



Attention: High-Level View

> Attention: At different steps, let a model "focus" on different parts of the input.



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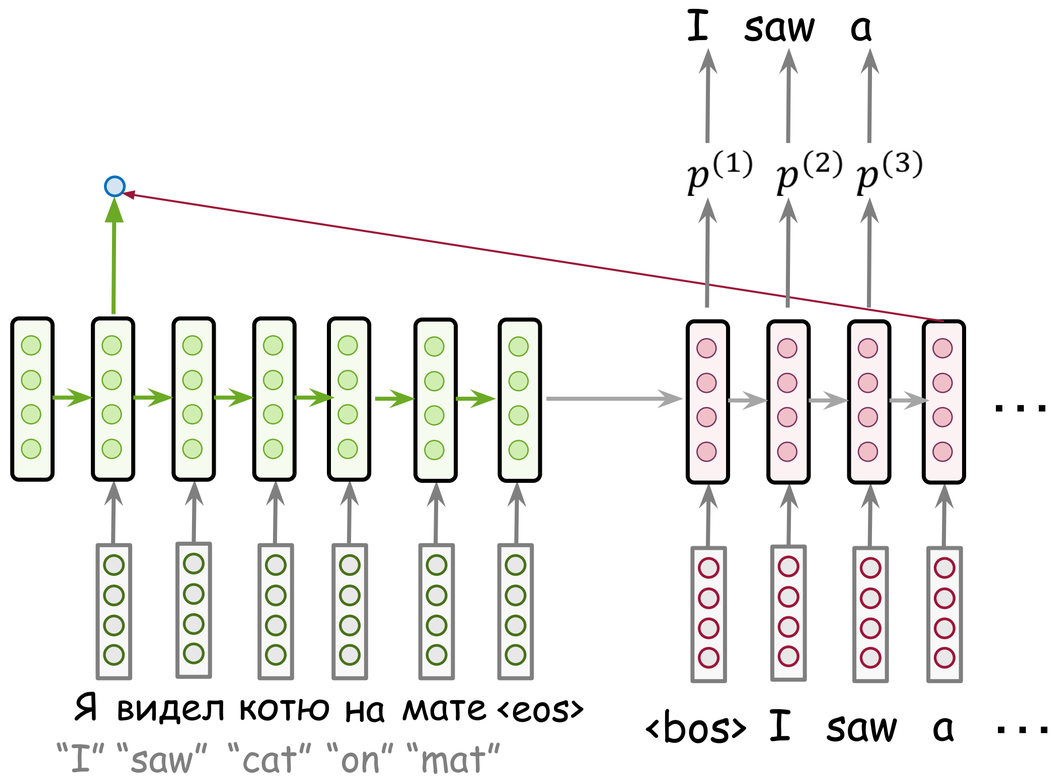

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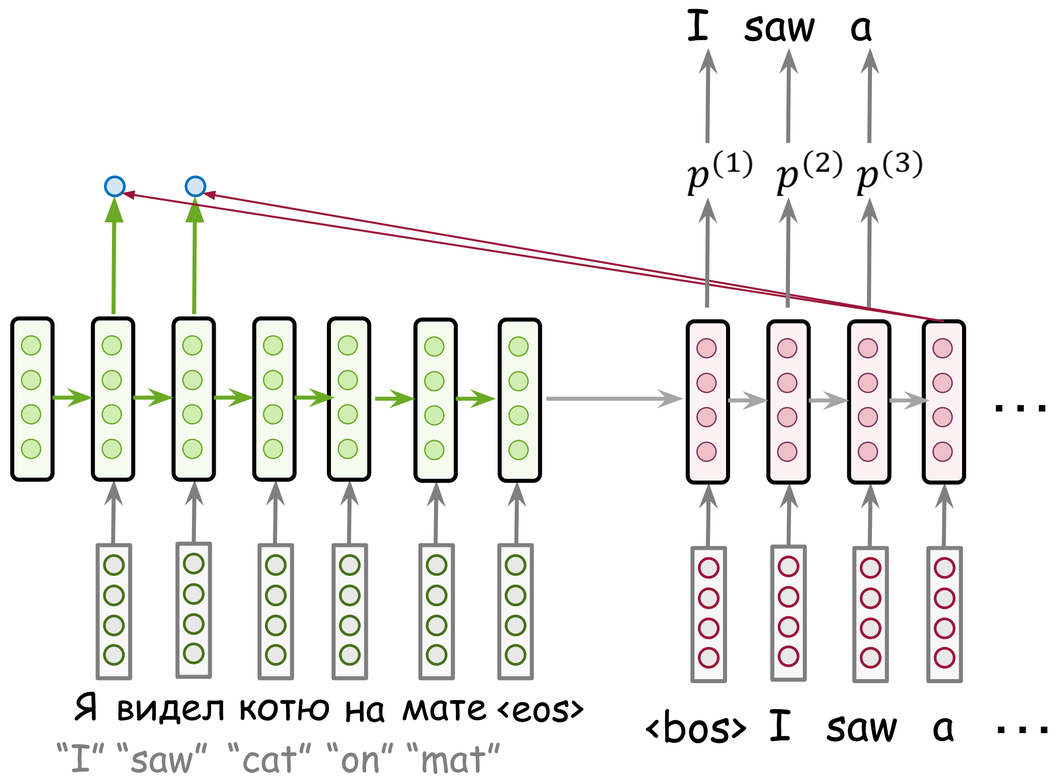

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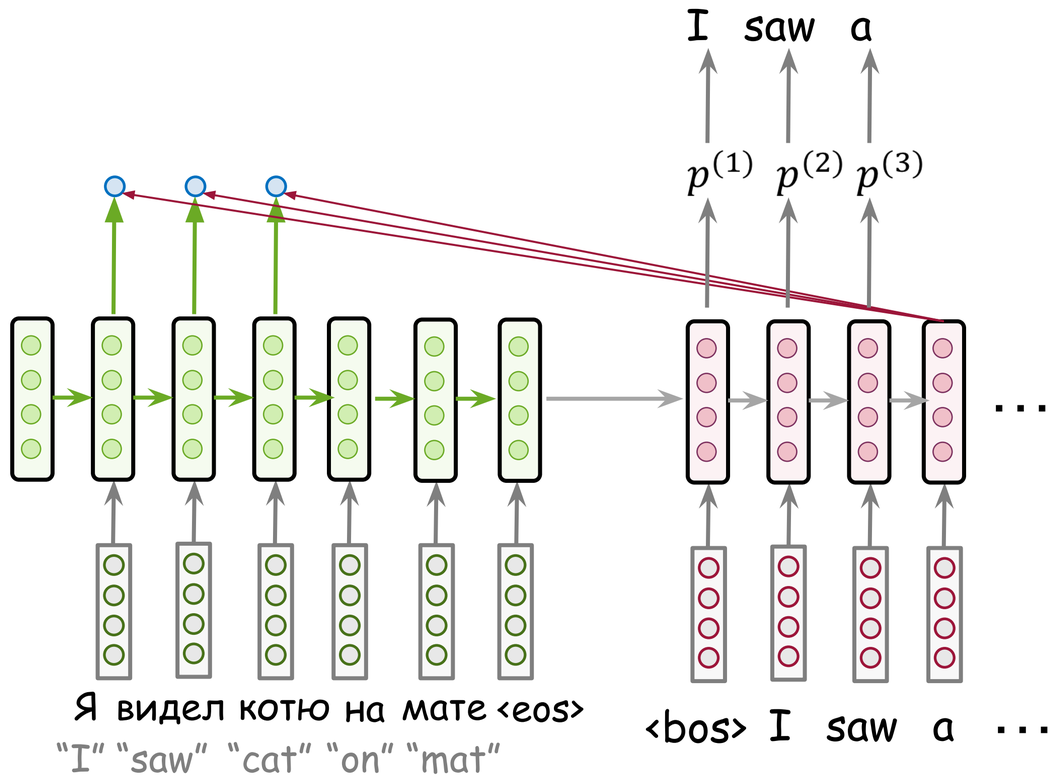

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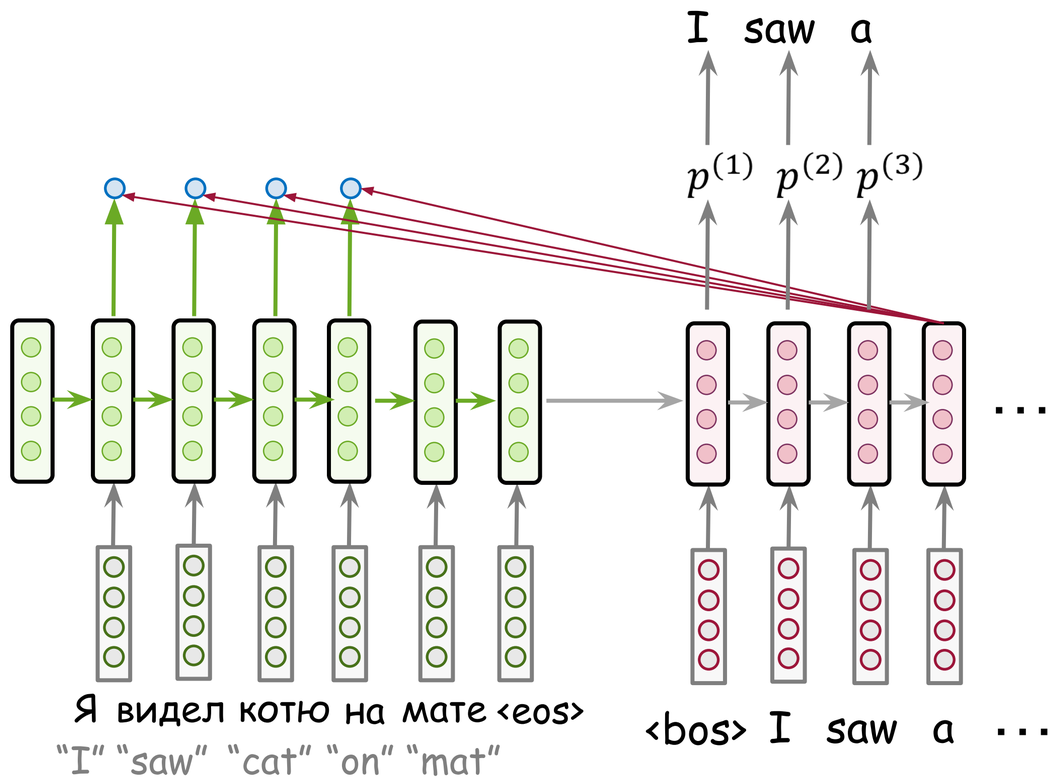

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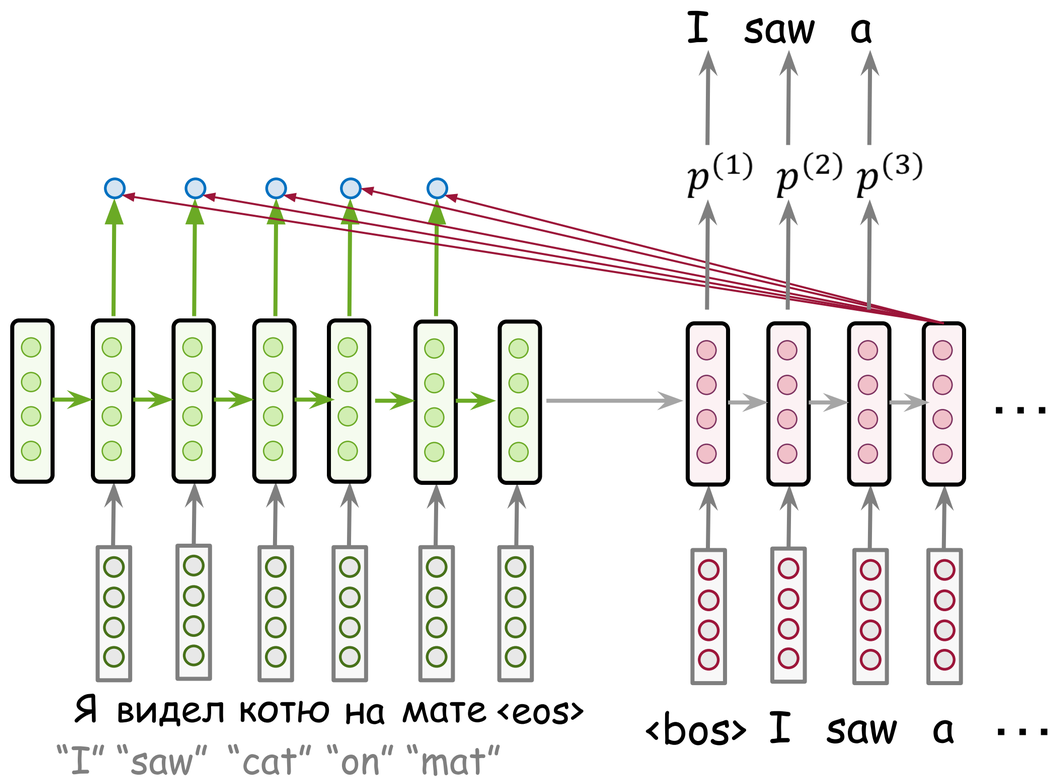

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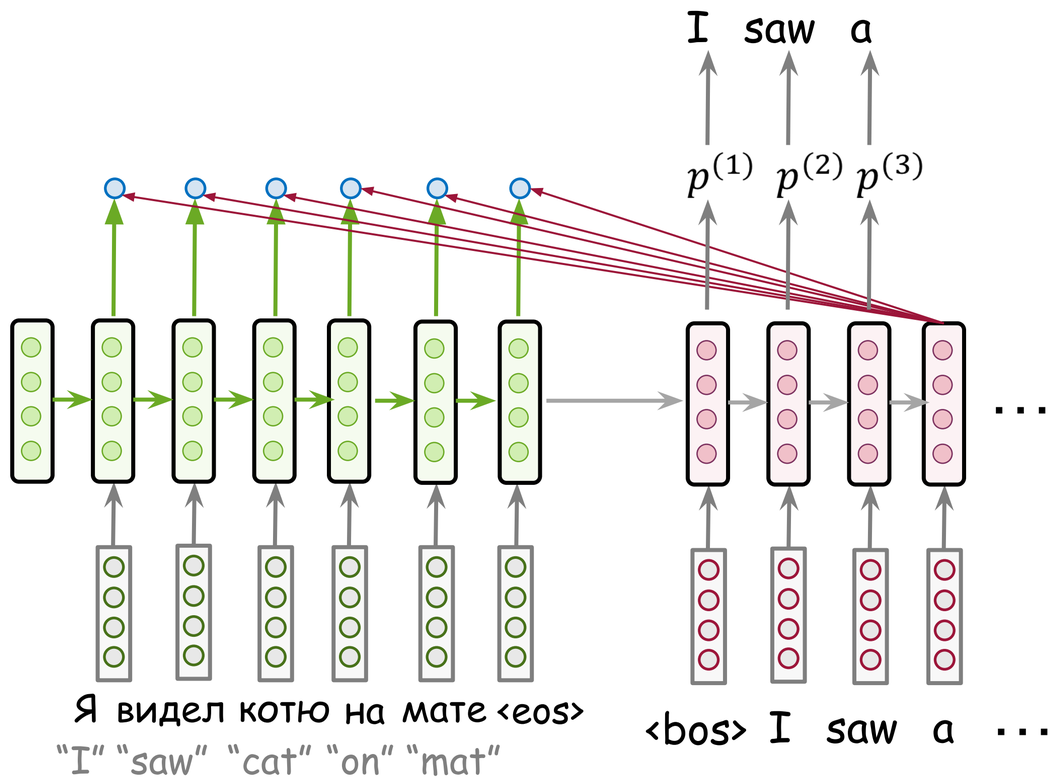

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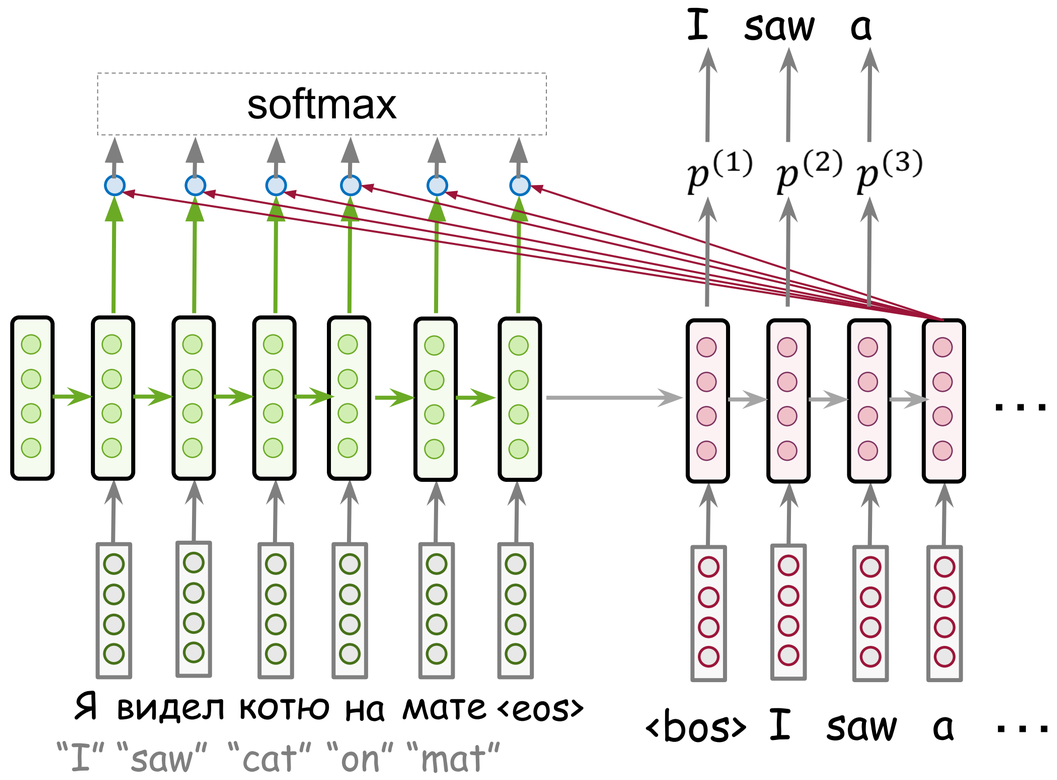

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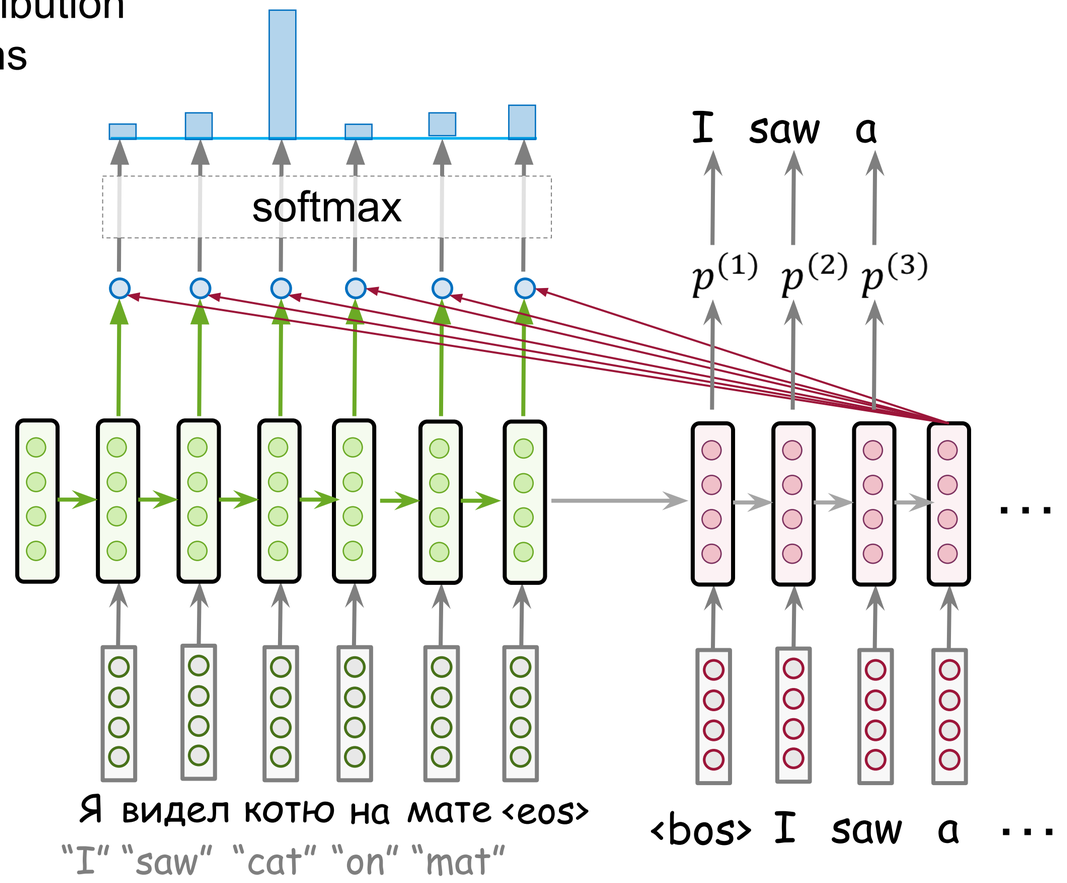
Attention weights: distribution over source tokens



Attention: High-Level View

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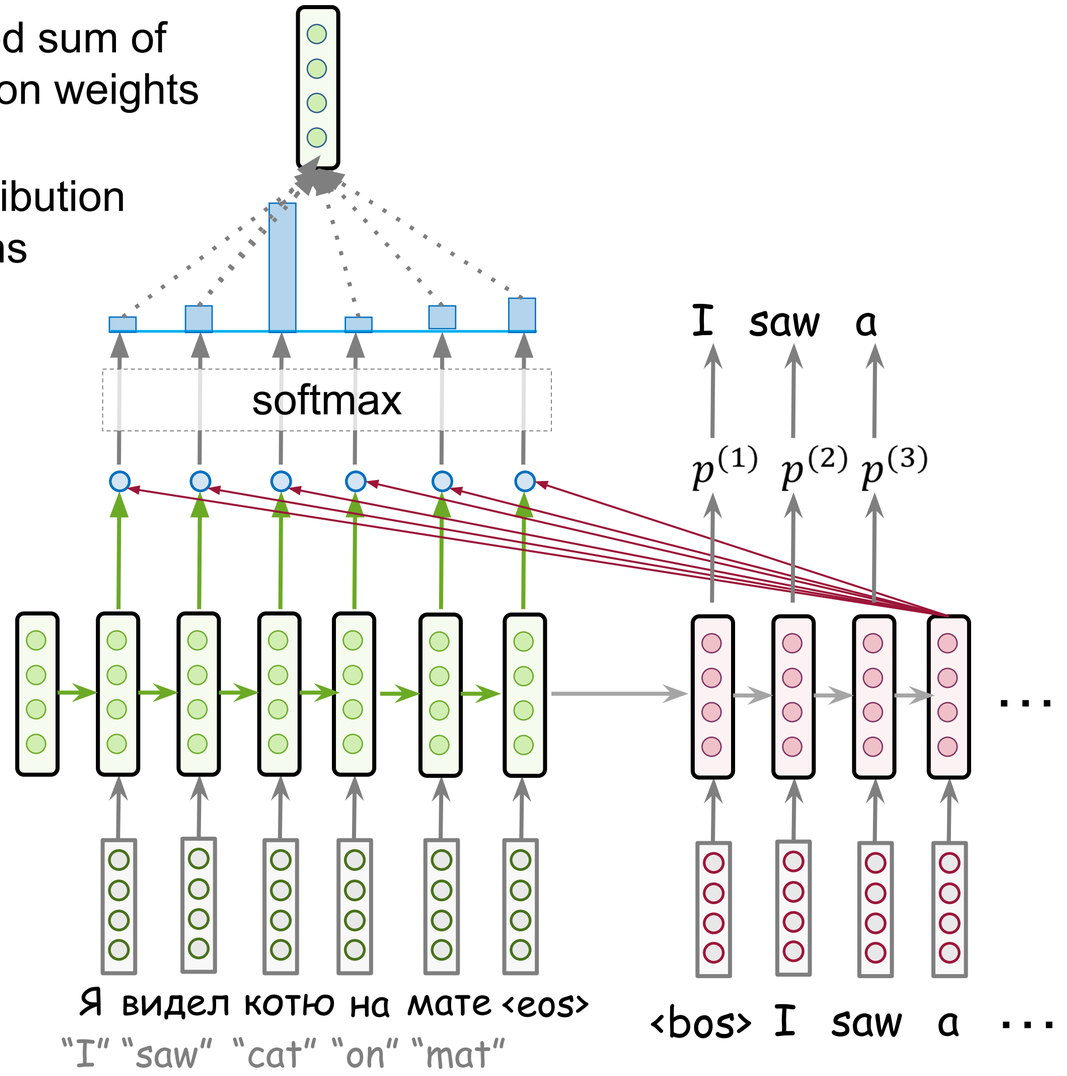
Attention output: weighted sum of encoder states with attention weights

Attention weights: distribution over source tokens



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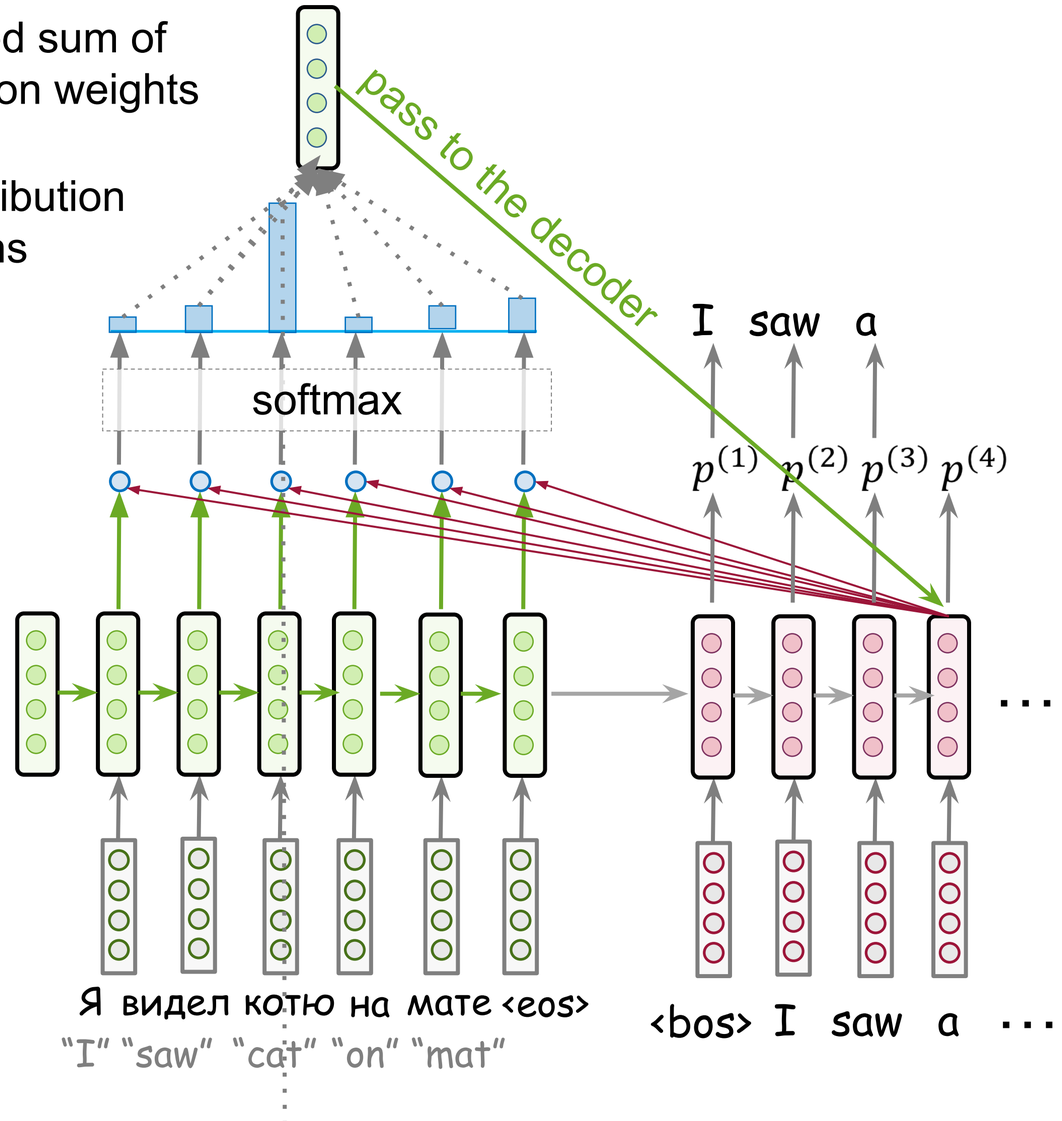

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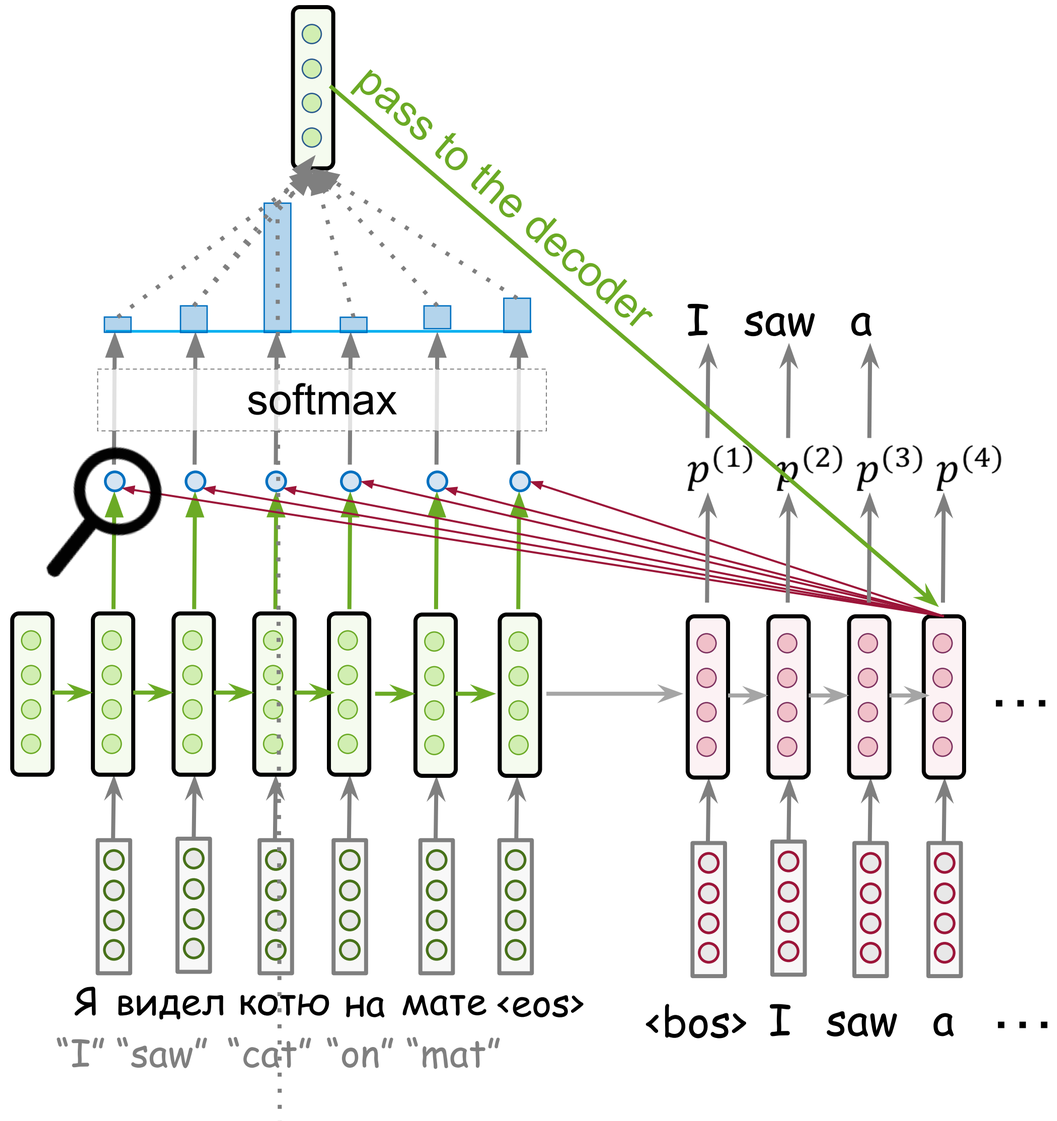

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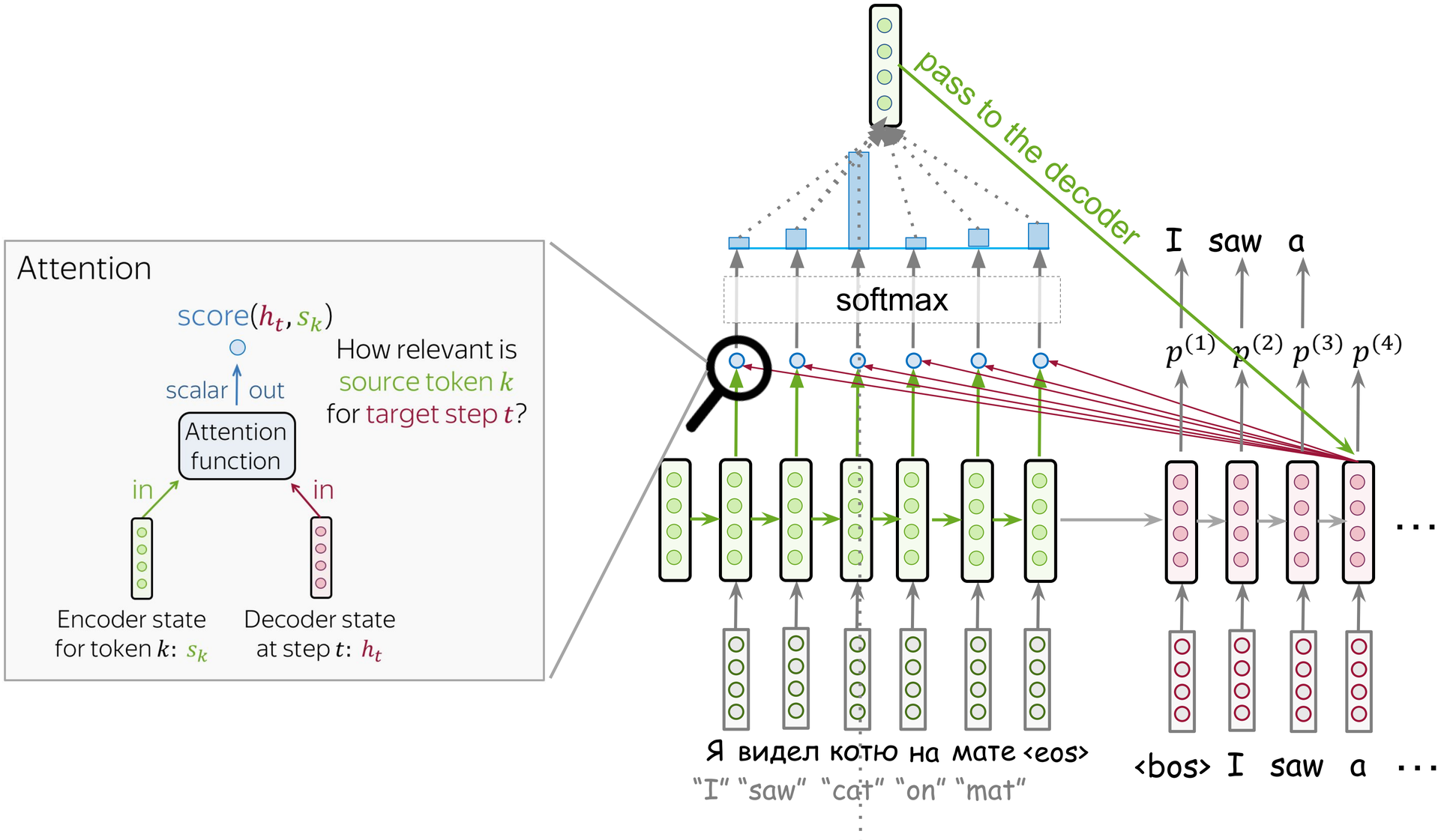
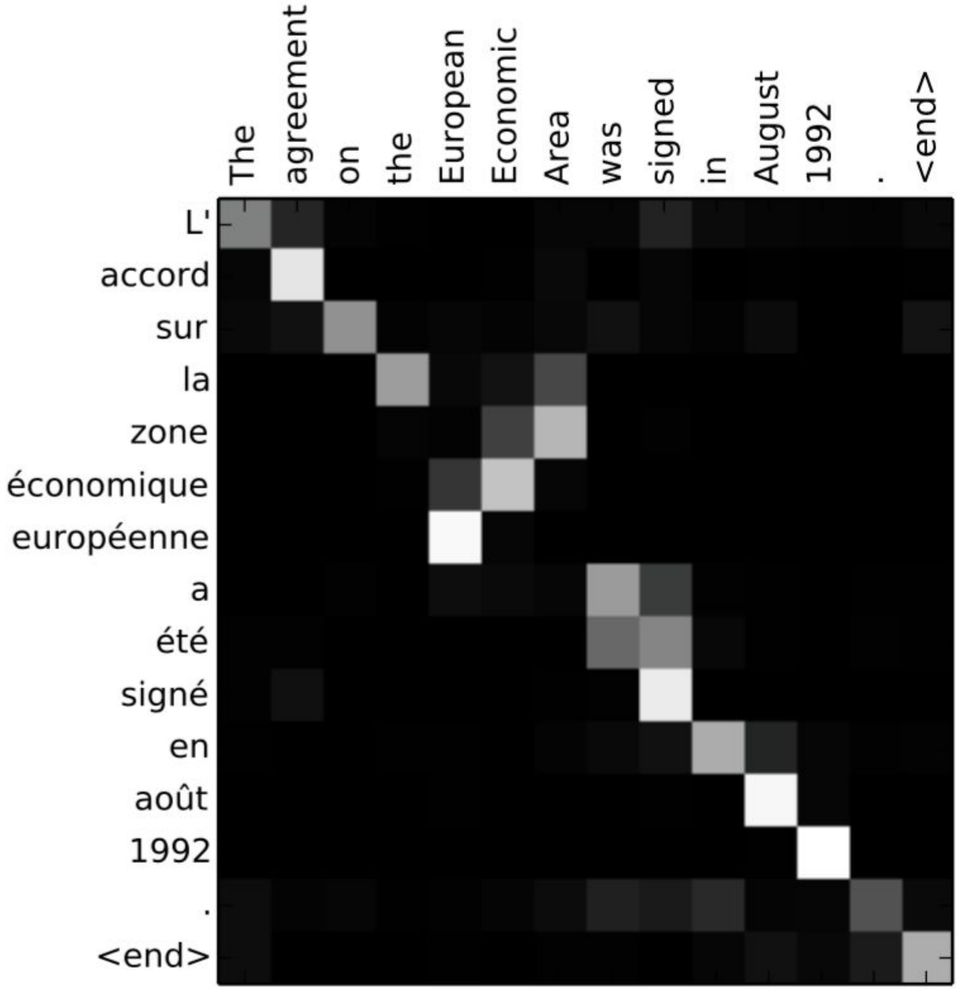
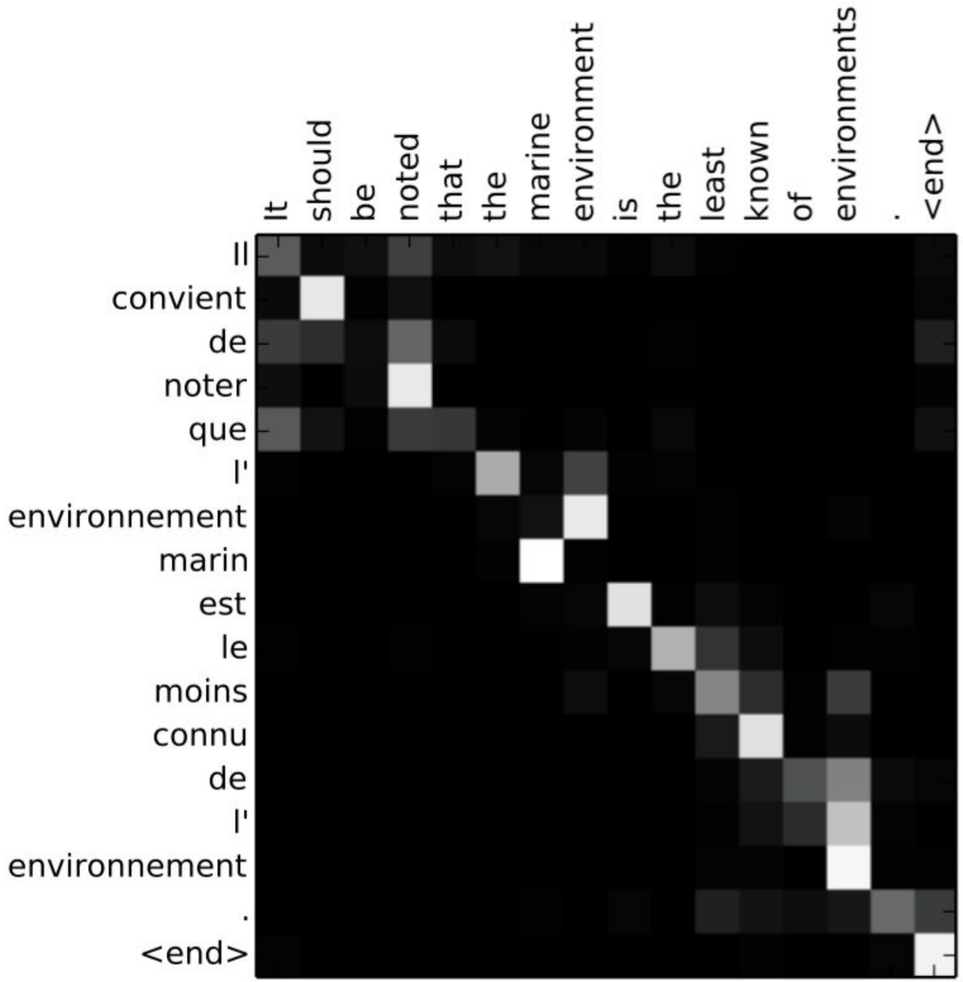


Attention Learned (almost) Alignment





The examples are from the paper Neural Machine Translation by Jointly Learning to Align and Translate.

Computation Pipeline

Attention input

$s_1, s_2, \ldots, s_m$

all encoder states

h_t

one decoder state

Computation Pipeline

Attention scores

$\text{score}(h_t, s_k), \text{k} = 1..\text{m}$

Attention input

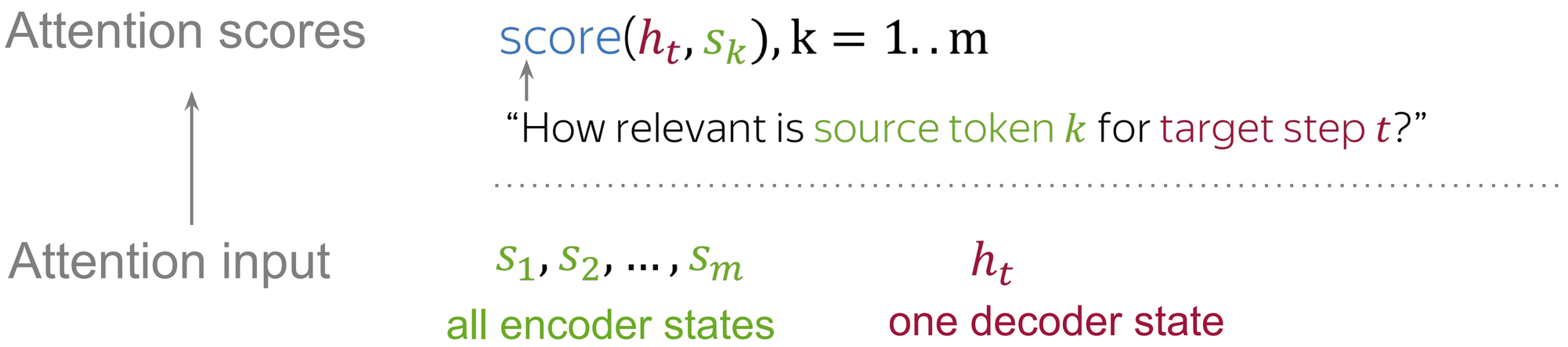
$s_1, s_2, \ldots, s_m$

all encoder states

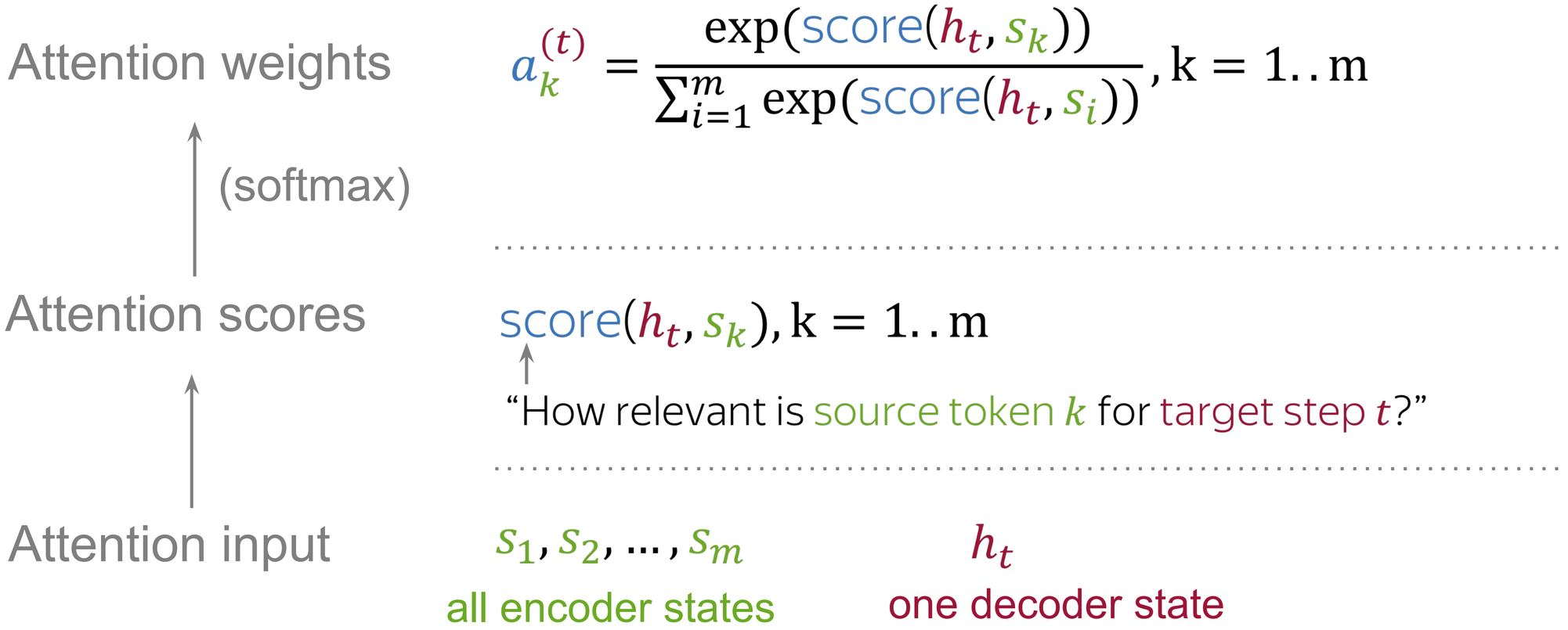
h_t

one decoder state

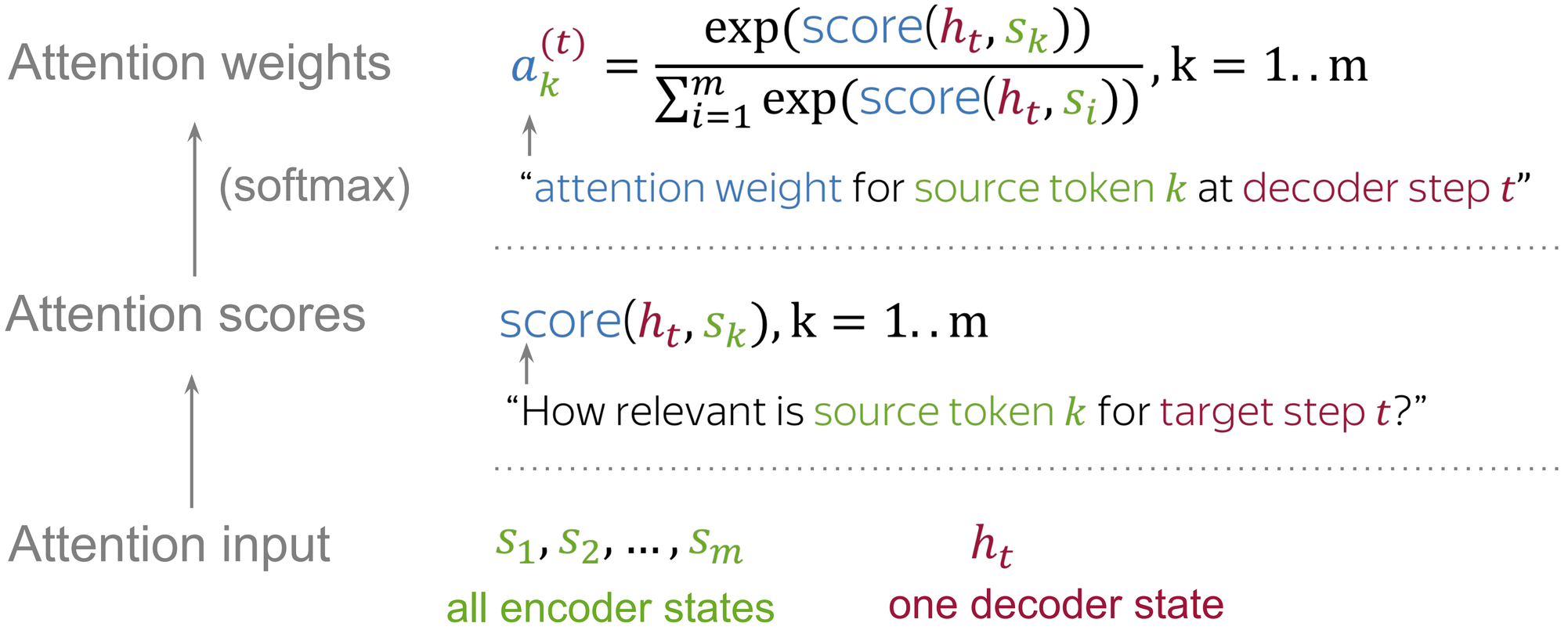
Computation Pipeline



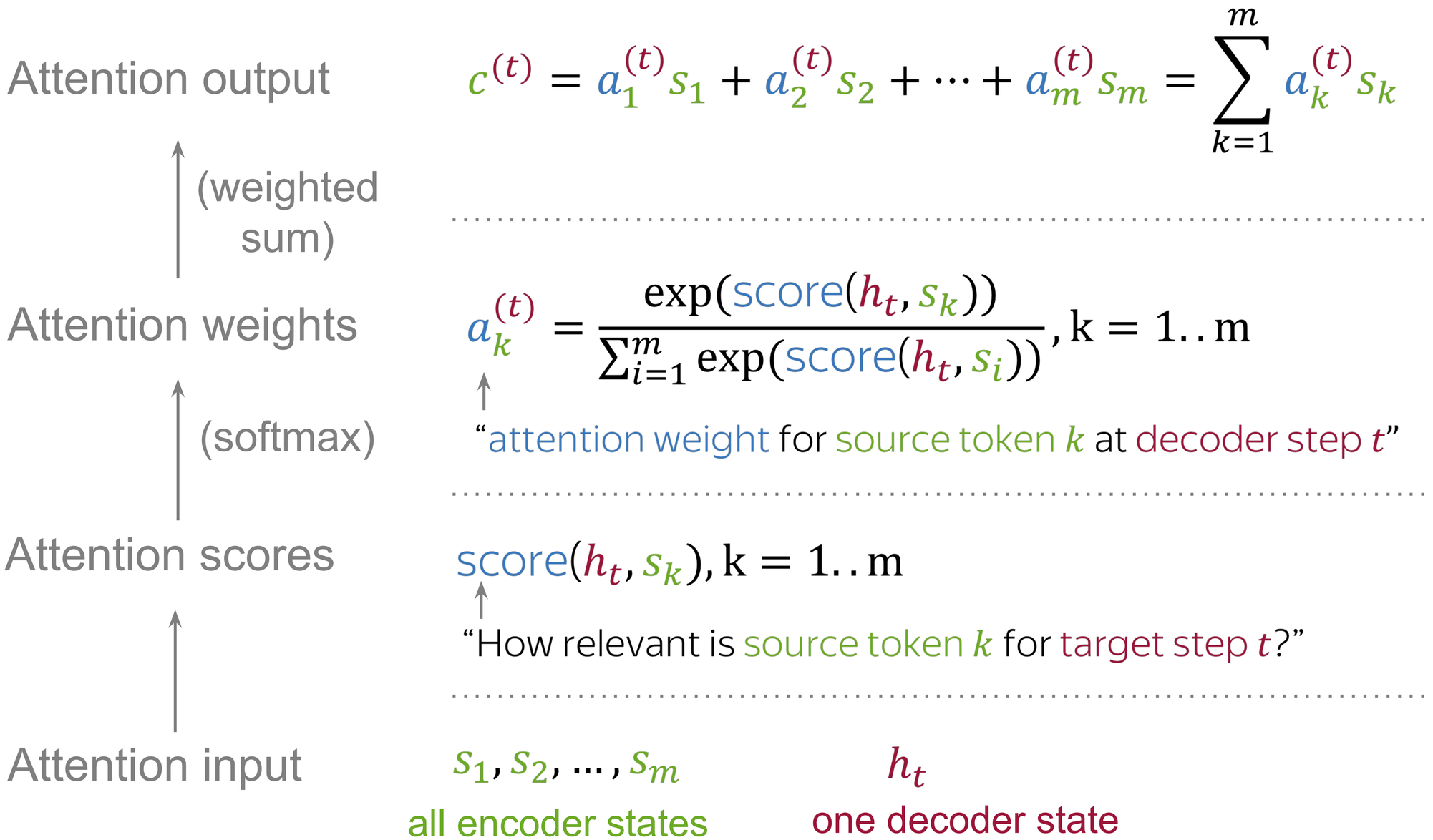
Computation Pipeline



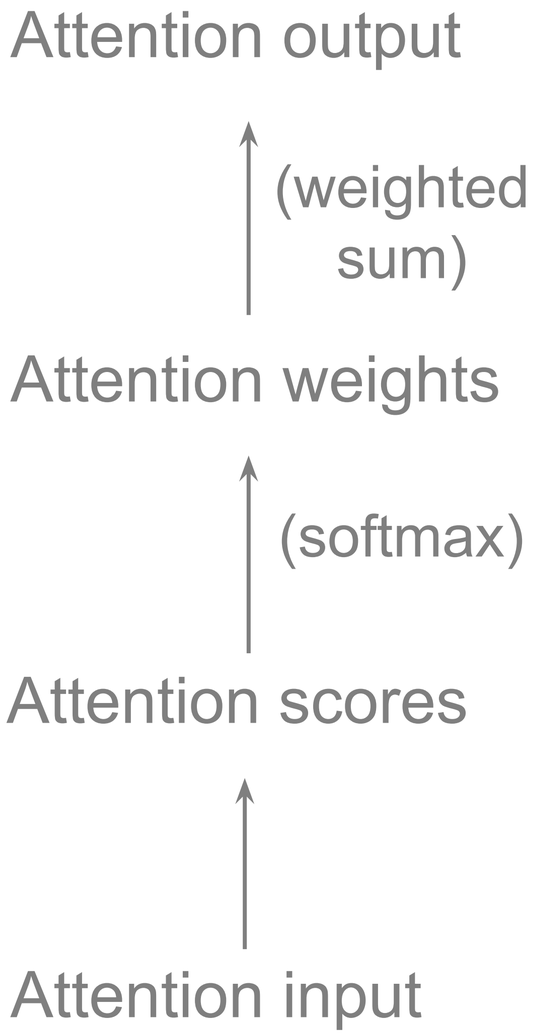
Computation Pipeline



Computation Pipeline



Computation Pipeline



$$c^{(t)} = a_1^{(t)} s_1 + a_2^{(t)} s_2 + \cdots + a_m^{(t)} s_m = \sum_{k=1}^{m} a_k^{(t)} s_k$$

"source context for decoder step t"

$$a_k^{(t)} = \frac{\exp(\text{score}(h_t, s_k))}{\sum_{i=1}^{m} \exp(\text{score}(h_t, s_i))}, \text{k} = 1..\text{m}$$

"attention weight for source token k at decoder step t"

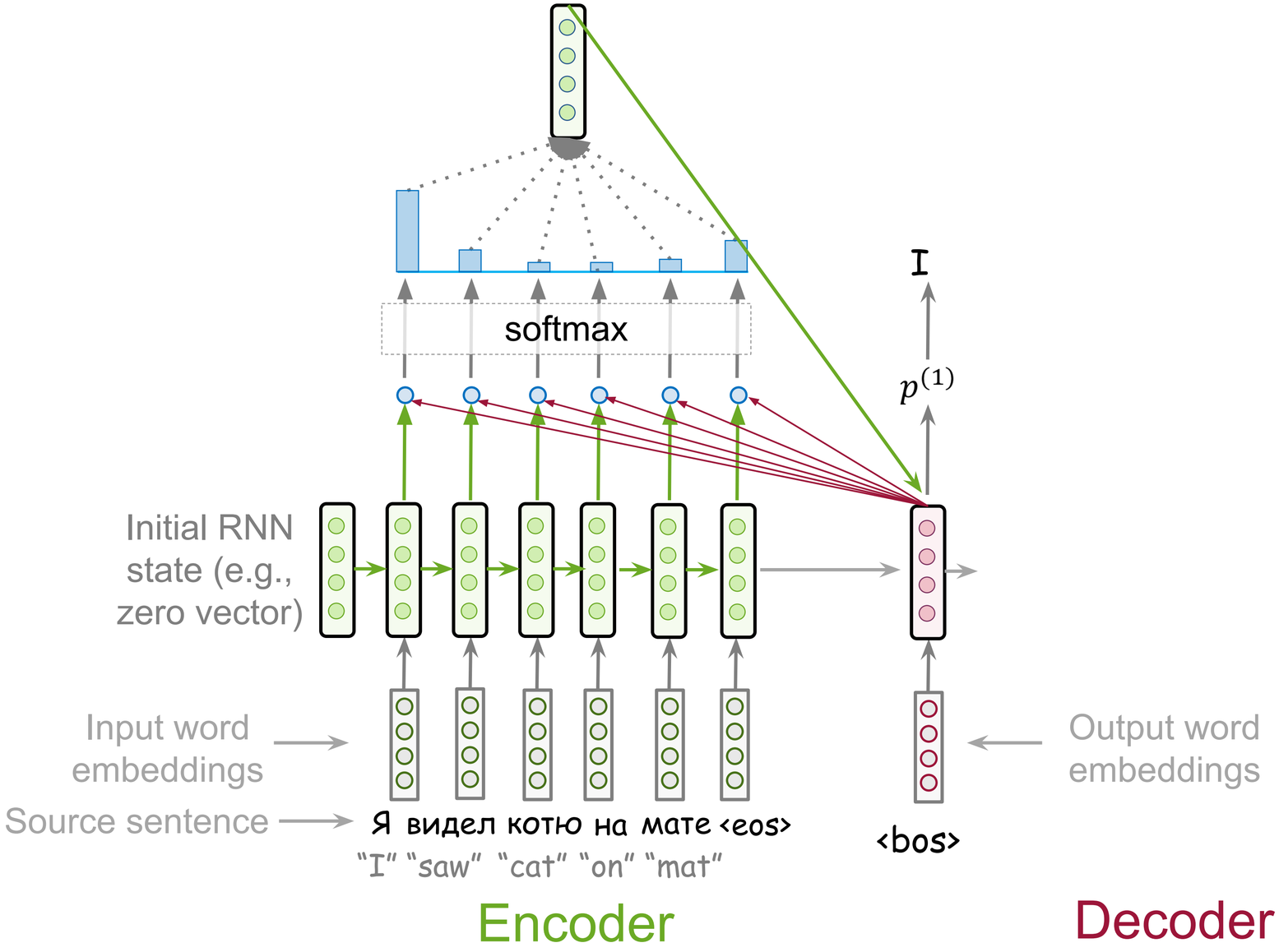
$$\text{score}(h_t, s_k), \text{k} = 1..\text{m}$$

"How relevant is source token k for target step t?"

$s_1, s_2, \ldots, s_m$ all encoder states

h_t one decoder state

Model Learns to Pick Relevant Tokens



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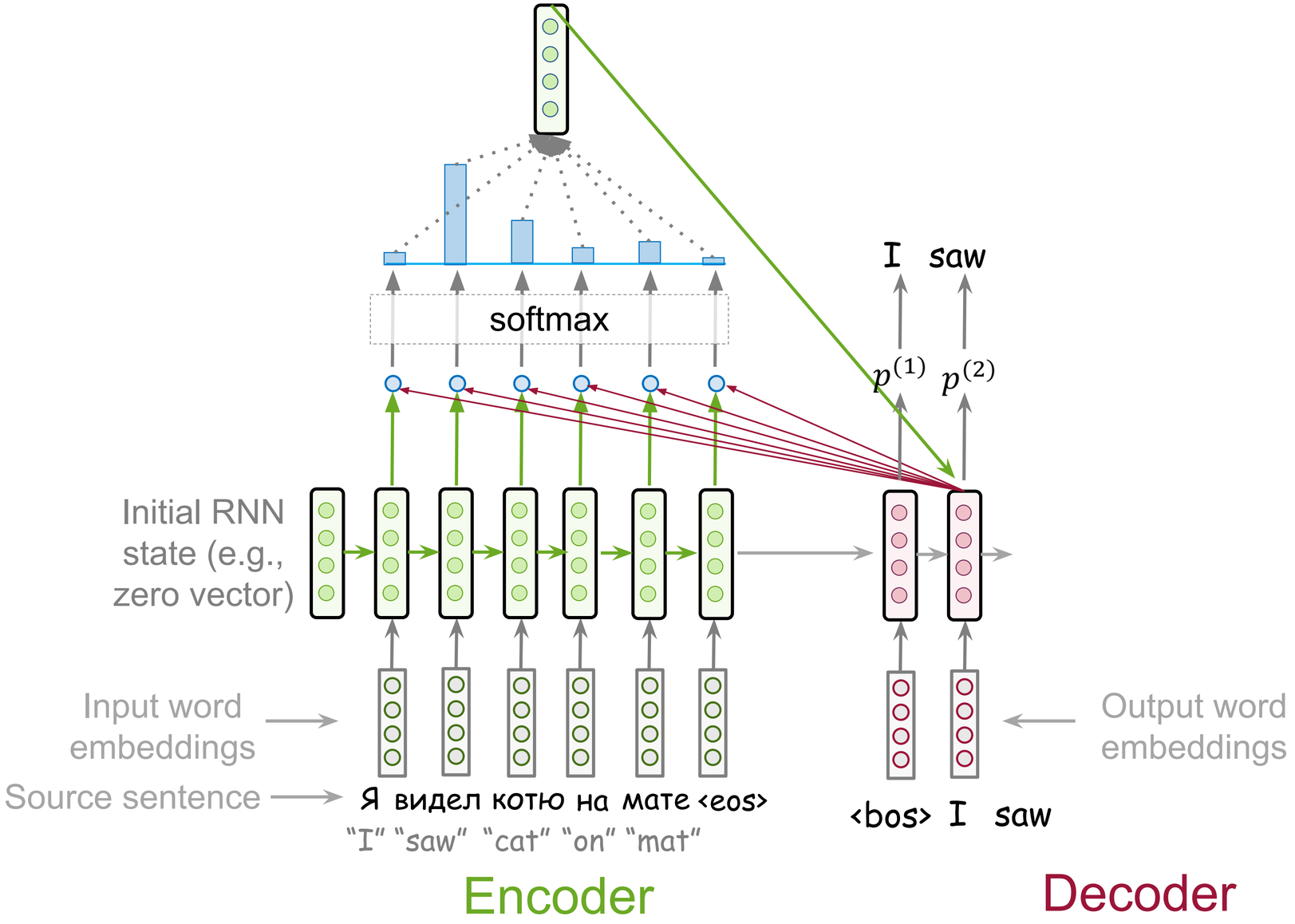

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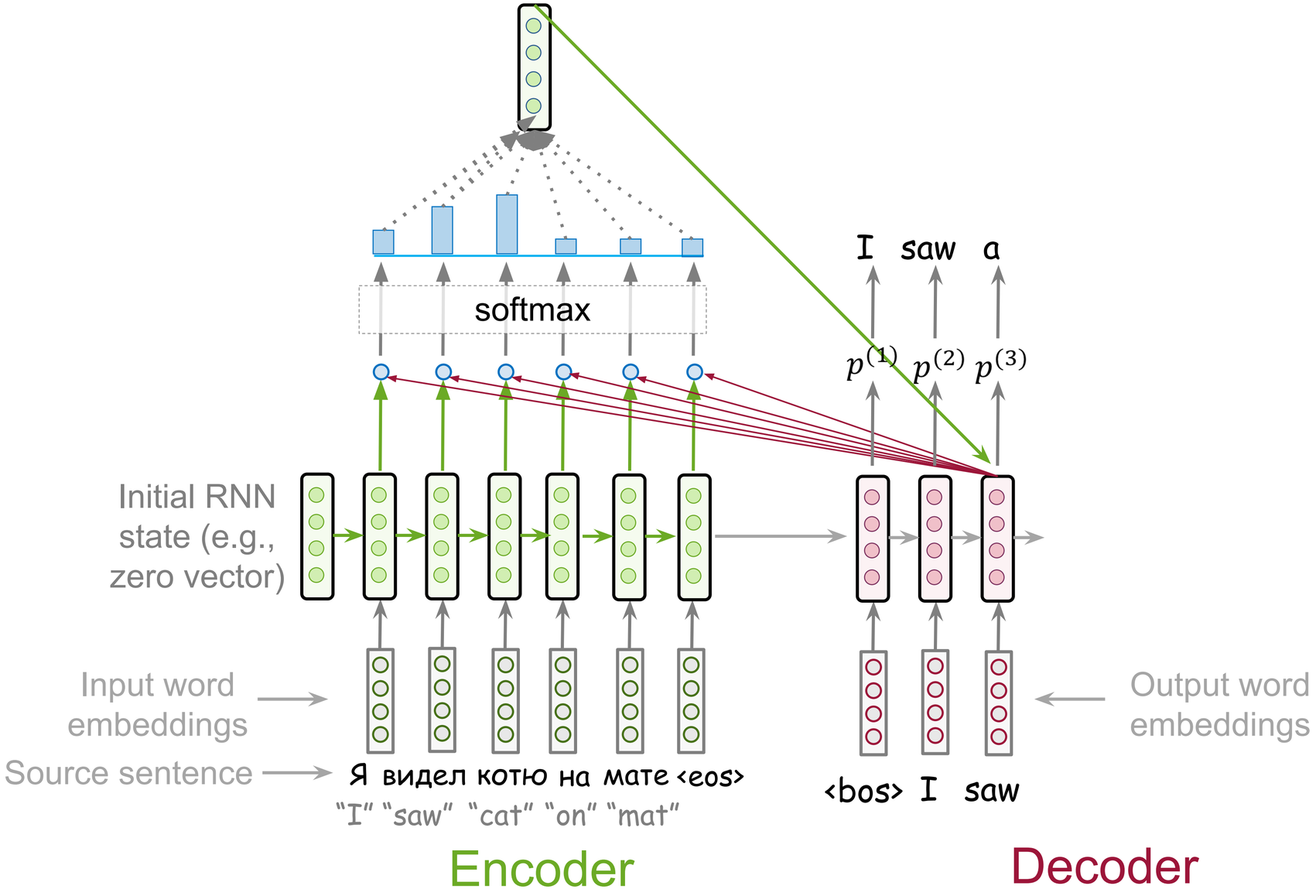

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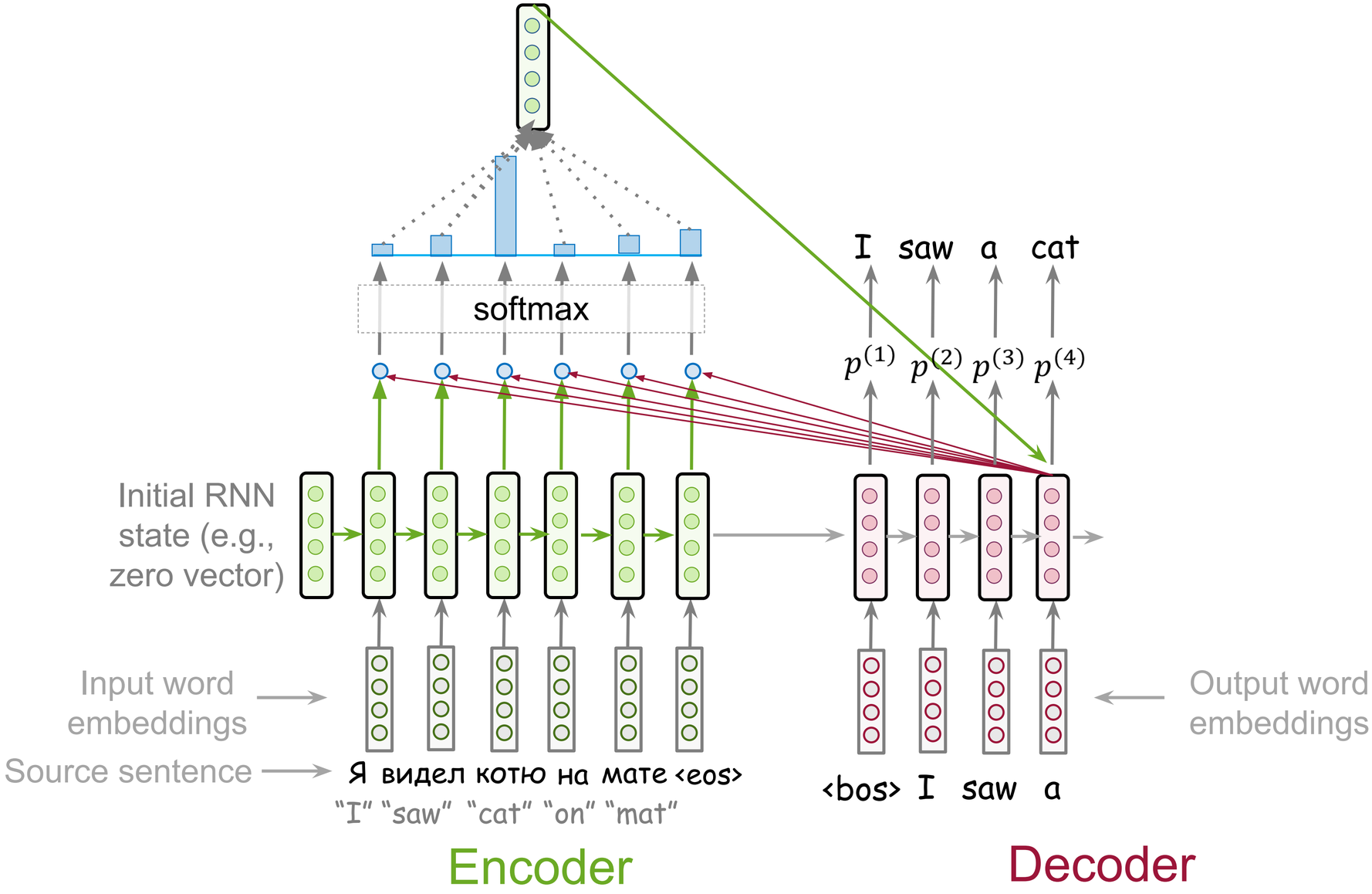

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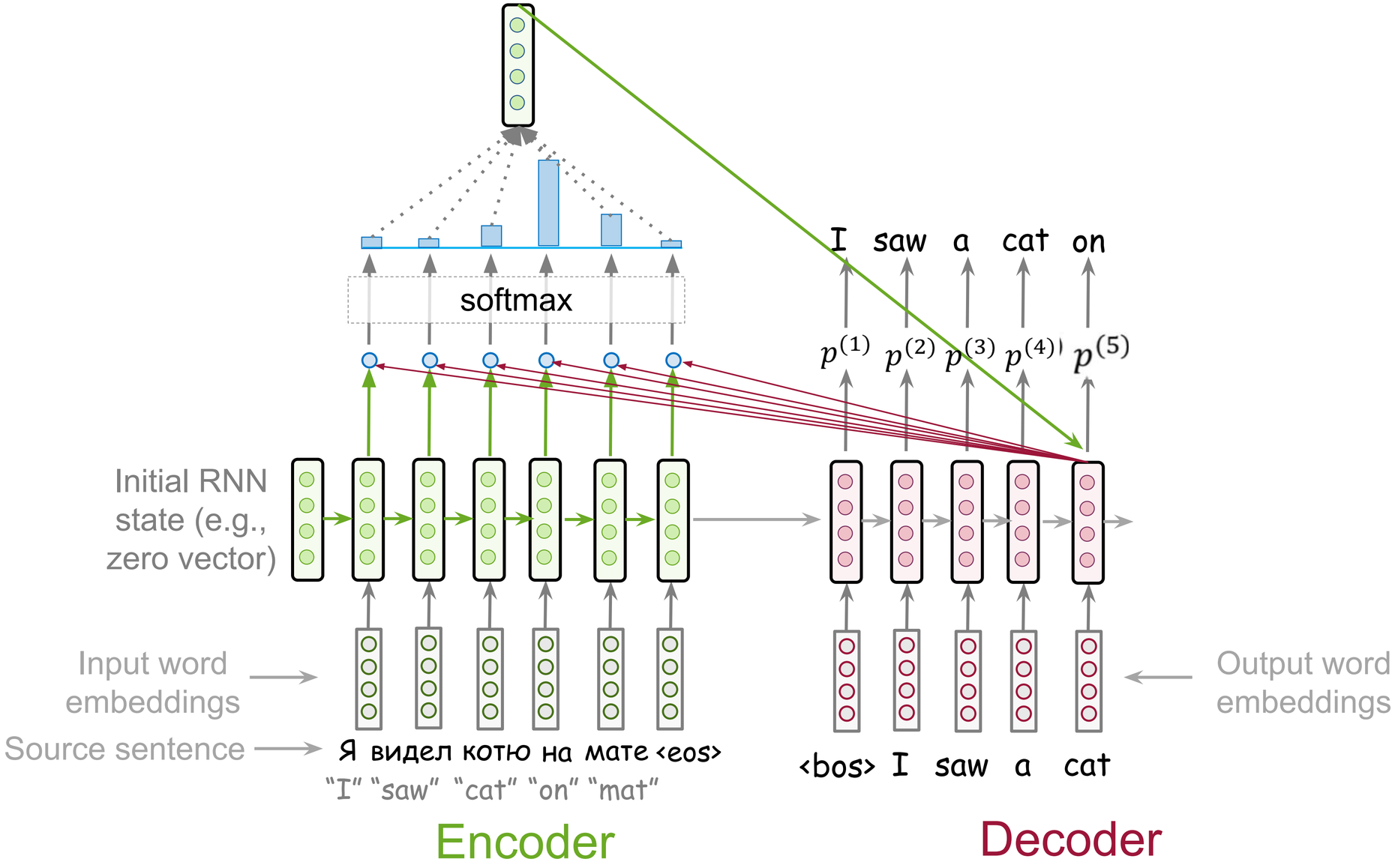

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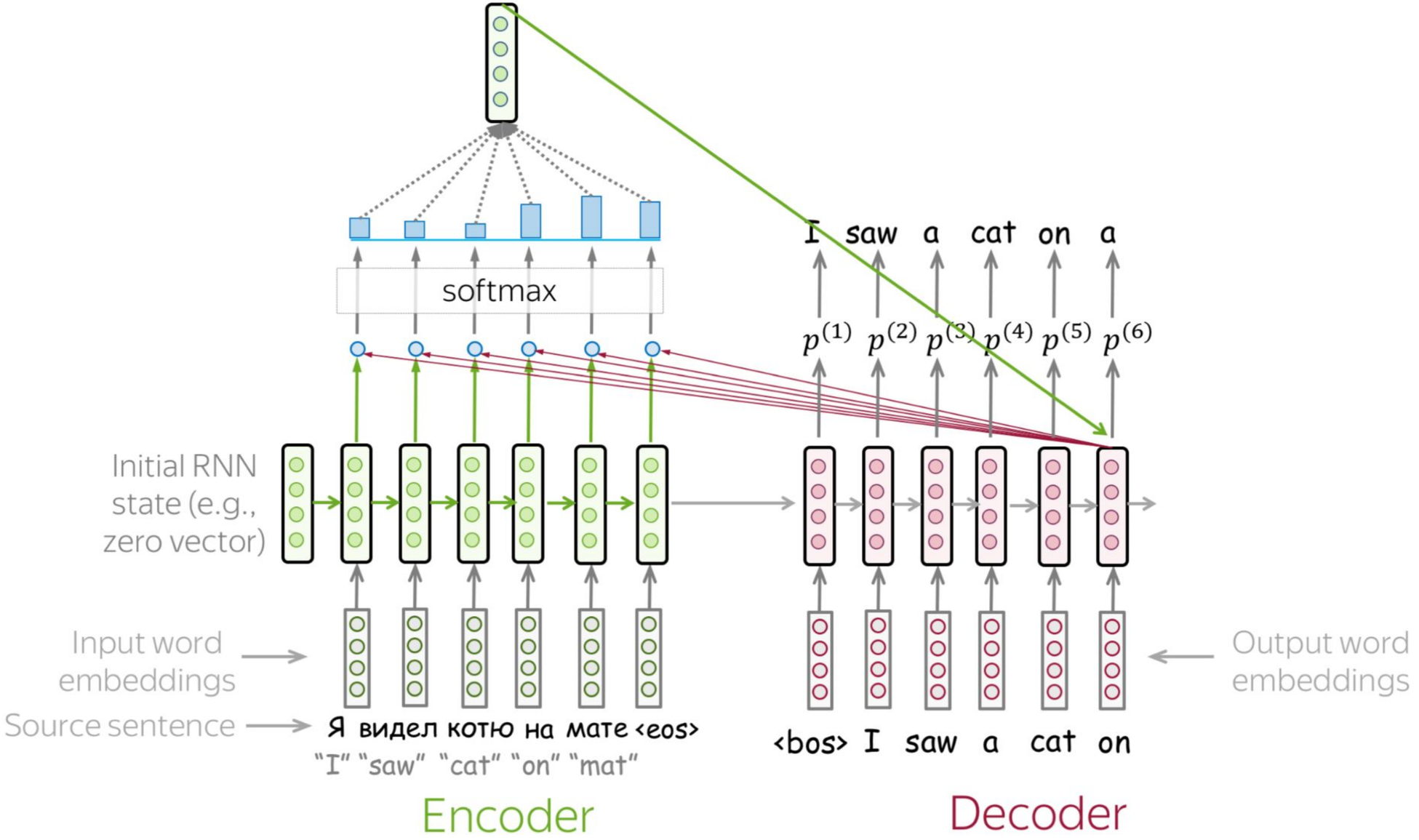

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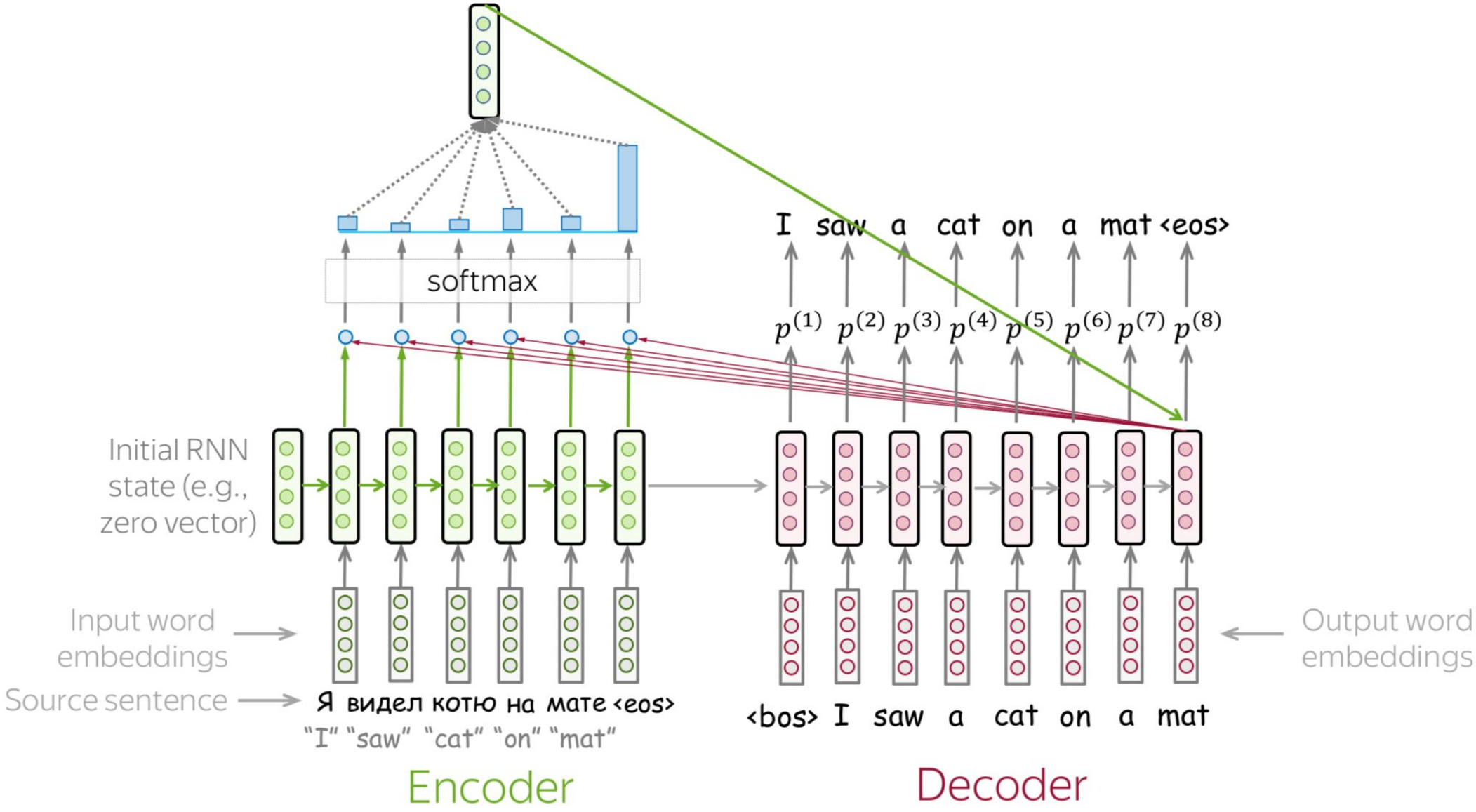
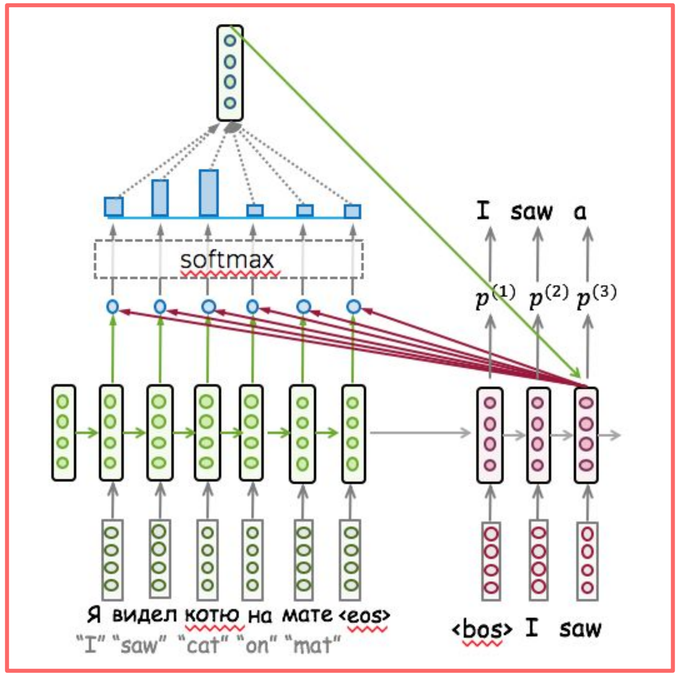


Attention Score Functions



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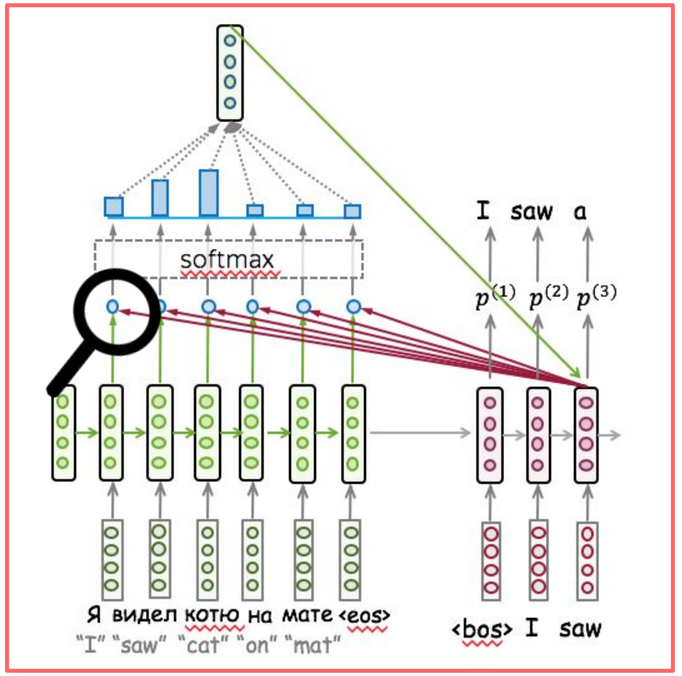

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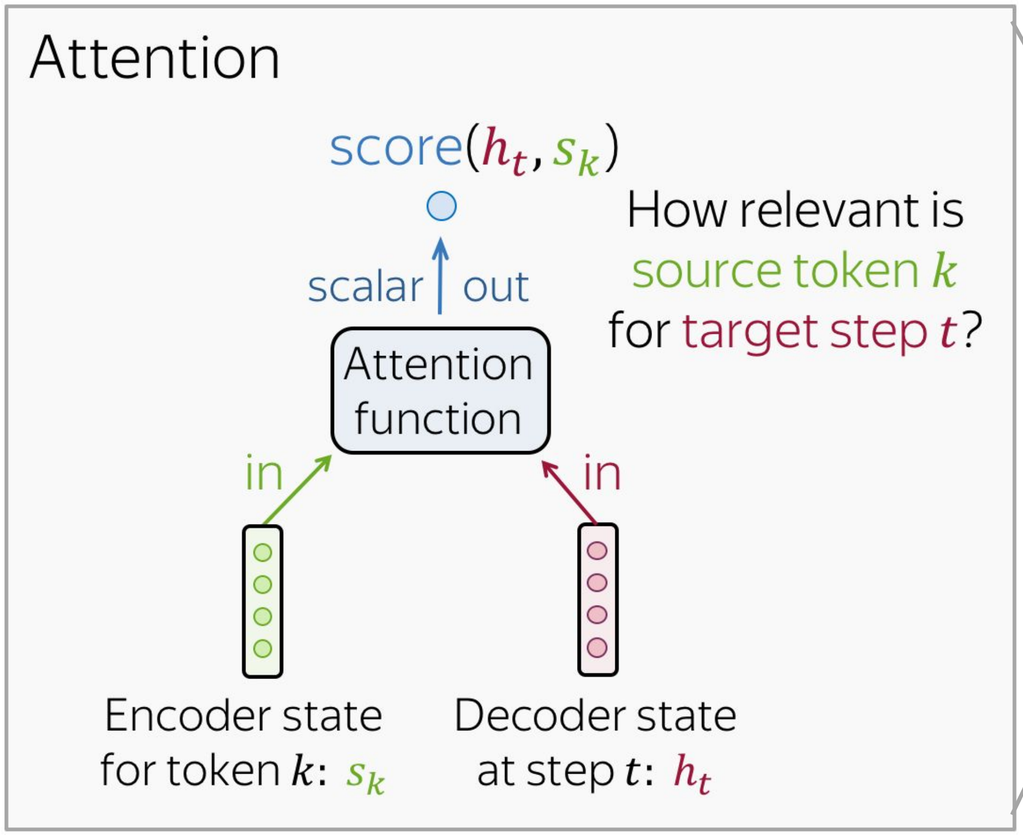
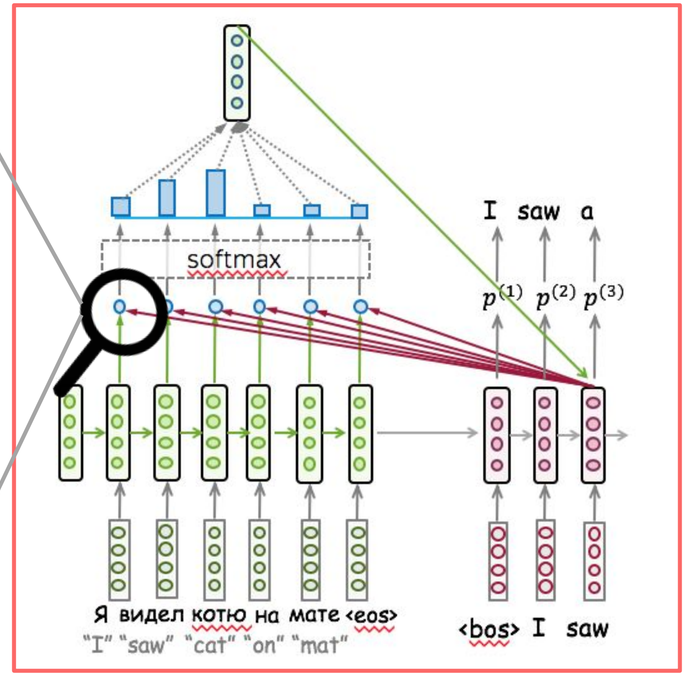
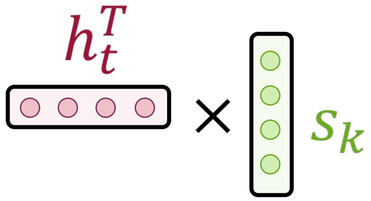
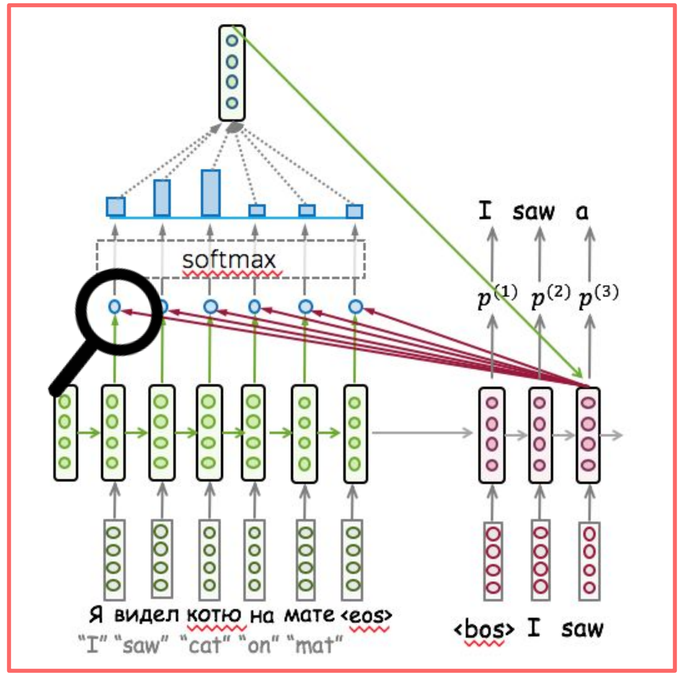




Attention Score Functions

- **Dot-product:** $\text{score}(h_t, s_k) = h_t^T \, s_k$





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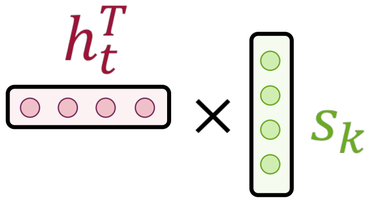
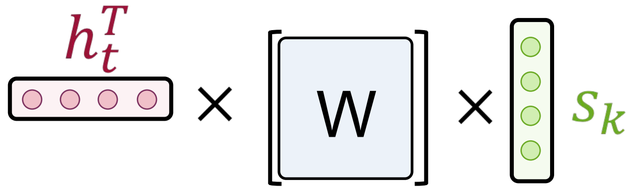
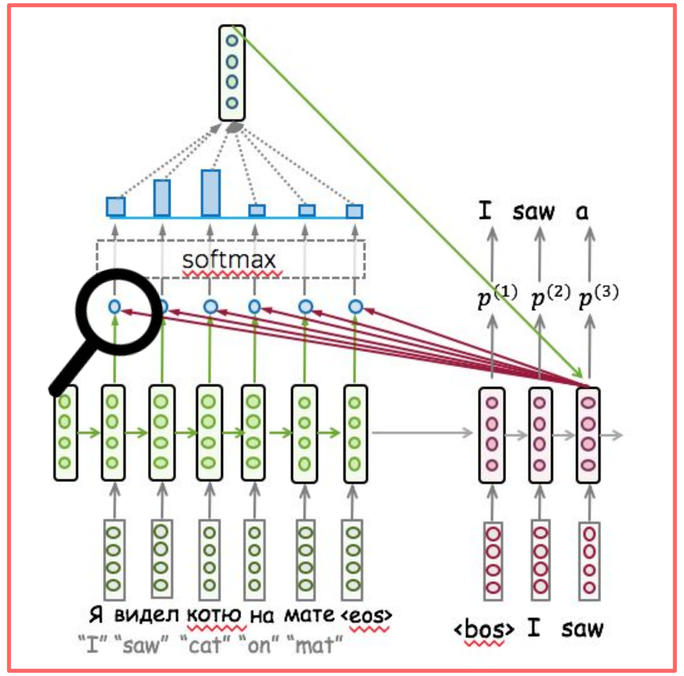


- **Bilinear:** $\text{score}(h_t, s_k) = h_t^T \, W \, s_k$





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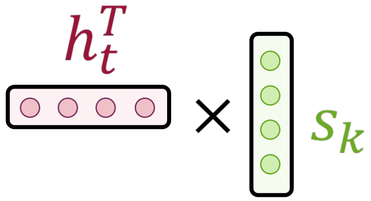

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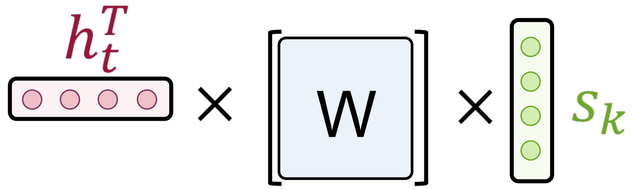
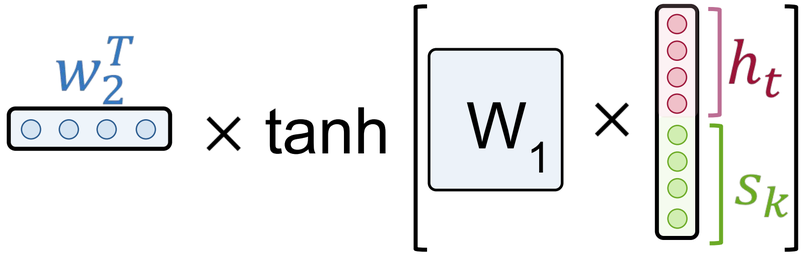
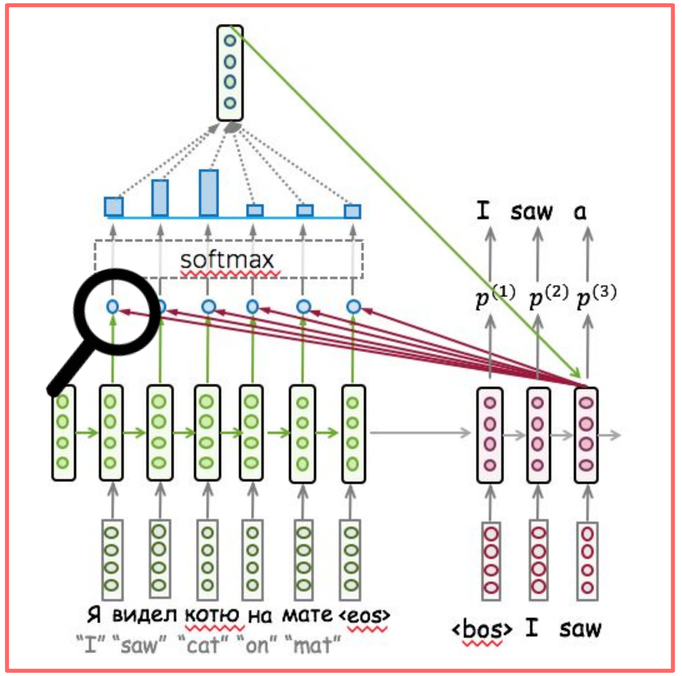


- **Multi-Layer Perceptron:** $\text{score}(h_t, s_k) = w_2^T \cdot \tanh(W_1[h_t, s_k])$





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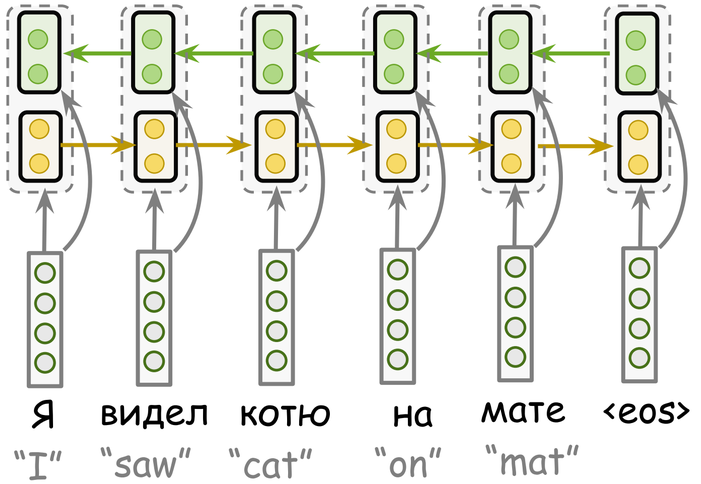
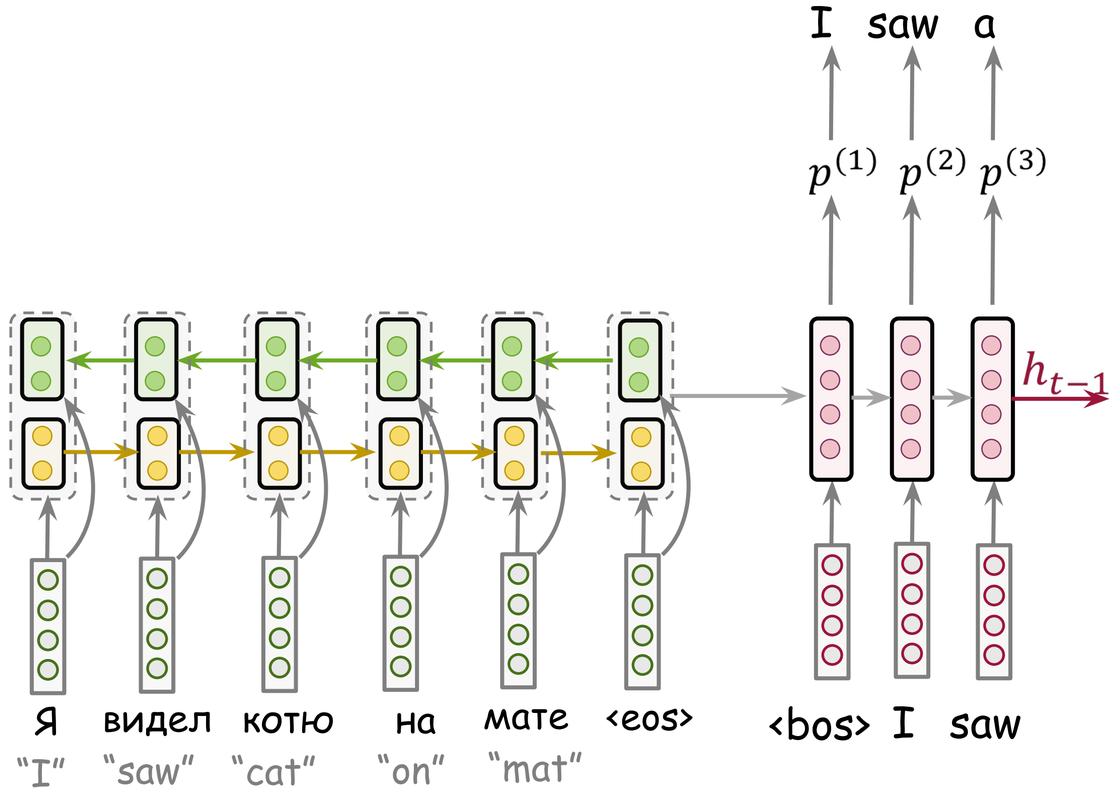
Bahdanau Model (the original attention model)

Bidirectional encoder
Concatenate states from
forward and backward RNNs



Bahdanau Model (the original attention model)

Bidirectional encoder
Concatenate states from
forward and backward RNNs



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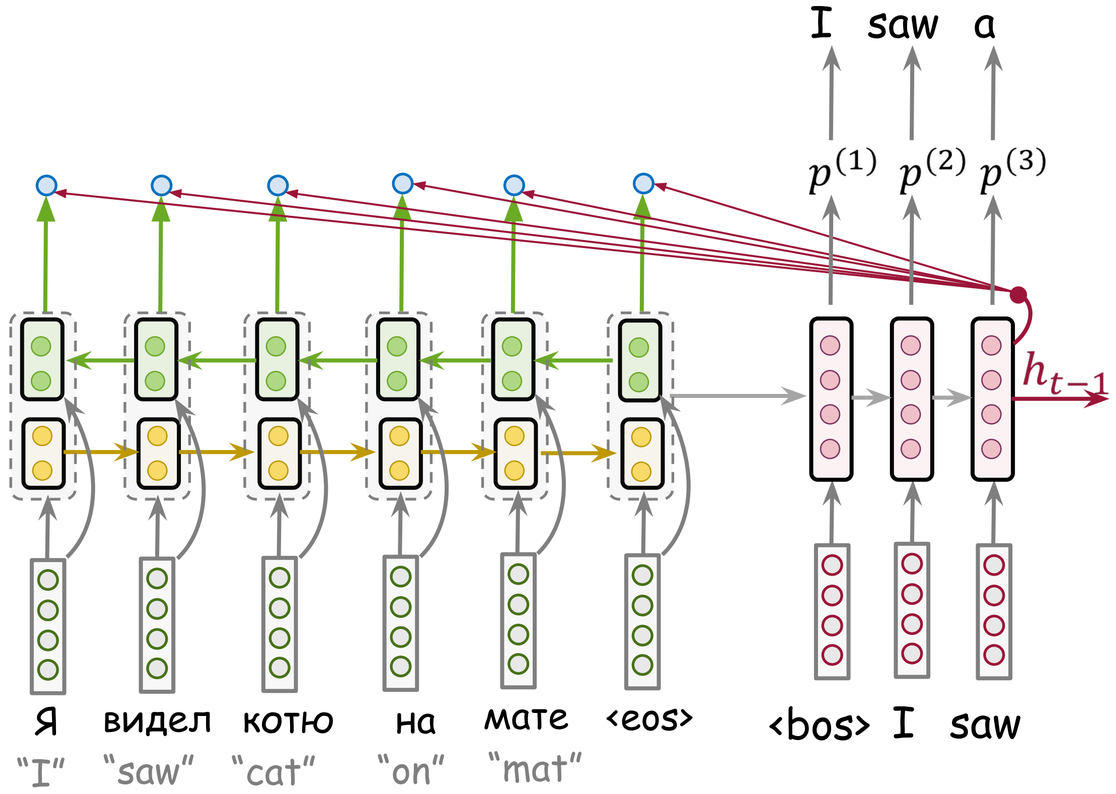

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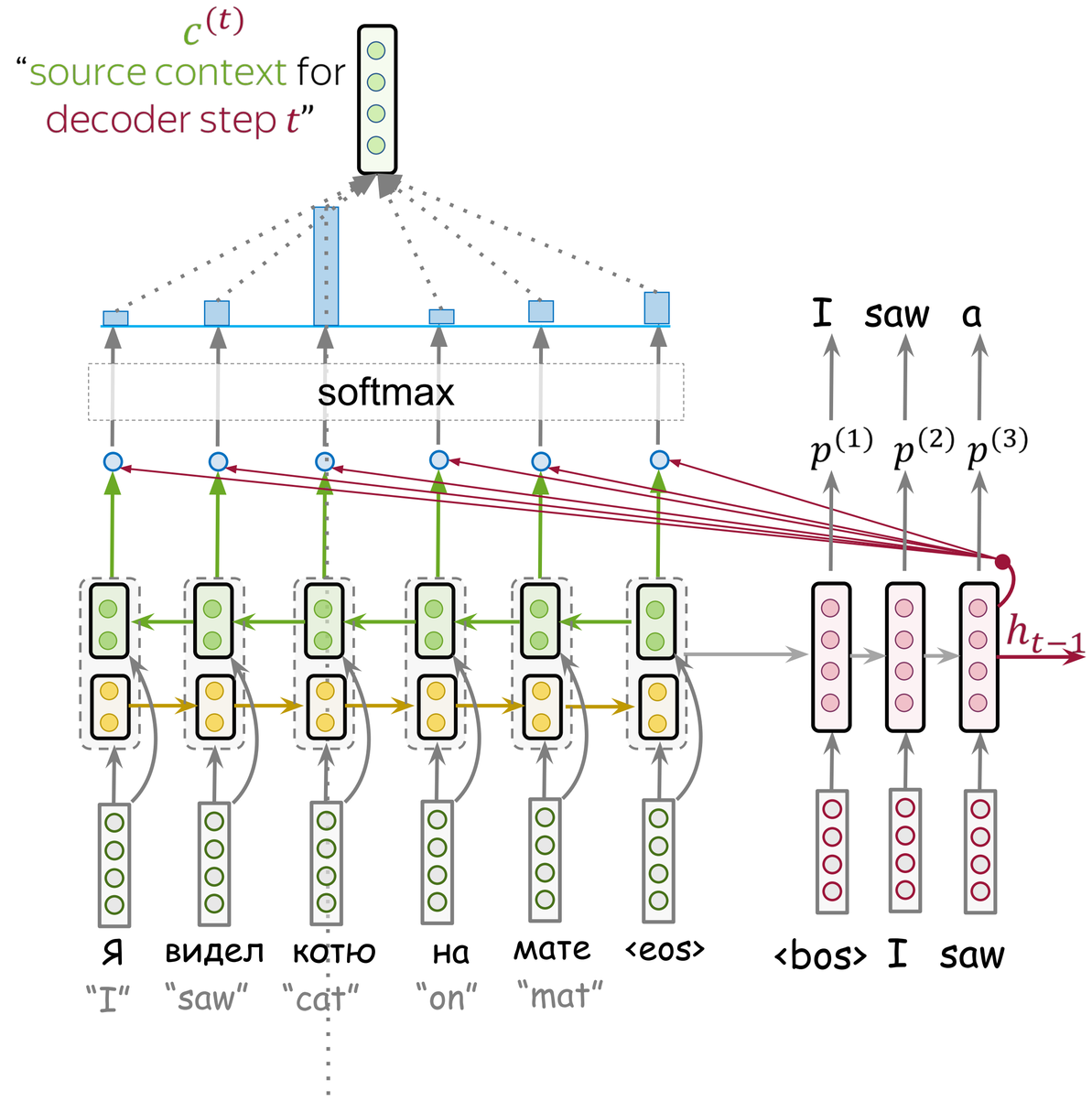

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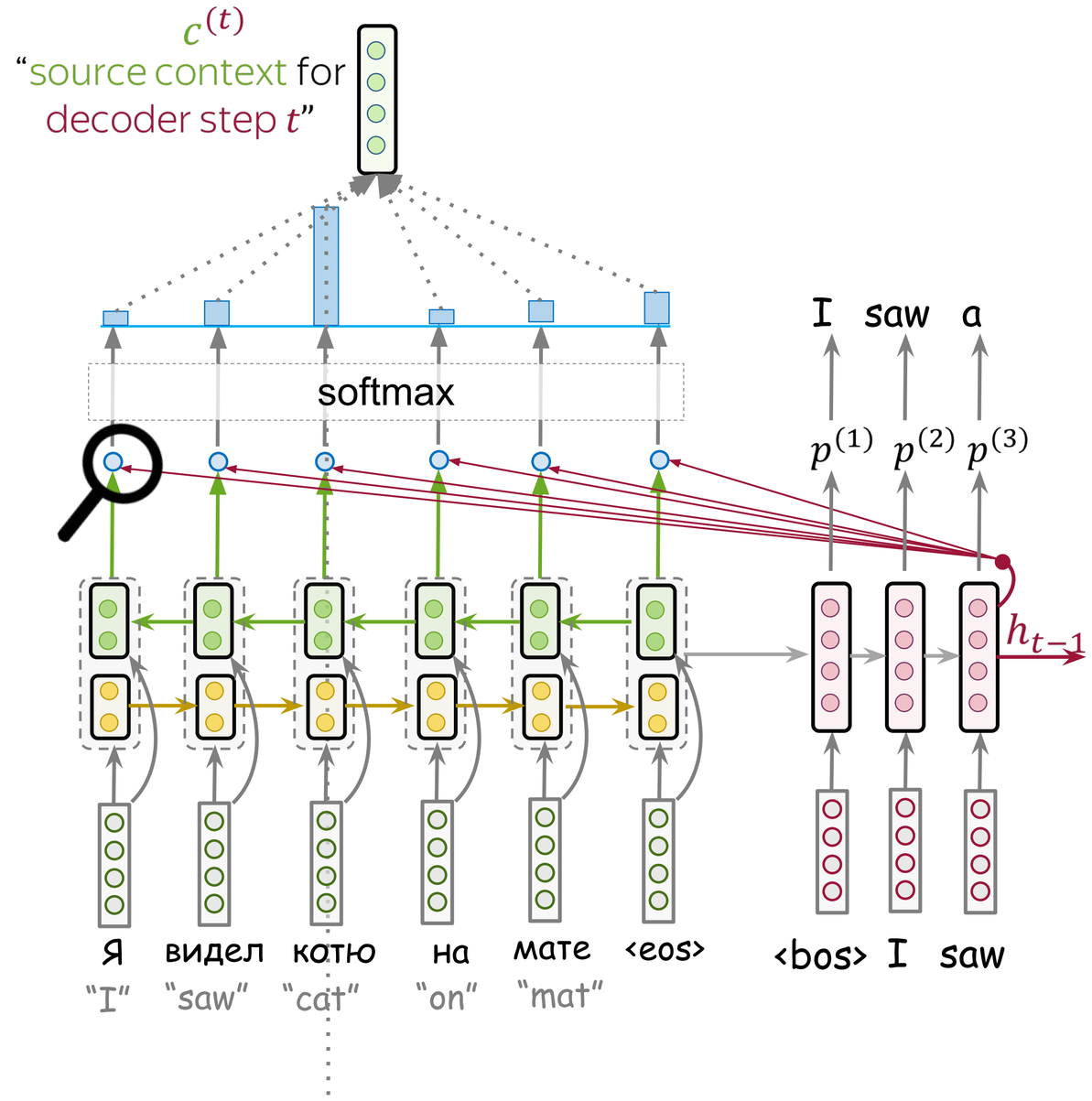

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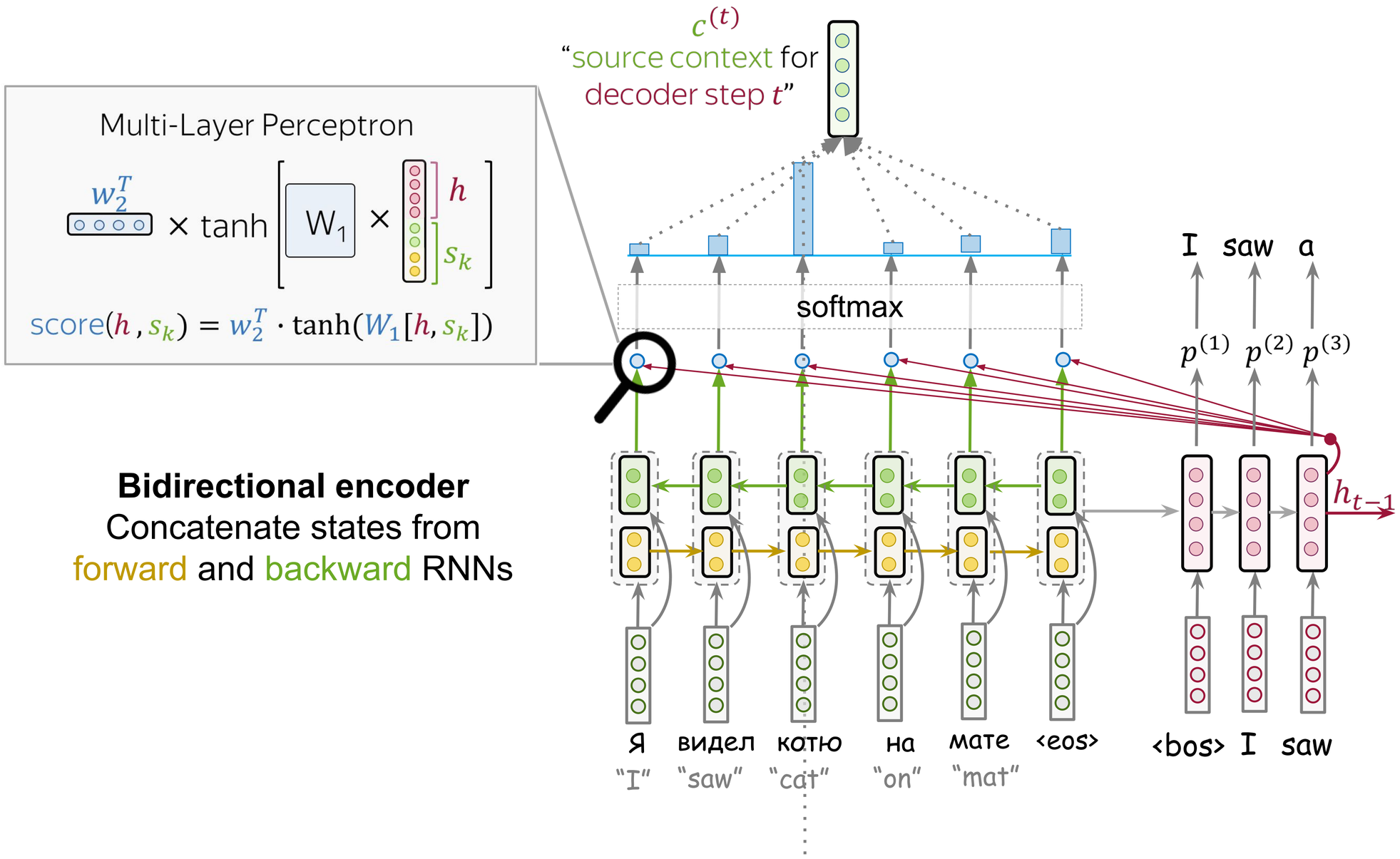

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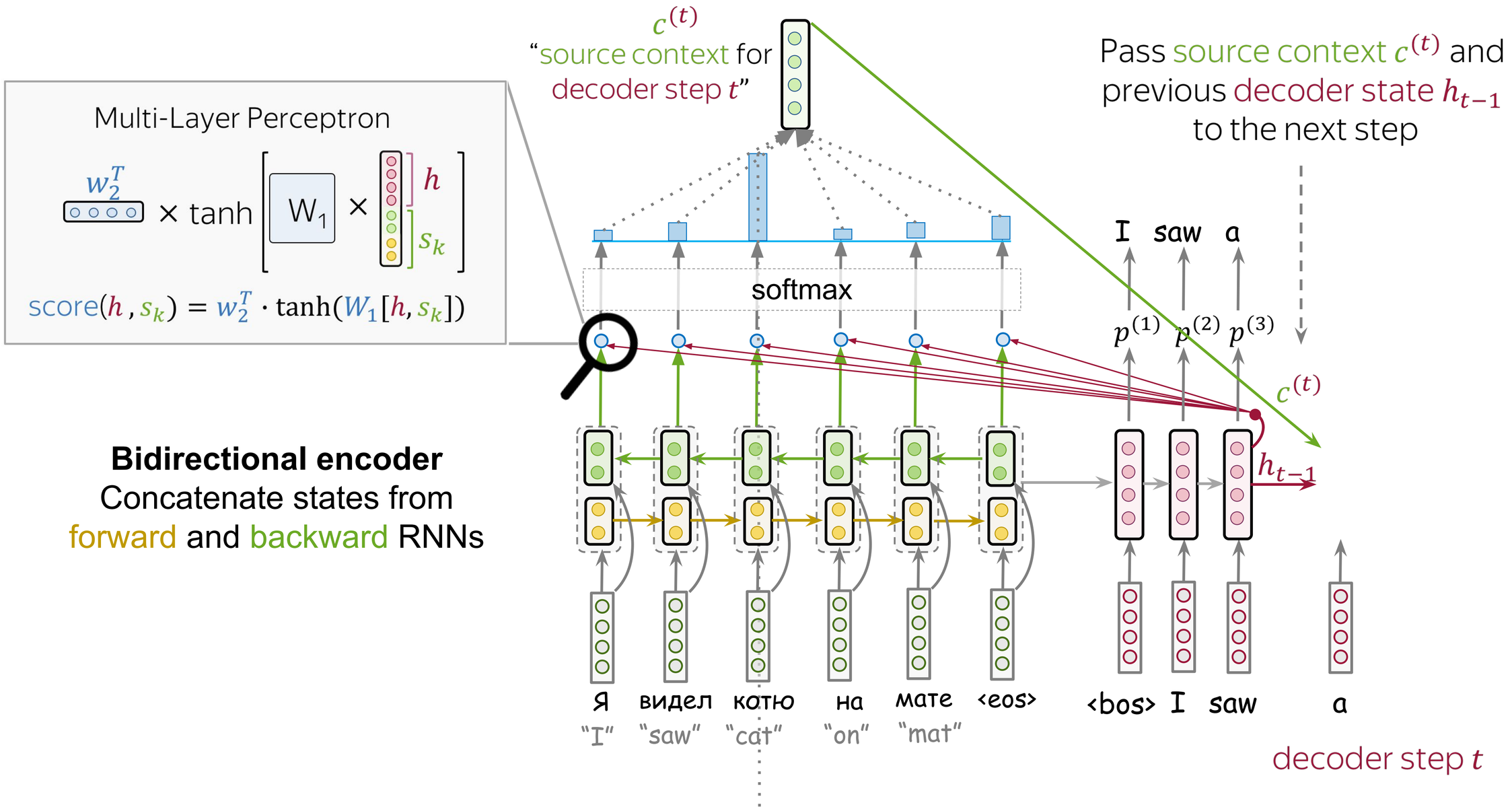

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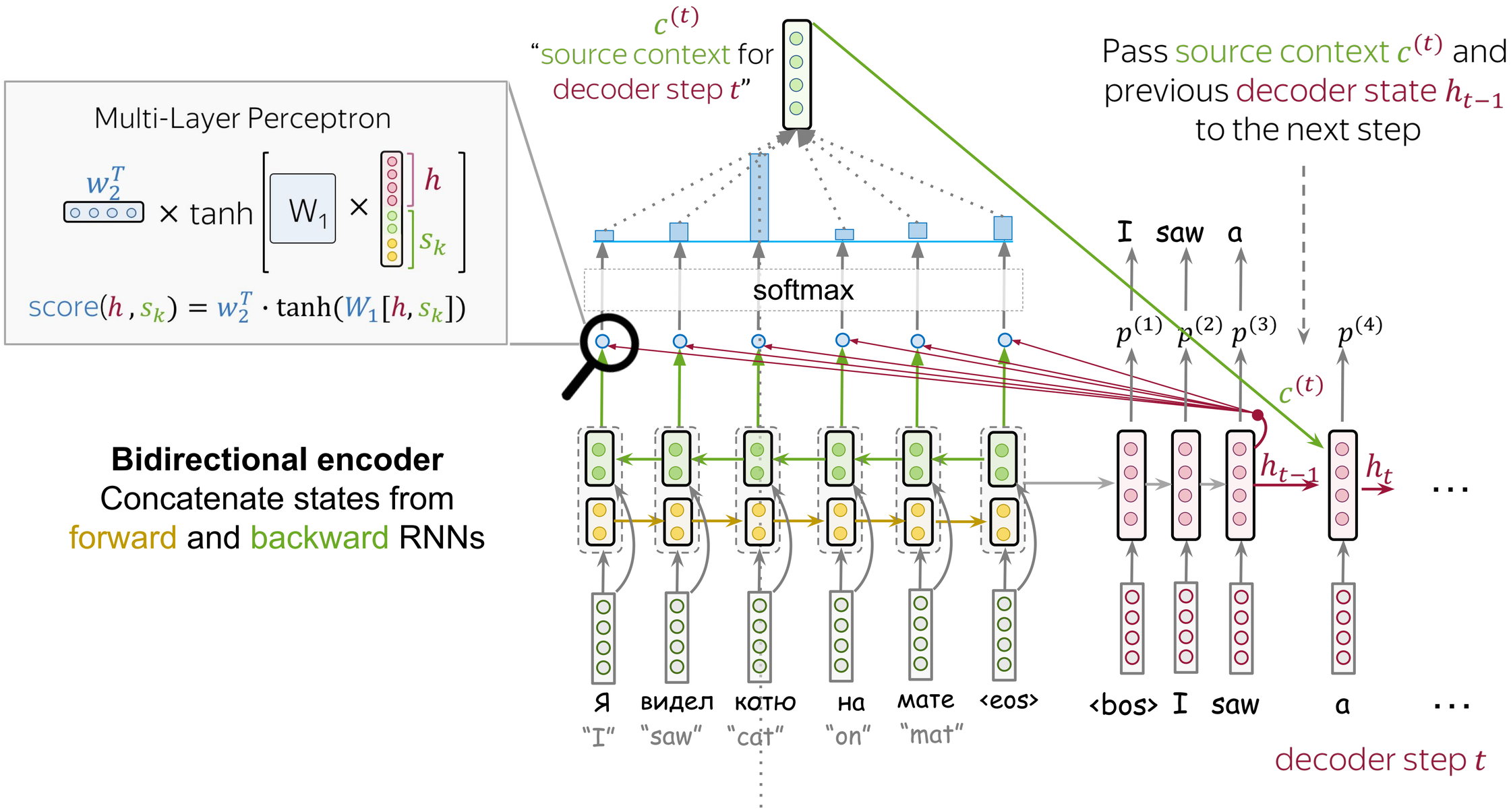

Bidirectional encoder
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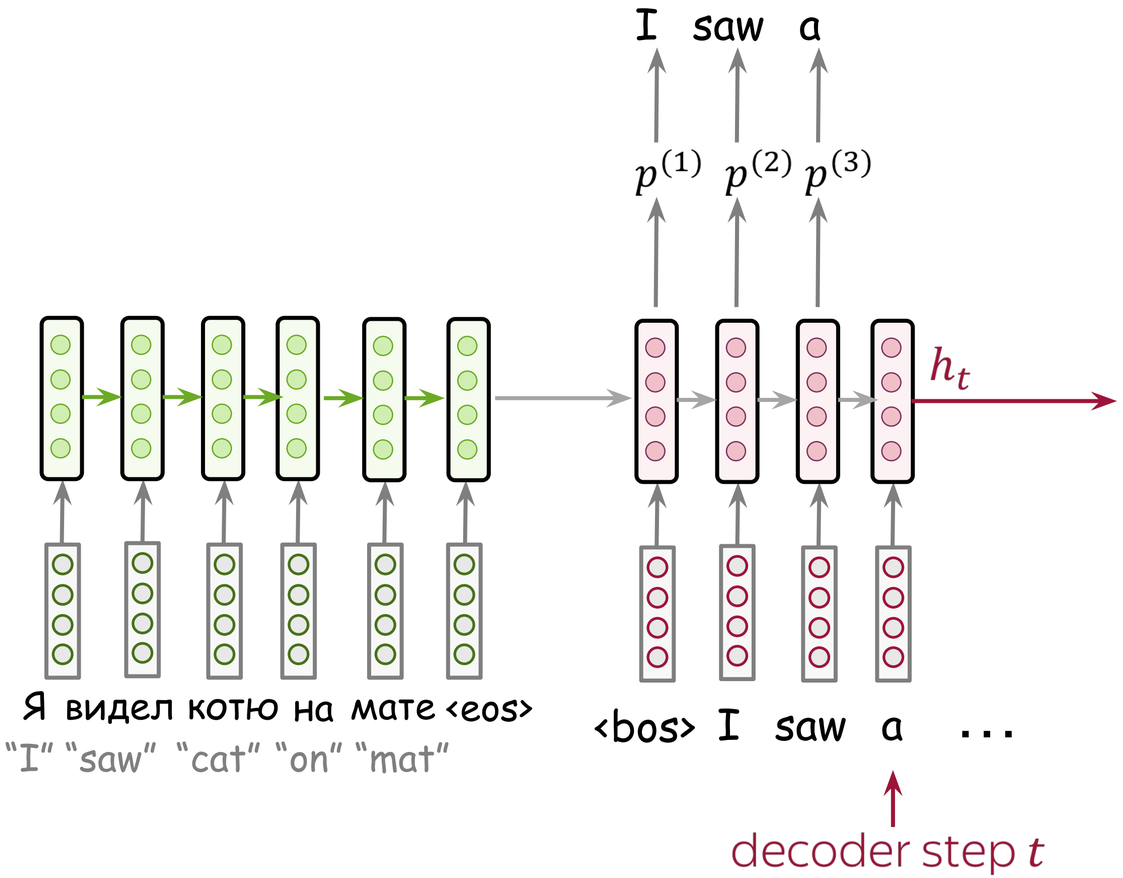
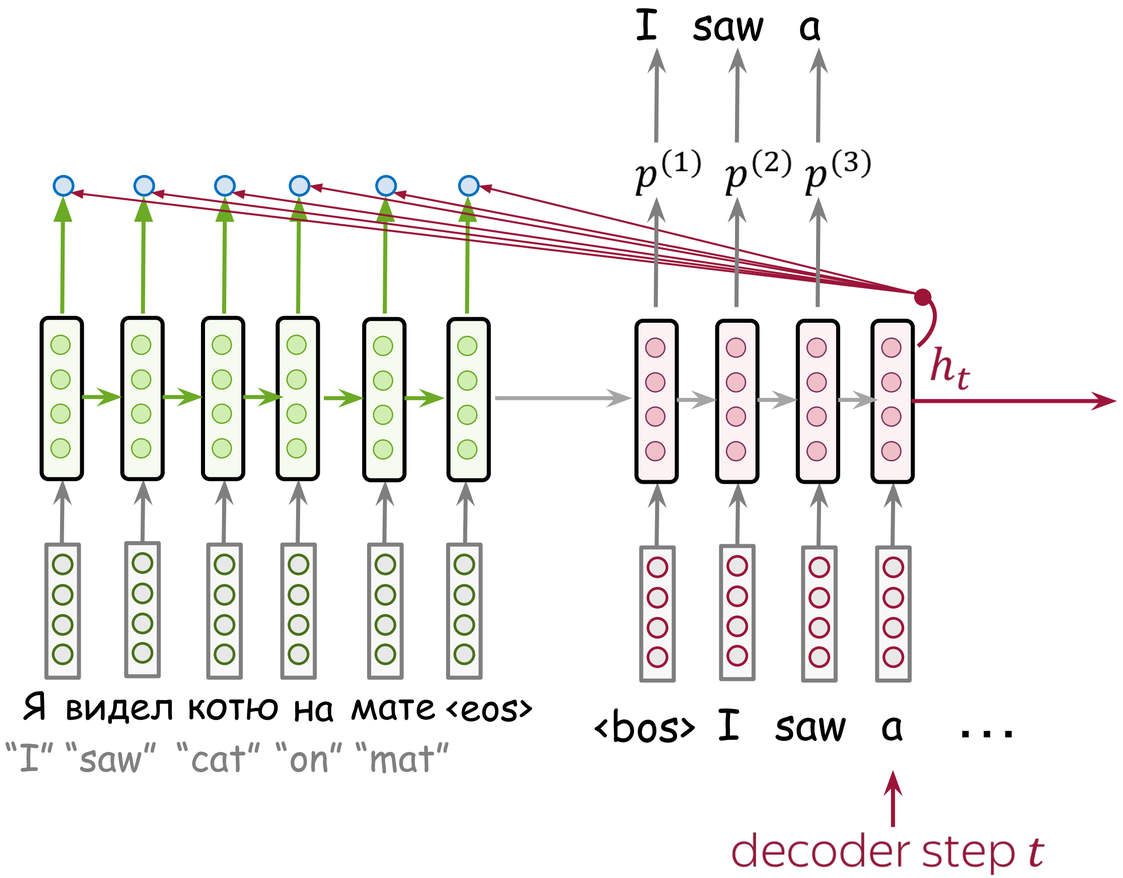
Luong Model


I saw a
$p^{(1)}$ $p^{(2)}$ $p^{(3)}$
h_t
Я видел котю на мате <eos>
"I" "saw" "cat" "on" "mat"
<bos> I saw a ...
decoder step t

Luong Model


I saw a
$p^{(1)}$ $p^{(2)}$ $p^{(3)}$
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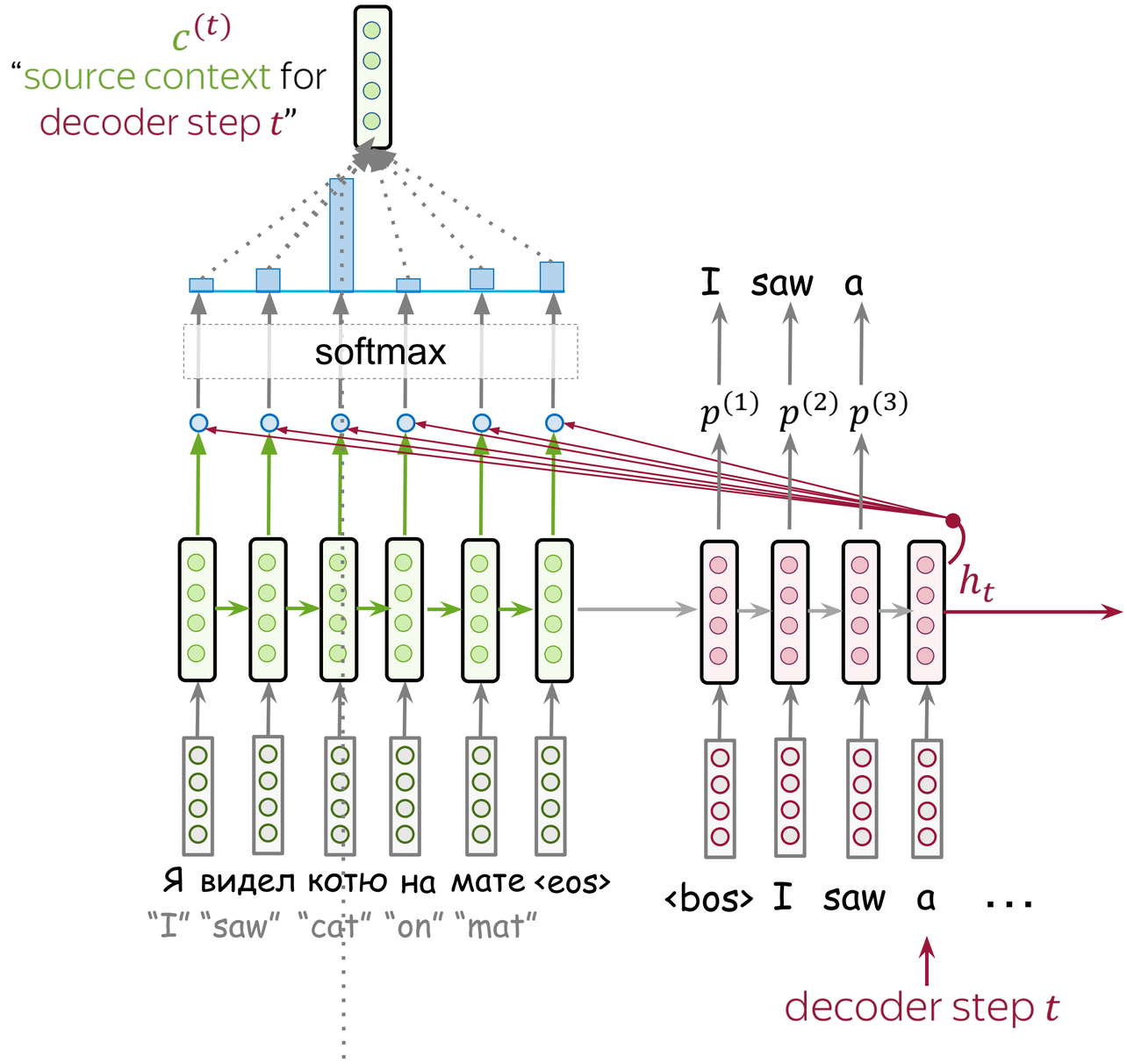
$c^{(t)}$
"source context for decoder step t"
softmax
I saw a
$p^{(1)}$ $p^{(2)}$ $p^{(3)}$
h_t
Я видел котю на мате <eos>
"I" "saw" "cat" "on" "mat"
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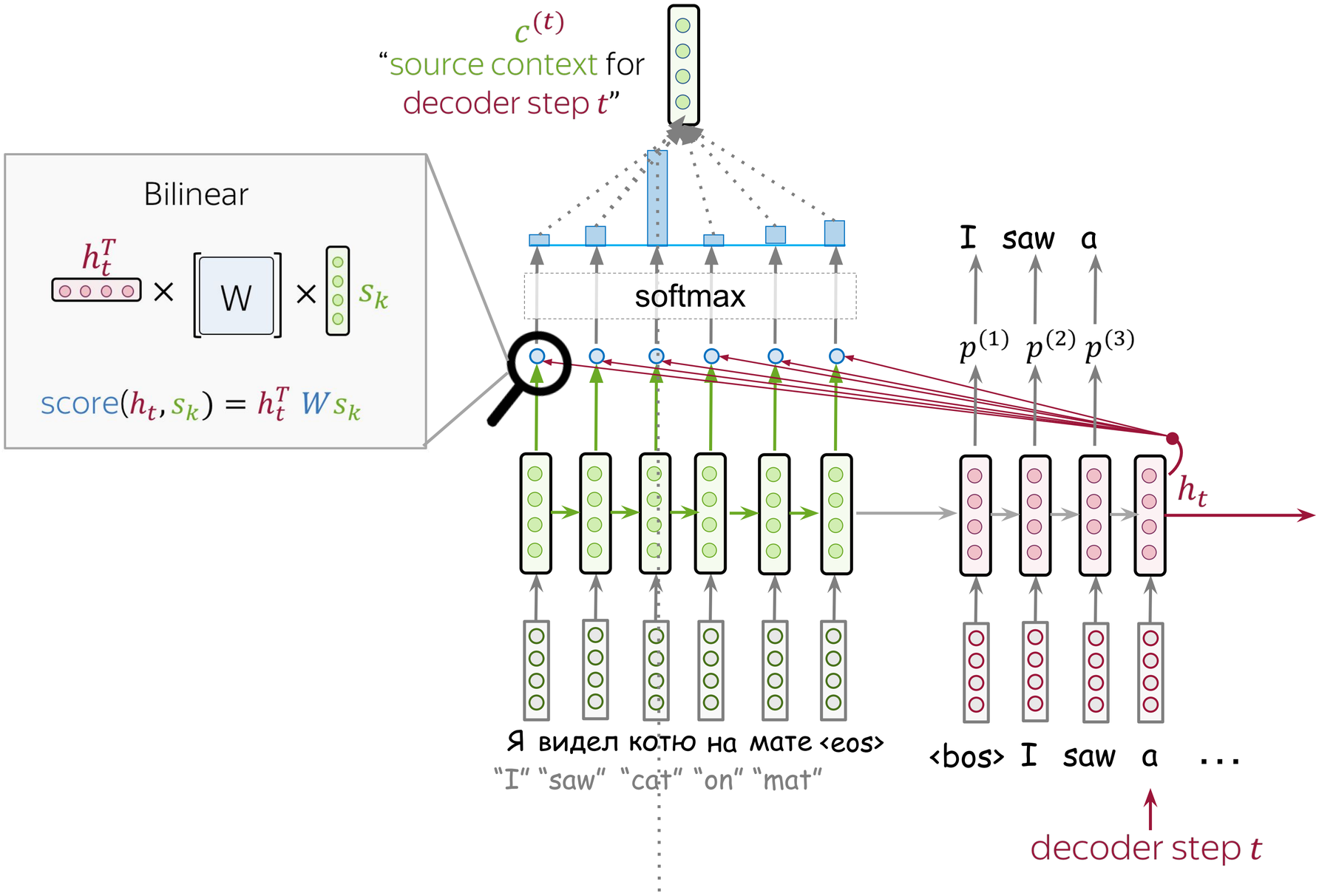
$c^{(t)}$
"source context for decoder step t"
Bilinear
h_t^T × W × s_k
score$(h_t, s_k) = h_t^T W s_k$
softmax
I saw a
$p^{(1)}$ $p^{(2)}$ $p^{(3)}$
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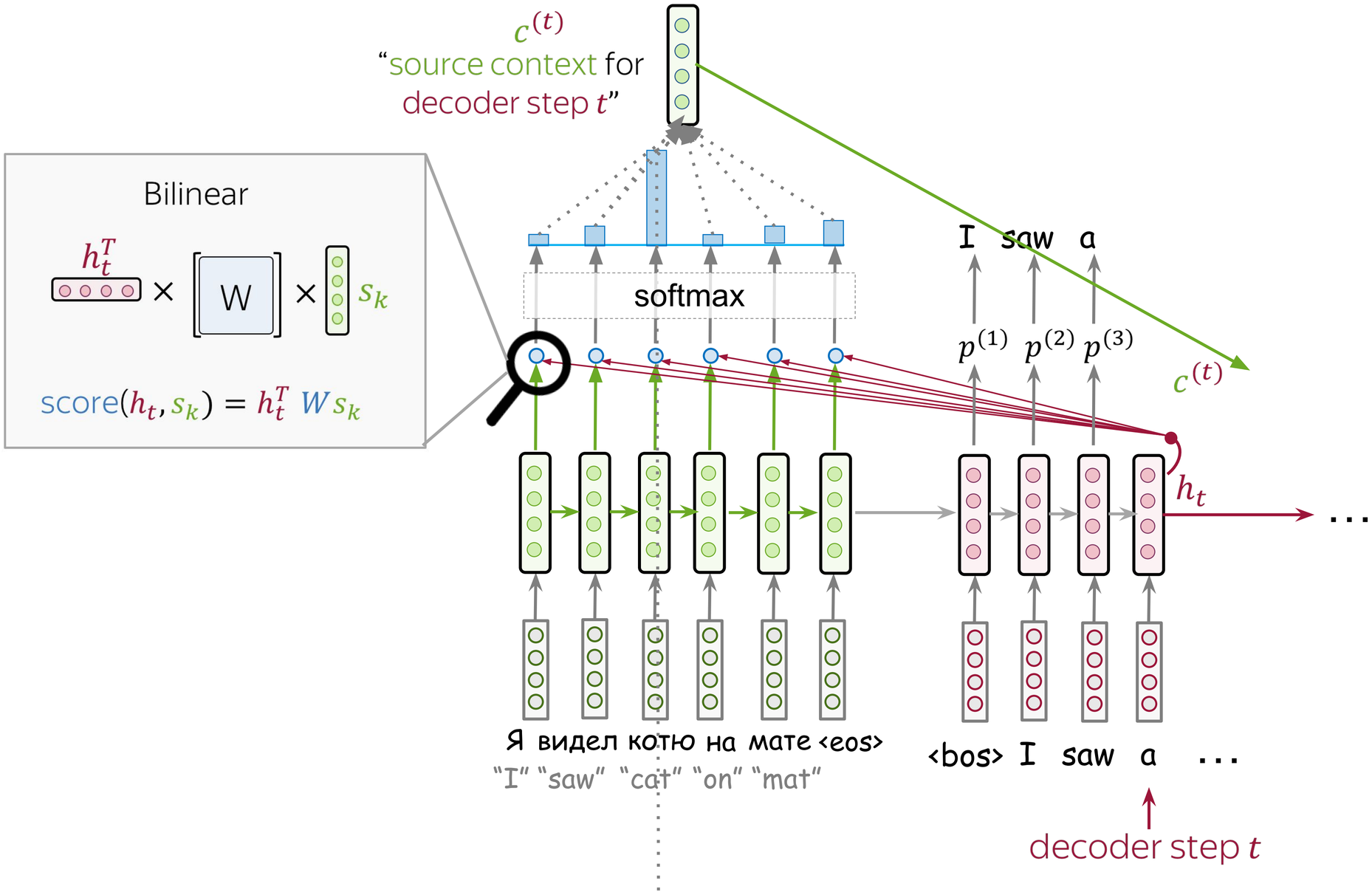
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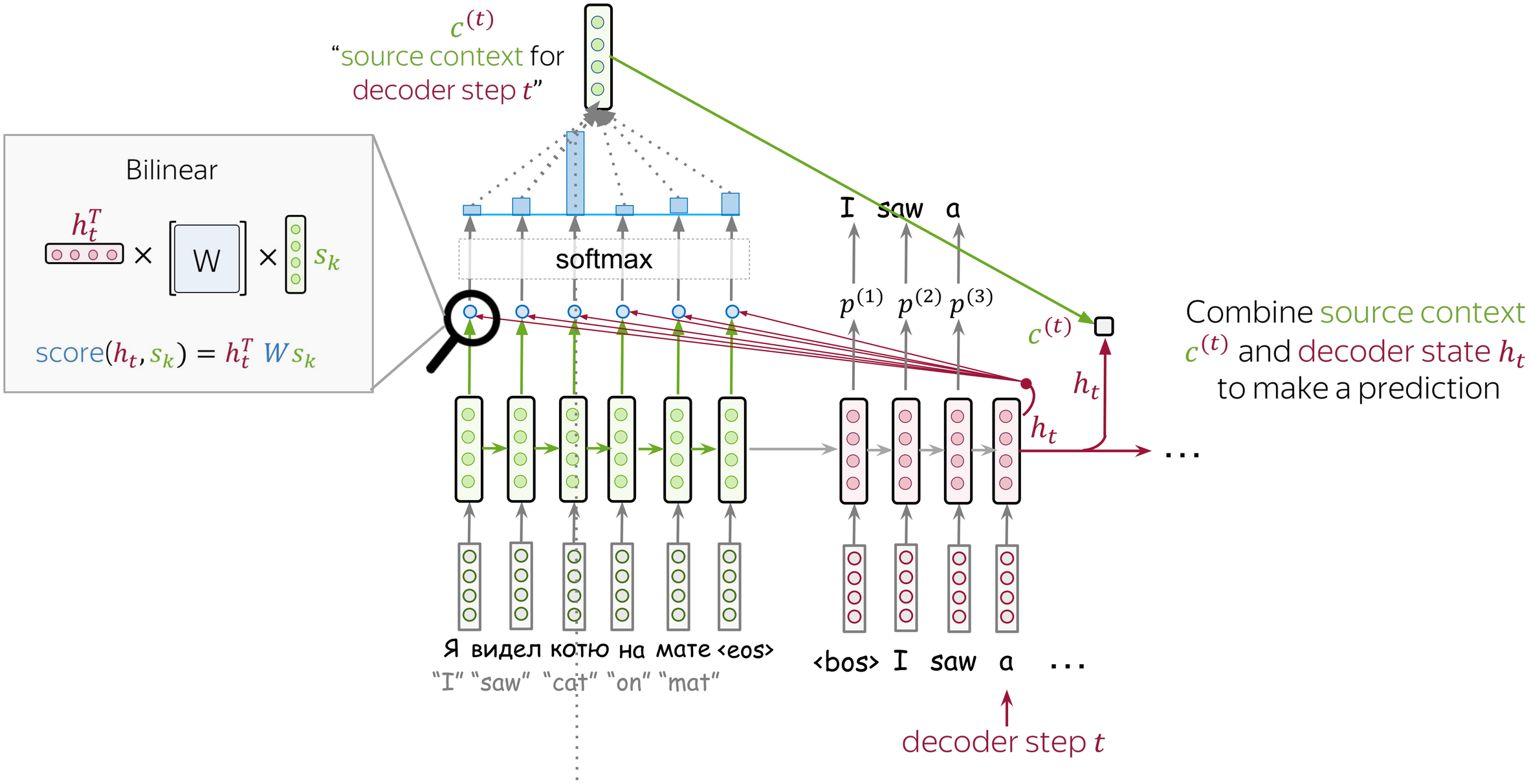
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softmax
$p^{(1)}$ $p^{(2)}$ $p^{(3)}$
I saw a
$c^{(t)}$
h_t
h_t
...
Combine source context $c^{(t)}$ and decoder state h_t to make a prediction
Я видел котю на мате <eos>
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<bos> I saw a ...
decoder step t

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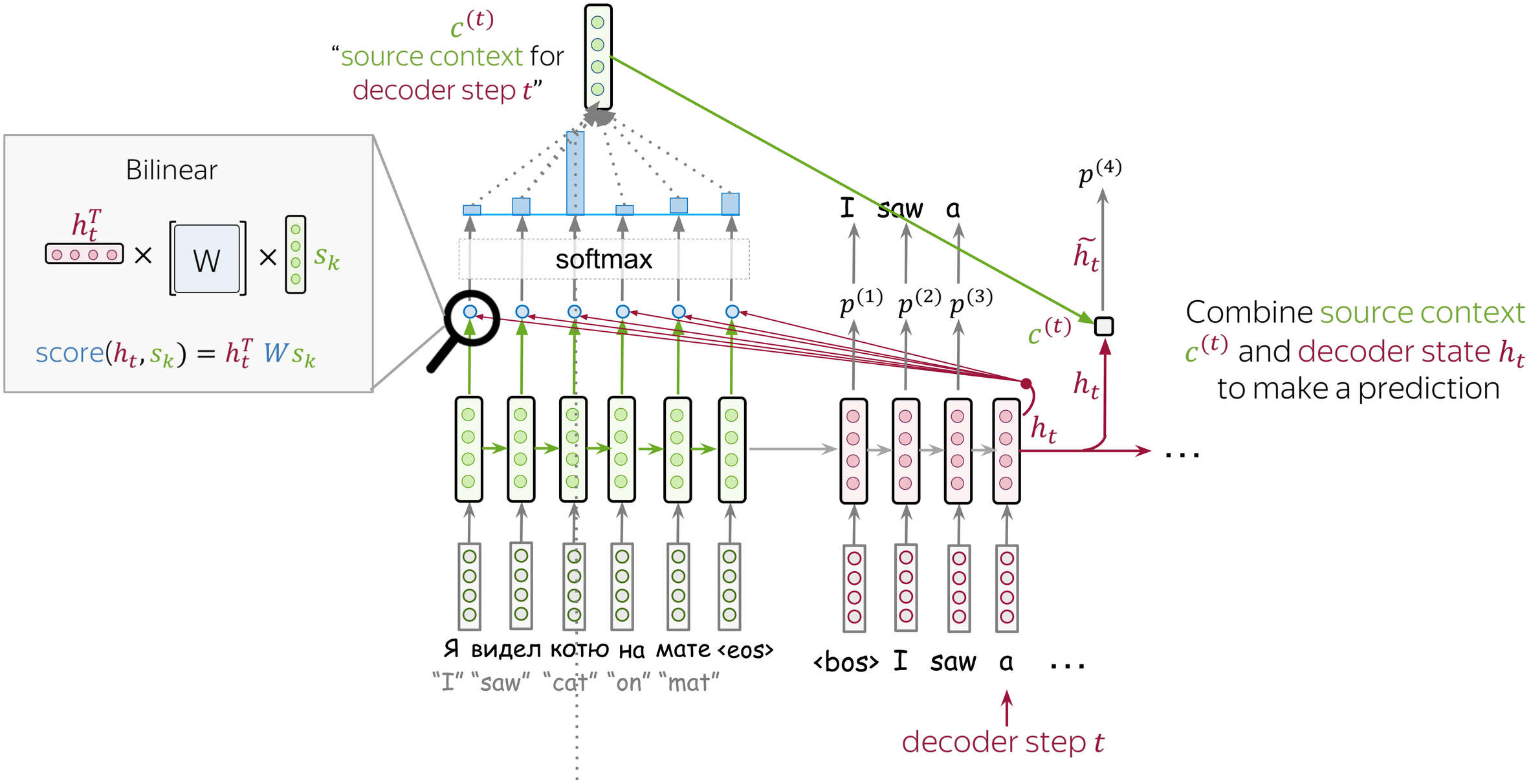
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$p^{(4)}$
$\tilde{h}_t$
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h_t
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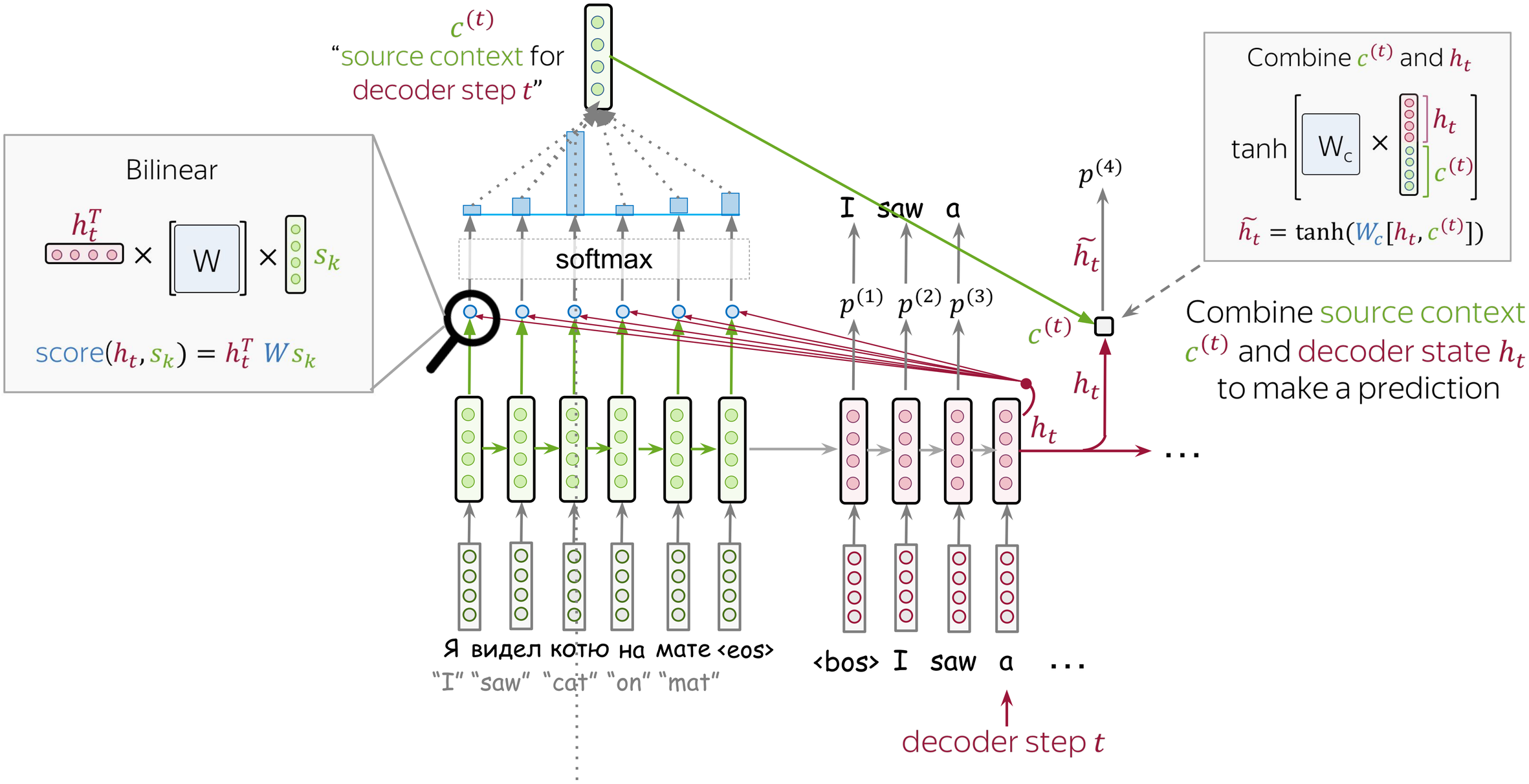
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I saw a
$p^{(1)}$ $p^{(2)}$ $p^{(3)}$
$p^{(4)}$
$\tilde{h}_t$
$c^{(t)}$
h_t
Combine $c^{(t)}$ and h_t
tanh [W$_c$ × h_t ; $c^{(t)}$]
$\tilde{h}_t = \tanh(W_c[h_t, c^{(t)}])$
Combine source context $c^{(t)}$ and decoder state h_t to make a prediction
...
Я видел котю на мате <eos>
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What is going to happen:

- Seq2seq Basics
- Attention
- Transformer
- Subword Segmentation: BPE
-  Analysis and Interpretability

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Idea: Attention is All You Need

	Seq2seq without attention	Seq2seq with attention
processing within encoder	RNN/CNN	RNN/CNN
processing within **decoder**	RNN/CNN	RNN/CNN
decoder -encoder interaction	static fixed-sized vector	attention

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Idea: Attention is All You Need

The animation is from the Google AI blog post.

Idea: Attention is All You Need

Encode
r

Who is doing:

What they are doing:

The animation is from the Google AI blog post.

Idea: Attention is All You Need

Encoder

Who is doing:

- all source tokens

What they are doing:

The animation is from the Google AI blog post.

Idea: Attention is All You Need

Encoder

Who is doing:

- all source tokens

What they are doing:

- look at each other
- update representations

repeat N times

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Idea: Attention is All You Need

Decoder

Who is doing:

What they are doing:

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Idea: Attention is All You Need

Decoder

Who is doing:

- target token at the current step

What they are doing:

The animation is from the Google AI blog post.

Idea: Attention is All You Need

Decoder

Who is doing:

- target token at the current step

What they are doing:

- looks at previous target tokens
- looks at source representations
- update representation

repeat N times

The animation is from the Google AI blog post.

Why can this be better than RNNs?

I arrived at the bank after crossing the …

…street?

…river?

What does bank mean in this sentence?

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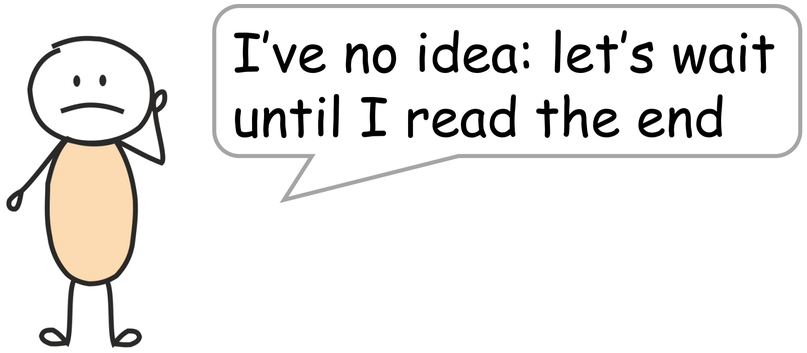

RNNs

O(N) steps to process a sentence with length N

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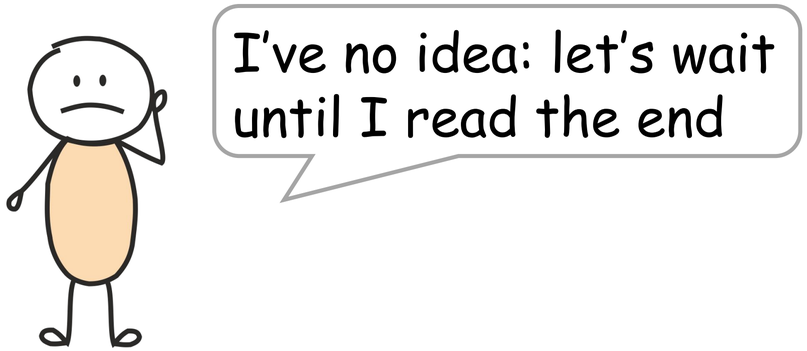

RNNs

O(N) steps to process a sentence with length N

Transformer

Constant number of steps to process any sentence

What is going to happen:

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Self-Attention: Why “Self”?

Decoder-encoder attention is looking

- **from**: one current decoder state
- **at**: all encoder states

Self-Attention: Why “Self”?

Decoder-encoder attention is looking

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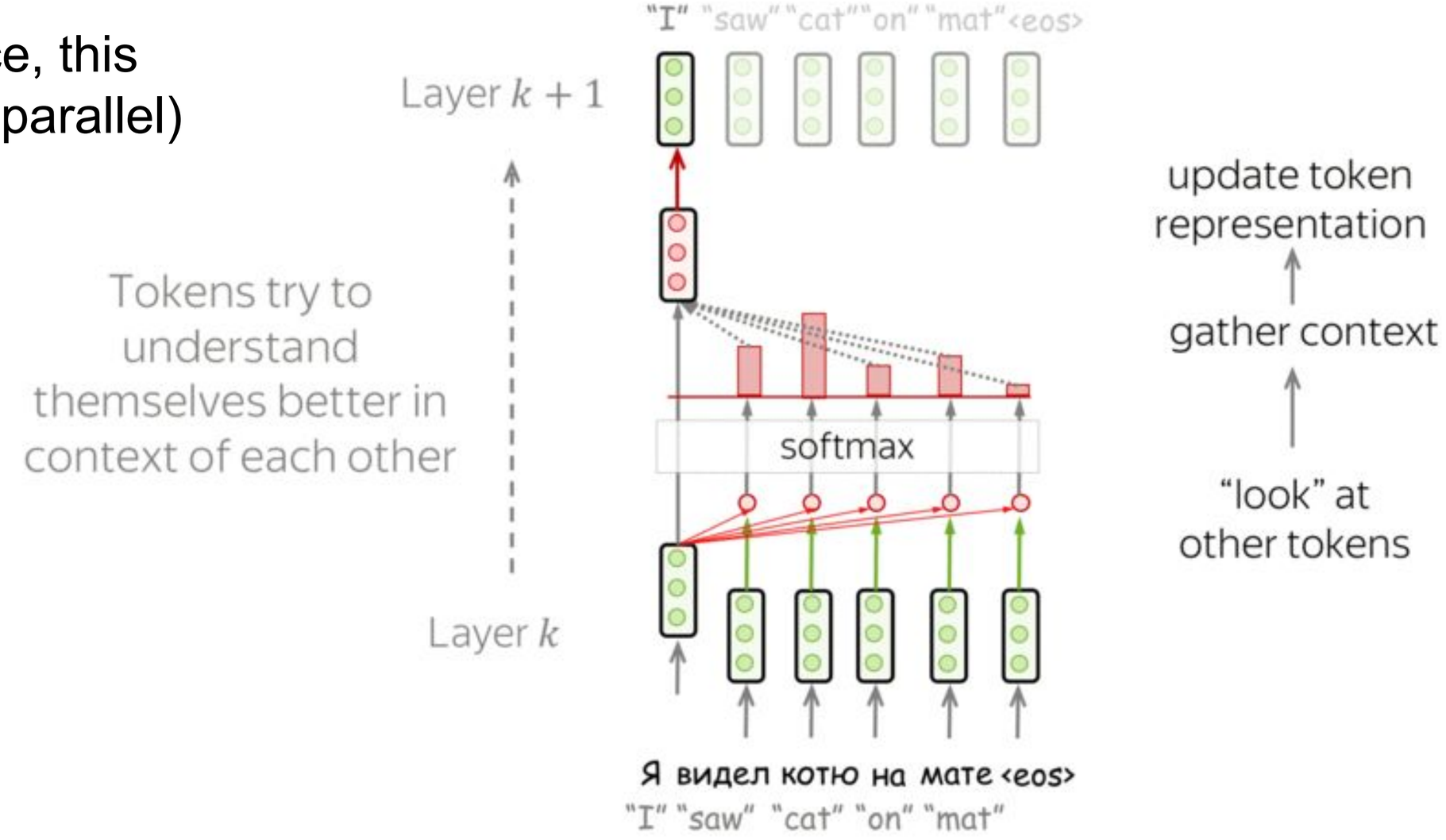
Self-attention is looking

- **from**: each state from a set of states
- **at**: all other states in the same set

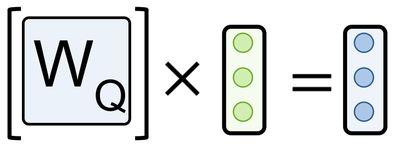
Self-Attention: The “Look at Each Other” Part

(in practice, this happens in parallel)



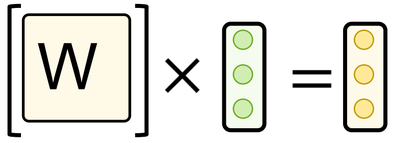
Query, Key, Value

Each vector receives three representations (“roles”)



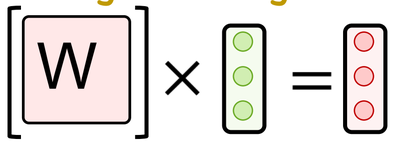
Query: vector **from** which the attention is looking

“Hey there, do you have this information?”



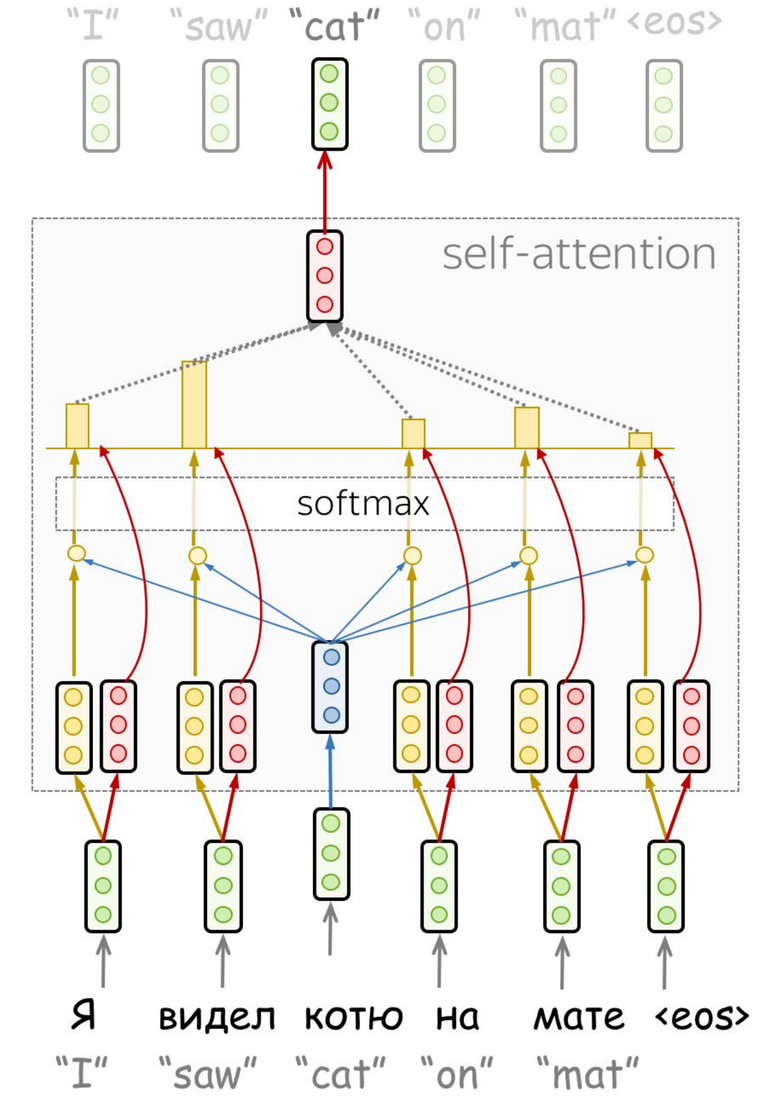
Key: vector **at** which the query looks to compute weights

“Hi, I have this information – give me a large weight!”



Value: their weighted sum is attention output

“Here’s the information I have!”



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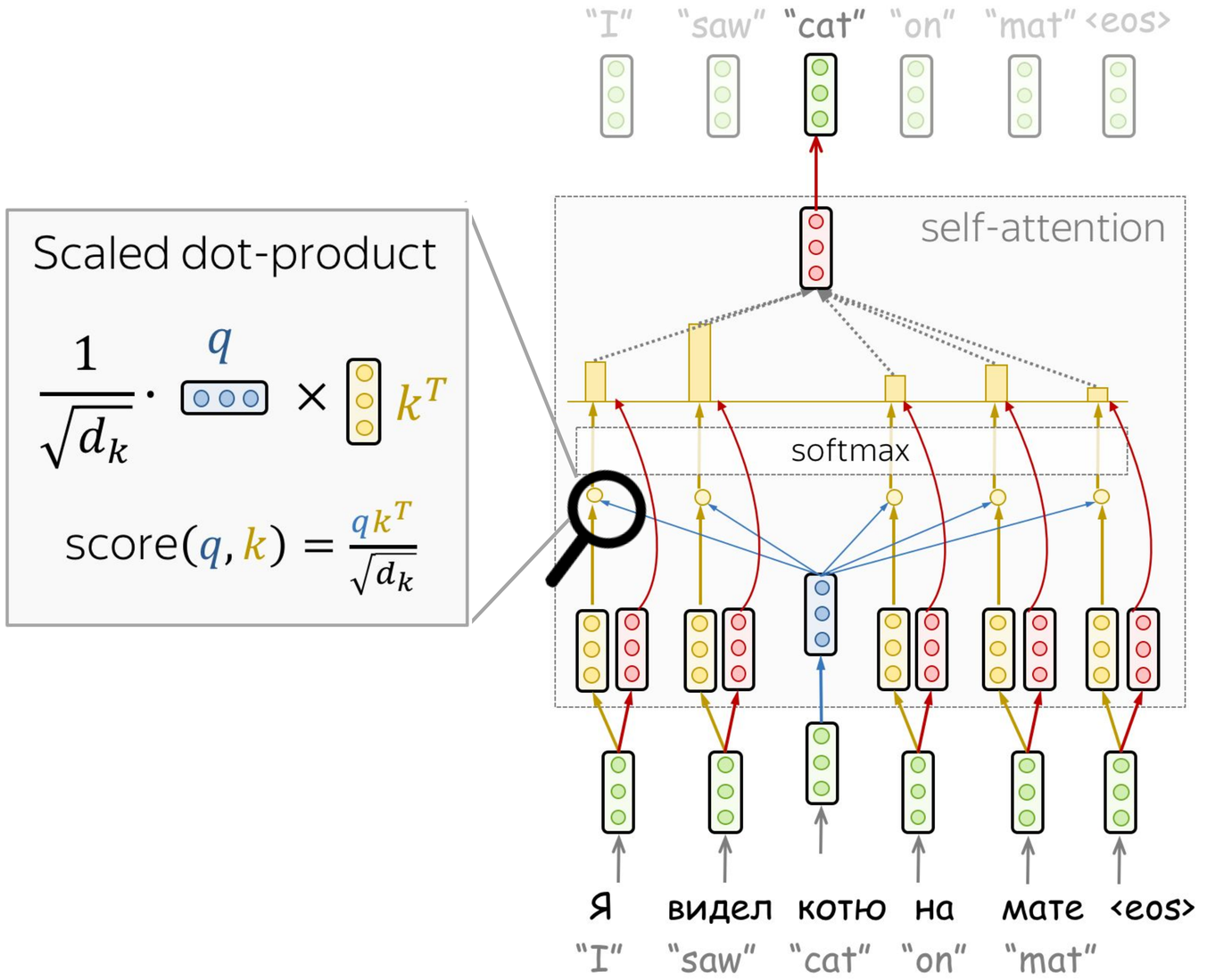
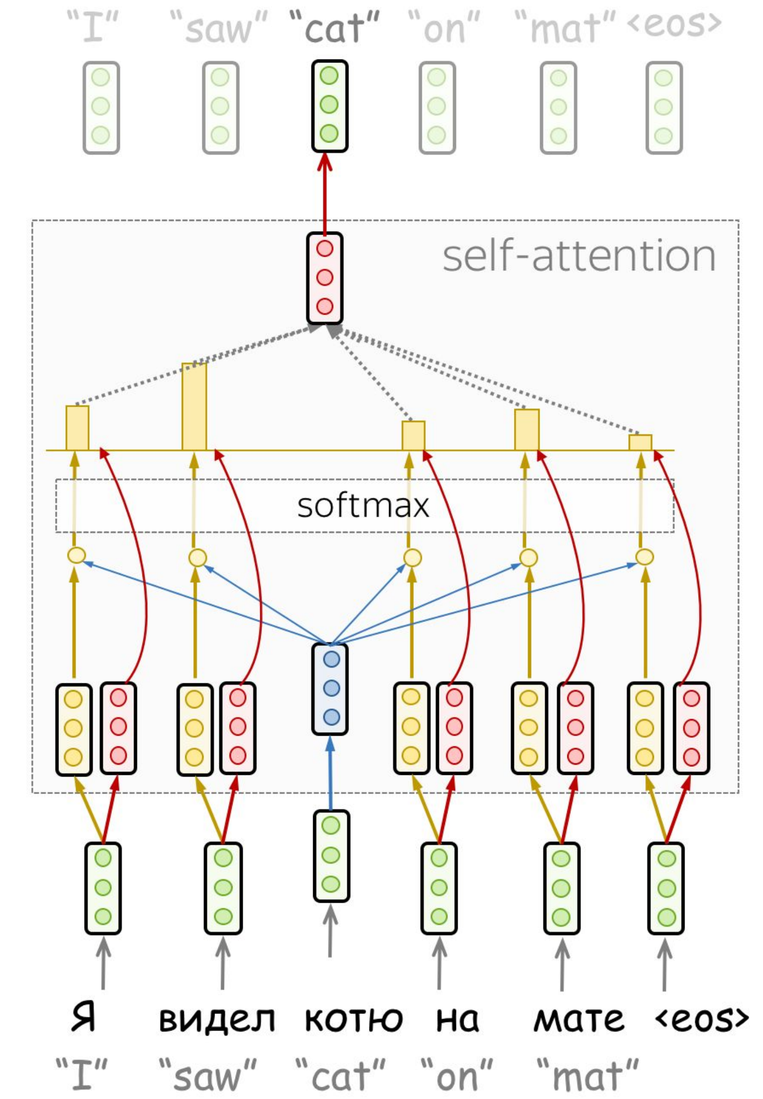


Query, Key, Value

$$Attention(q, k, v) = \overbrace{softmax\left(\frac{qk^T}{\sqrt{d_k}}\right)}^{\text{Attention weights}} v$$

from — to

vector dimensionality of K, V



$\sqrt{d_k}$?

Assumption: $0 = \mathbb{E}[k_i] = \mathbb{E}[q_i],\ 1 = Var[k_i] = Var[q_i]$

Can we assume this?

$$Var[XY] = \mathbb{E}[X^2Y^2] - \mathbb{E}^2[XY] \overset{why?}{=} \mathbb{E}[X^2]\mathbb{E}[Y^2] - \mathbb{E}^2[X]\mathbb{E}^2[Y]$$
$$= Var[X]Var[Y] + \mathbb{E}[X^2]\mathbb{E}^2[Y] + \mathbb{E}^2[X]\mathbb{E}[Y^2] - 2\mathbb{E}^2[X]\mathbb{E}^2[Y]$$
$$= Var[X]Var[Y] + Var[X]\mathbb{E}^2[Y] + Var[Y]\mathbb{E}^2[X]$$

Hence

$$Var[k^T q] = \sum Var[k_i q_i] = \sum Var[k_i]Var[q_i] = d_k$$

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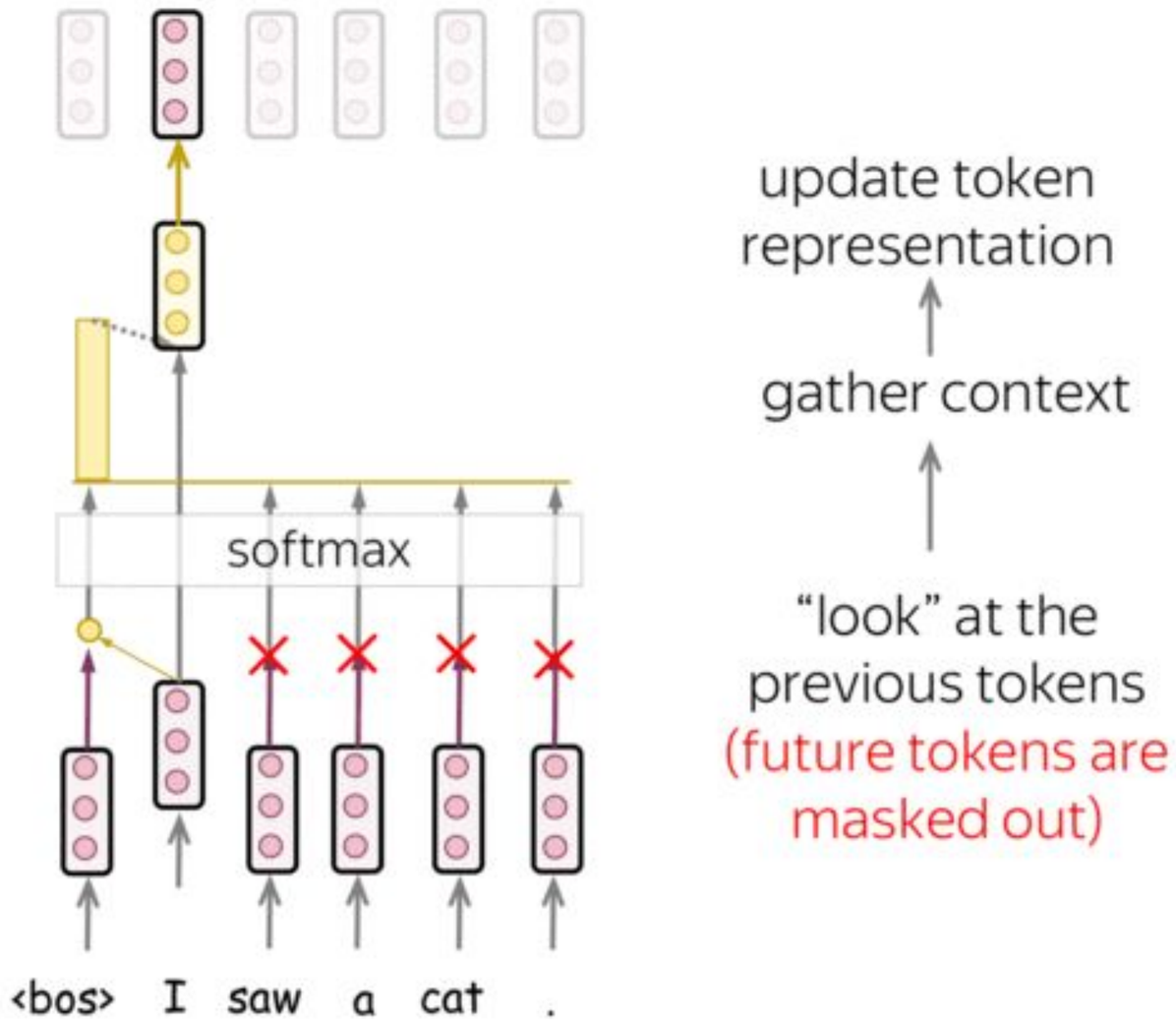
Masked Self-Attention: “Don’t Look Ahead”

In the decoder, we forbid looking at future tokens – we don’t know them

Note: in training, decoder processes all target tokens at once – without masks, it would see future



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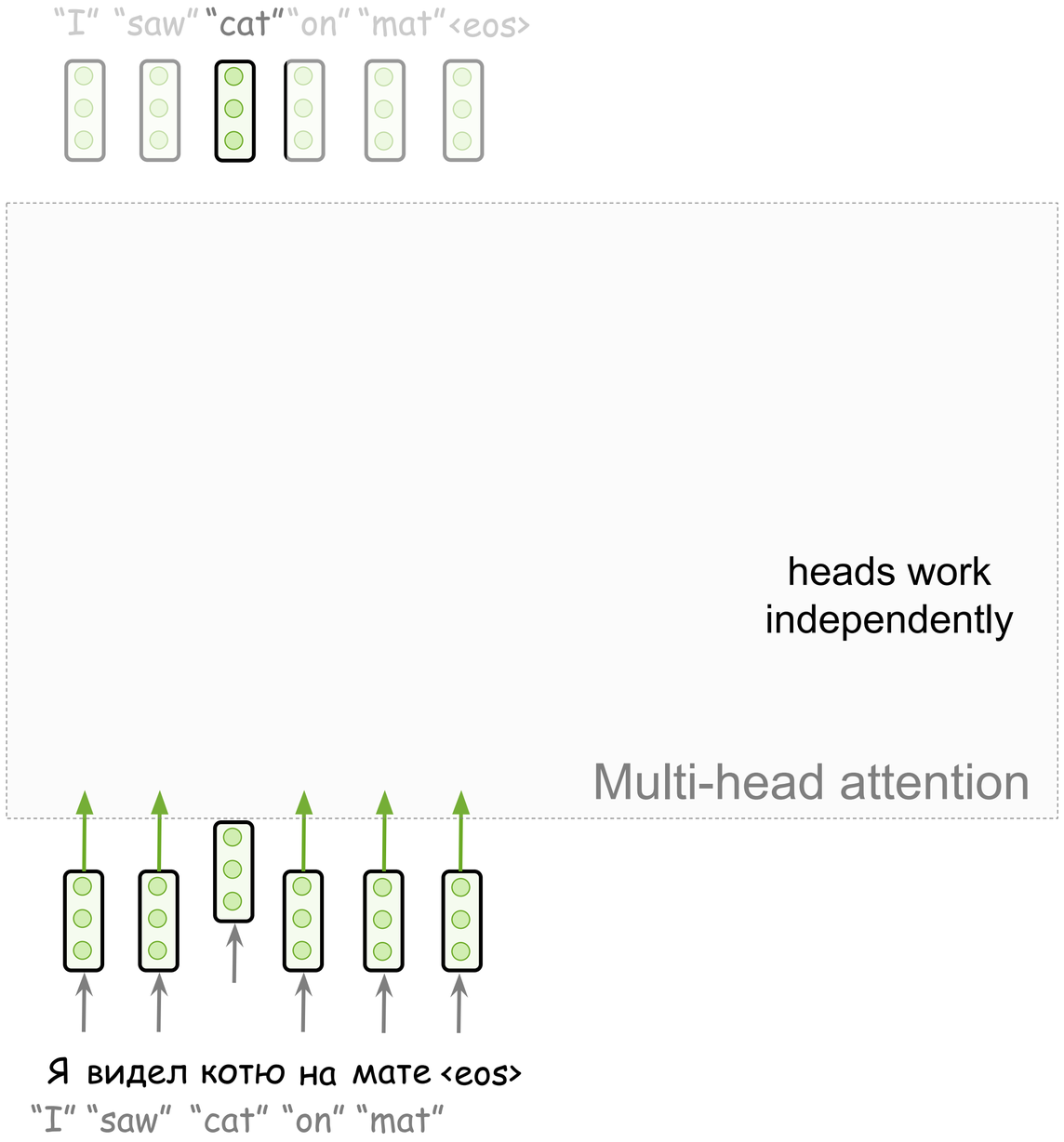
Multi-Head Attention

We need to track many different things at once!

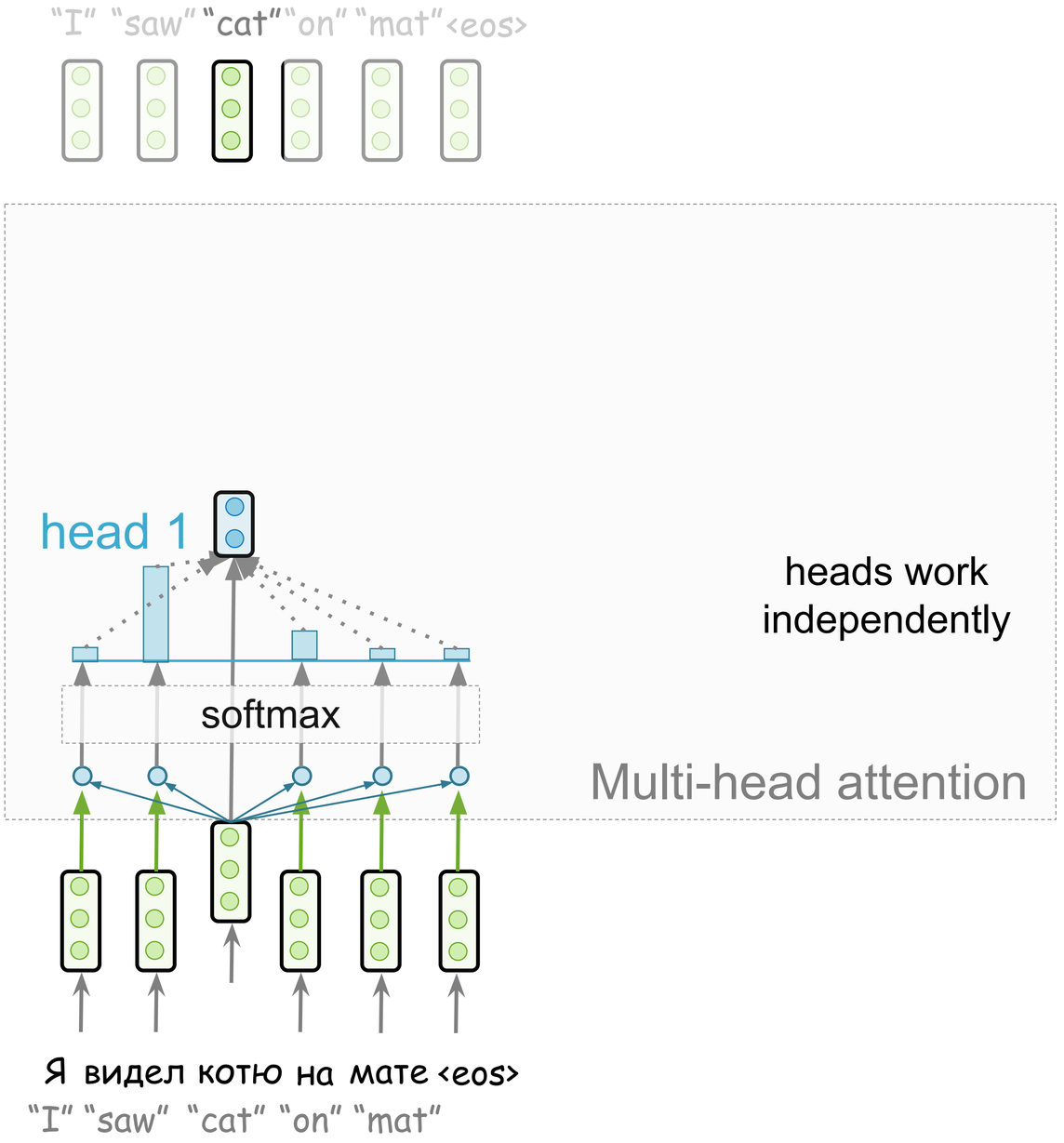
- Gender agreement
- Case government
- Lexical preferences
- ...

The example is from David Talbot

Multi-Head Attention



Multi-Head Attention



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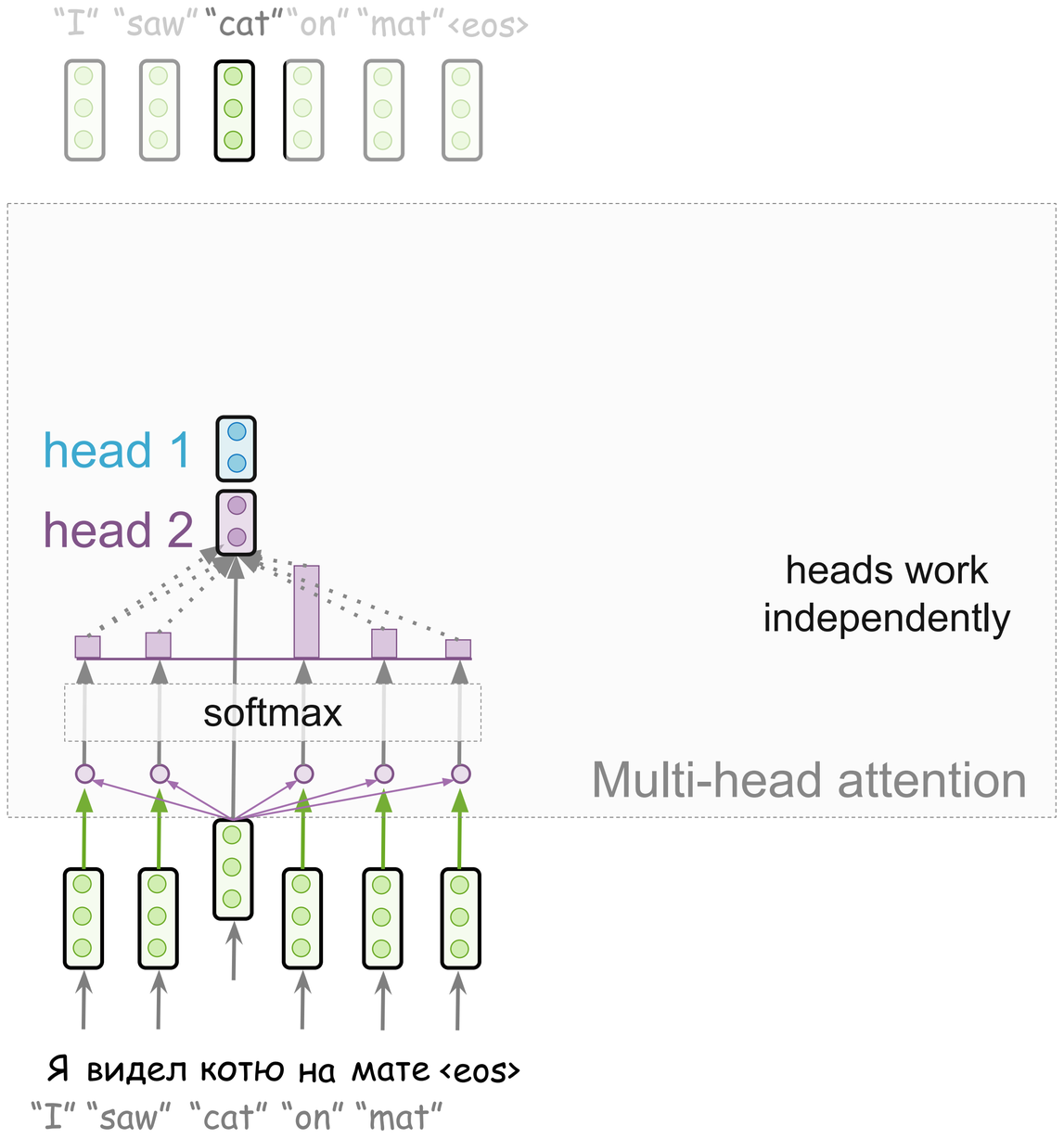

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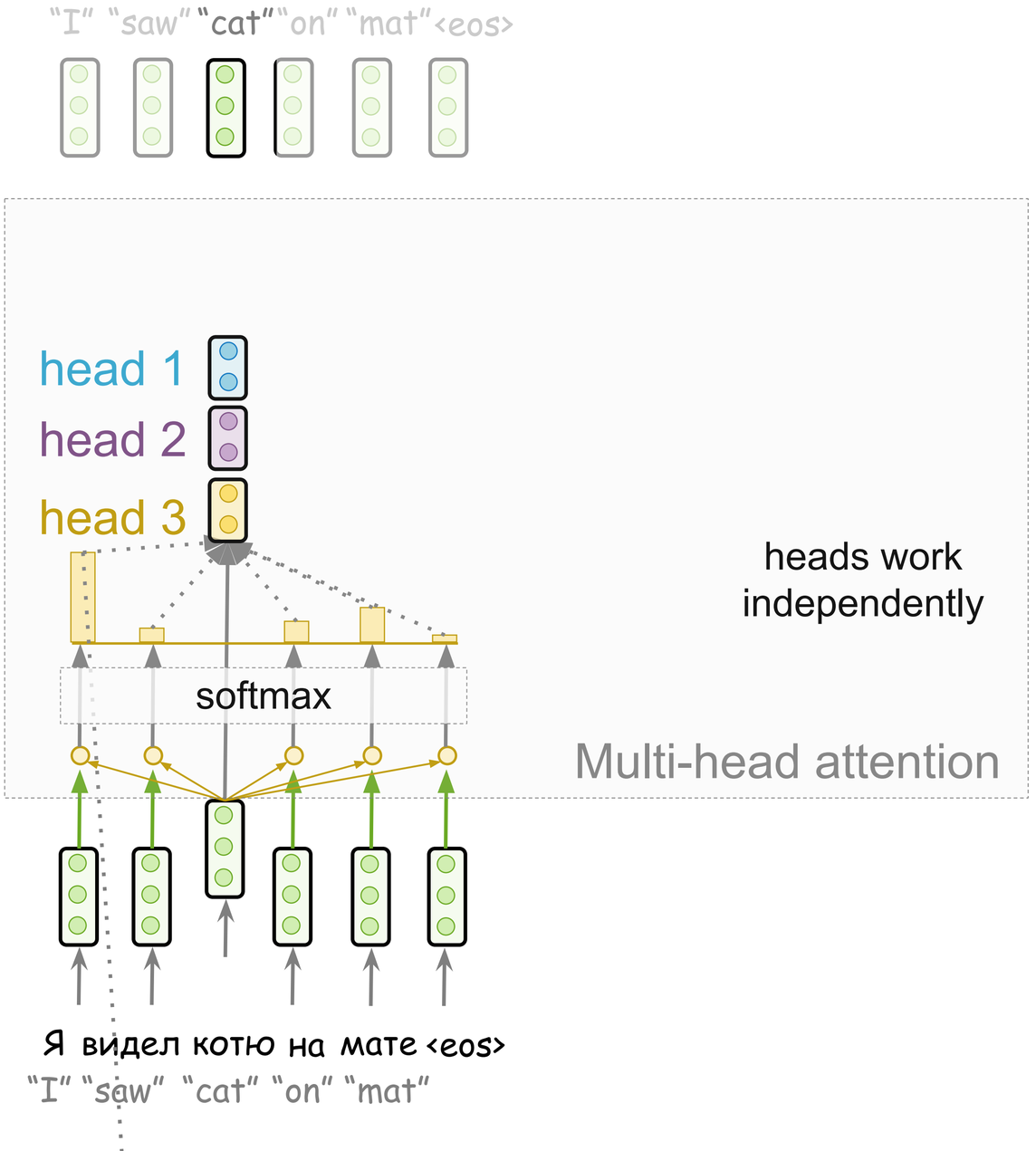

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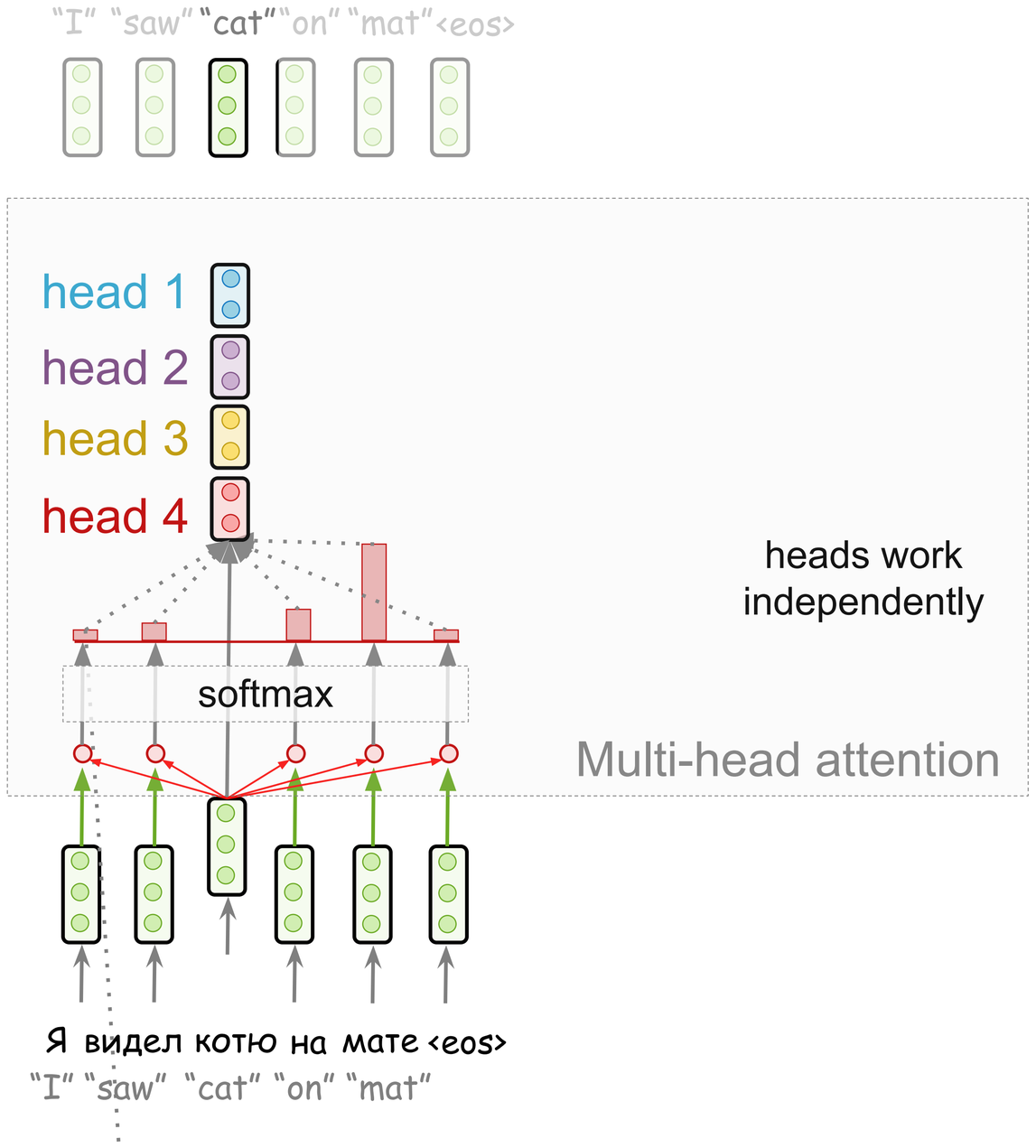
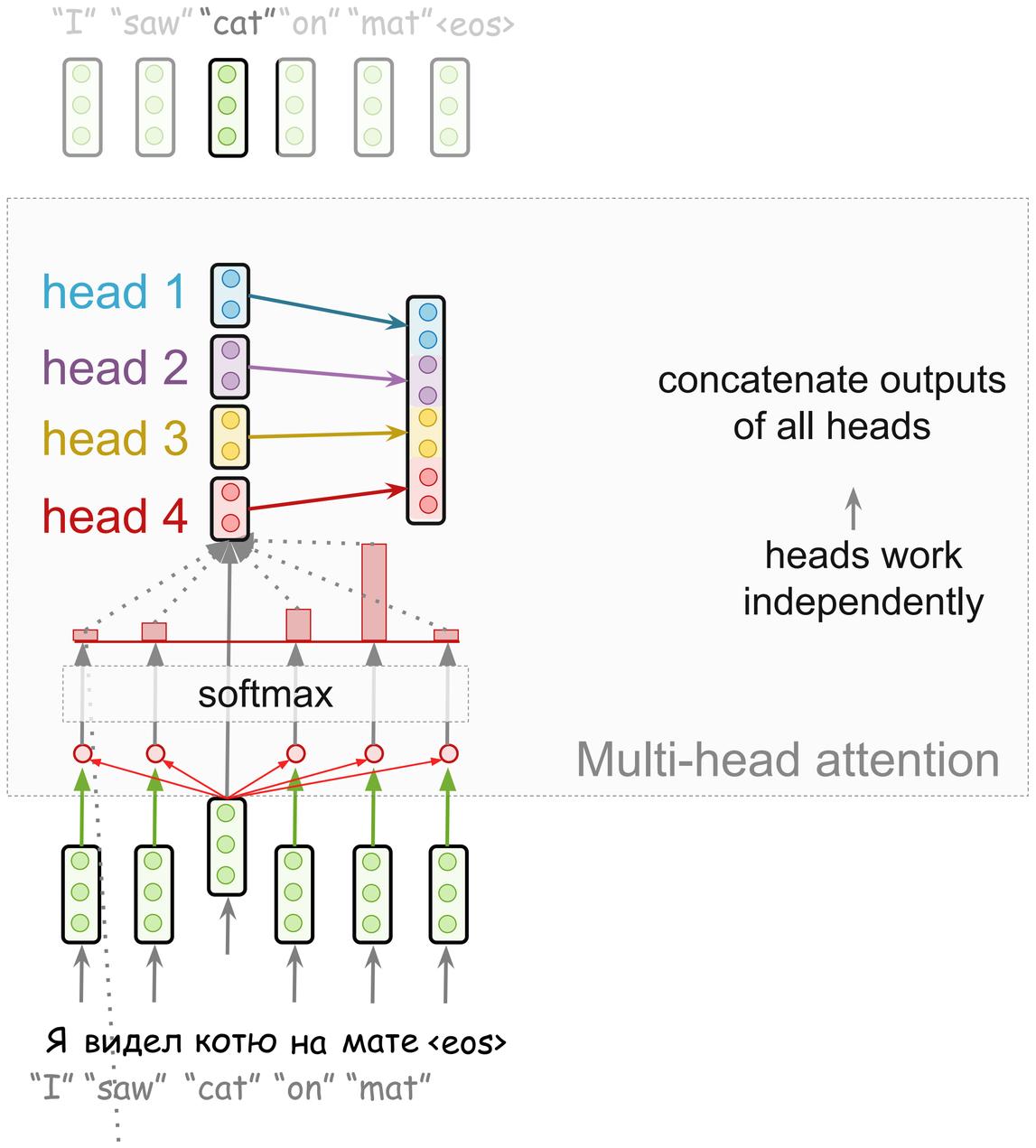


Multi-Head Attention



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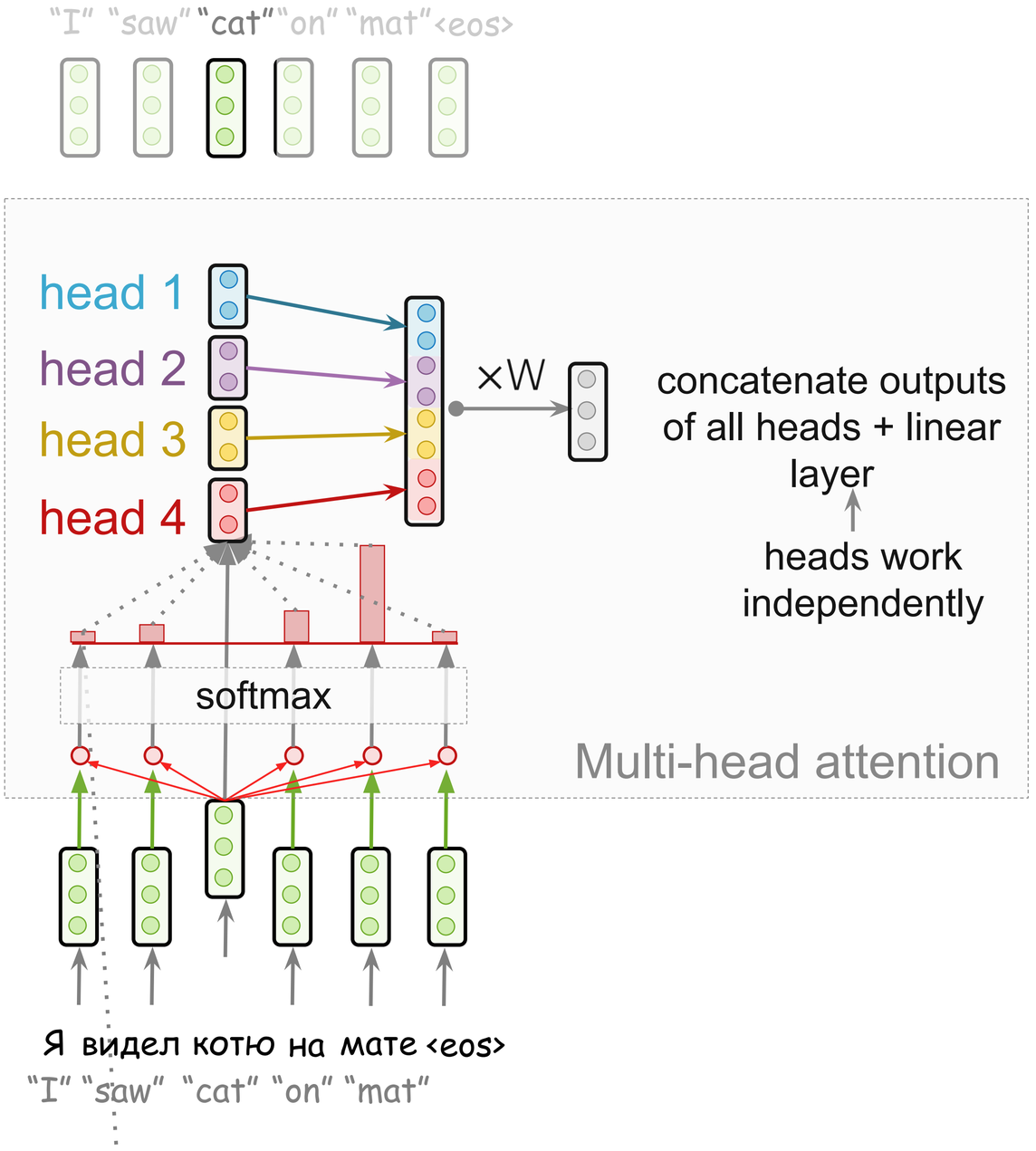

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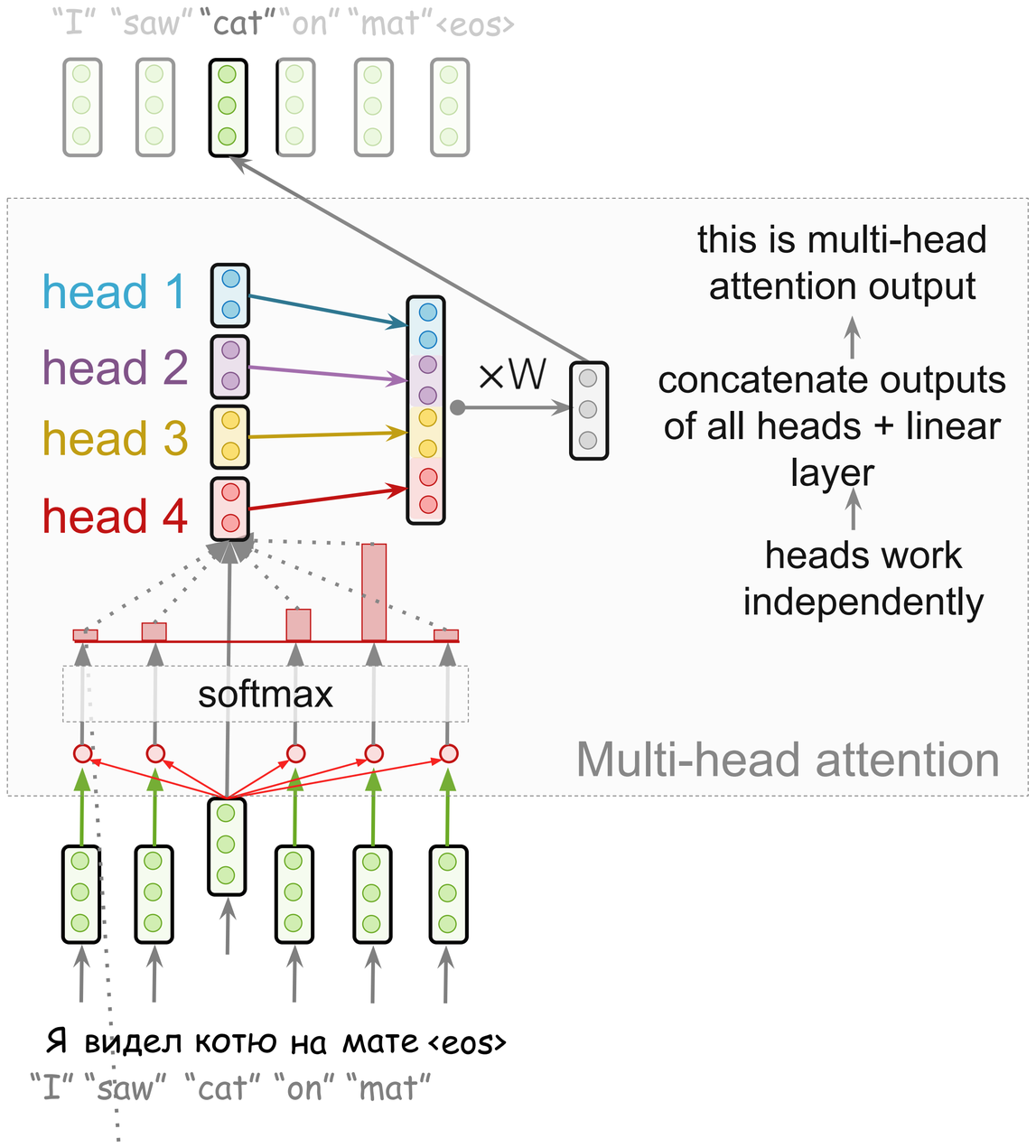
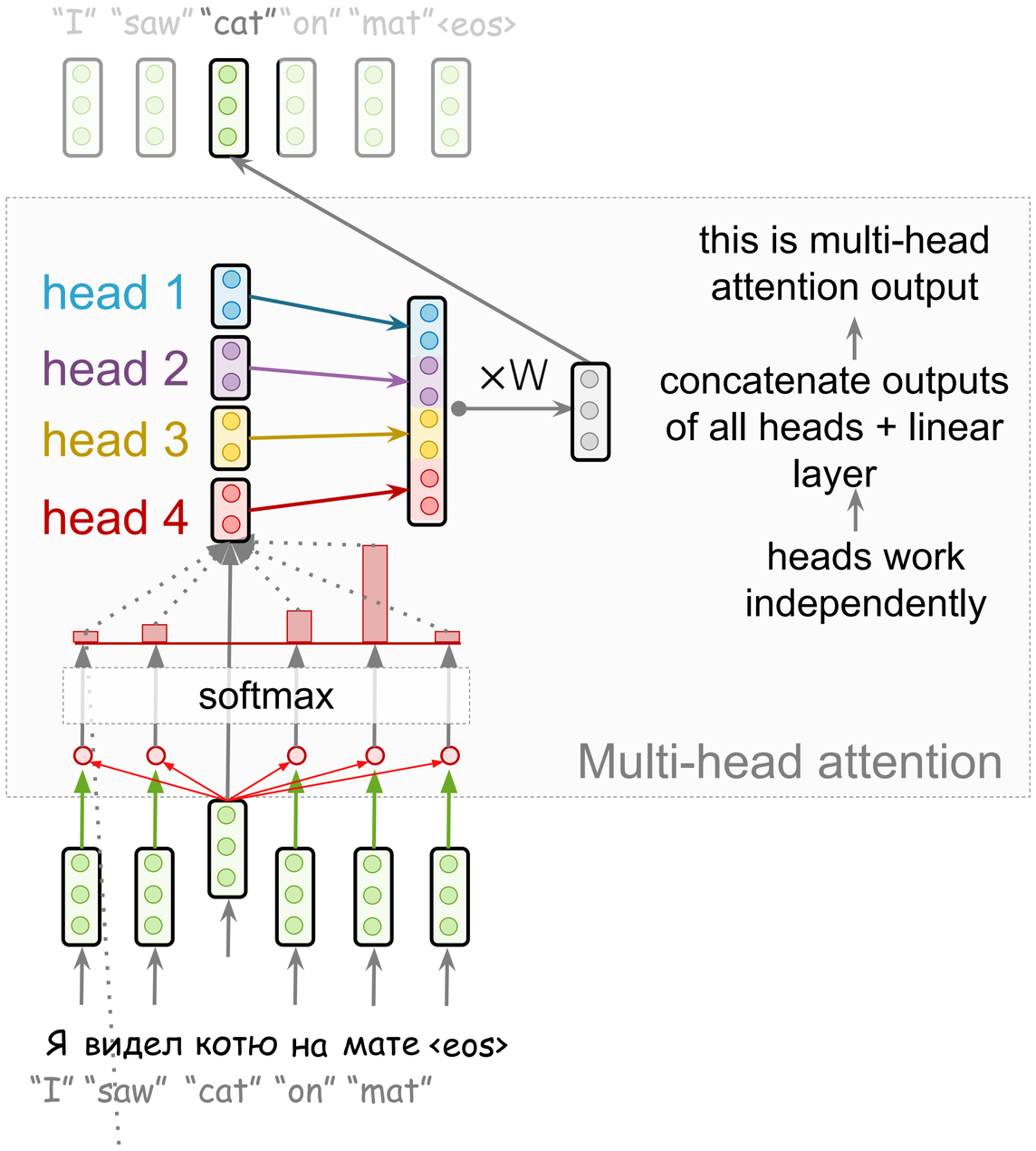


Multi-Head Attention

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_n)W_o,$$

$$\text{head}_i = \text{Attention}(QW_Q^i, KW_K^i, VW_V^i)$$



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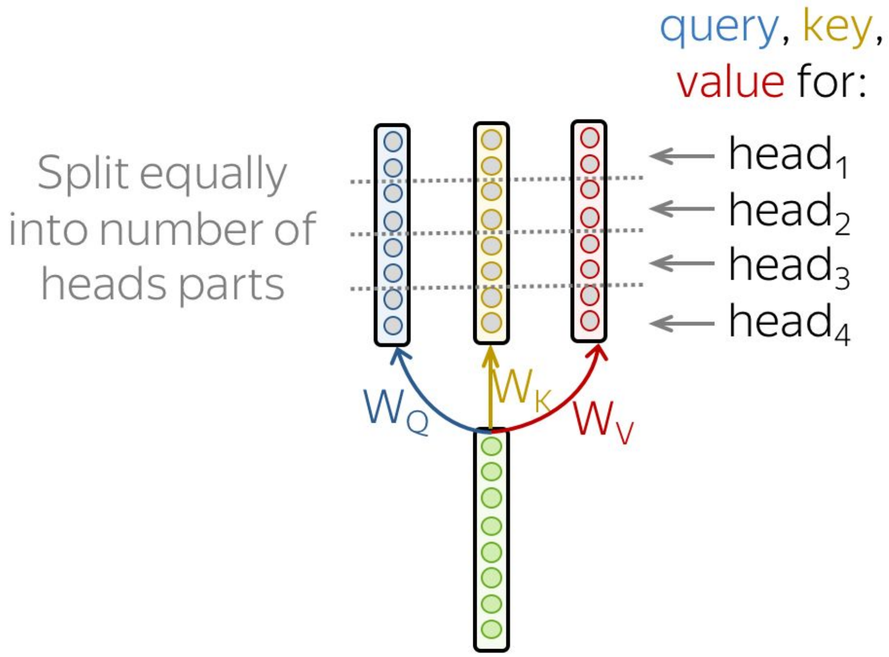

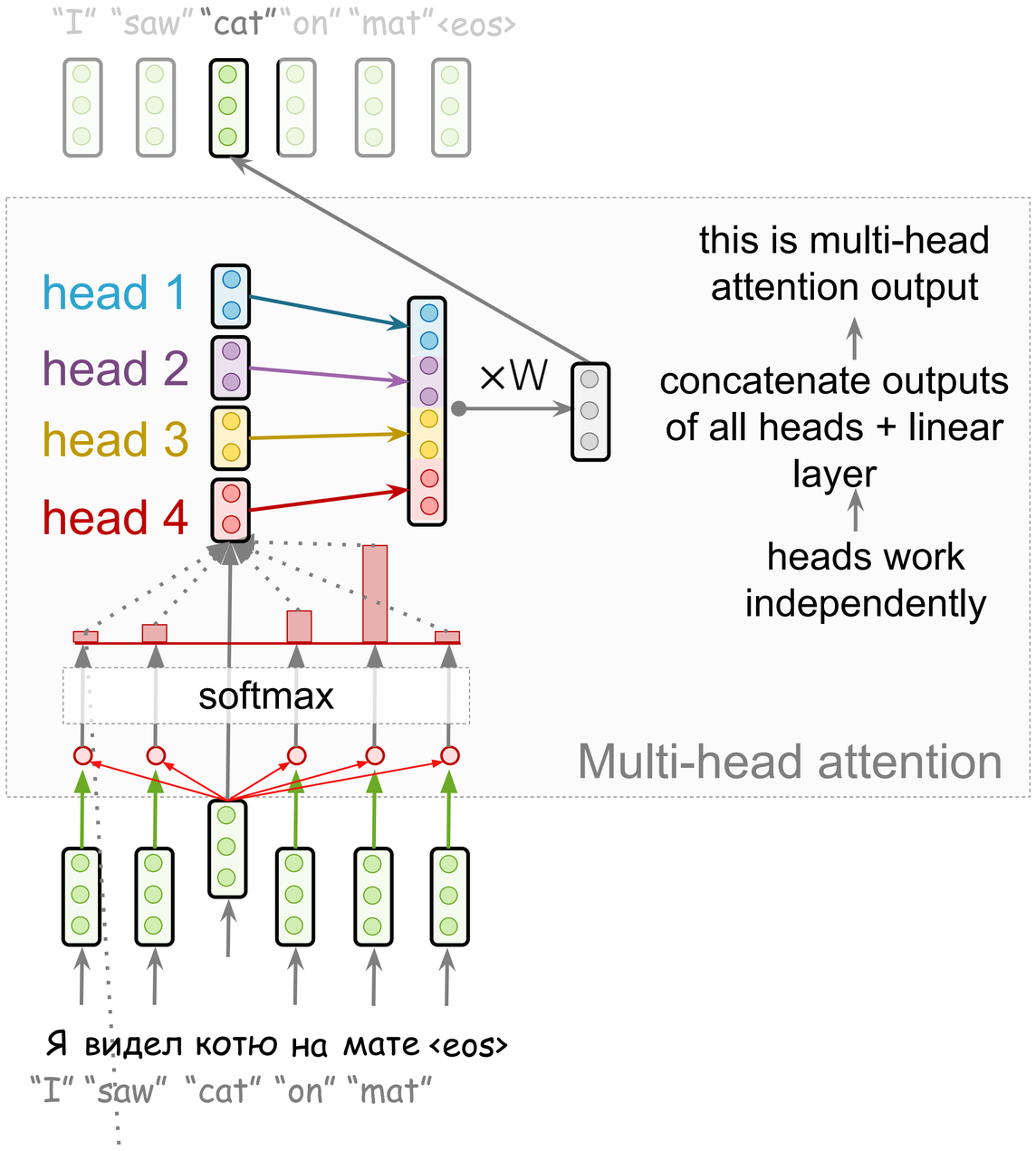

What is going to happen:

- Seq2seq Basics
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- Transformer
  - Idea
  - Self-Attention
  - Masked Self-Attention
  - Multi-Head Attention
  - KV caching
  - Model architecture
- Subword Segmentation: BPE
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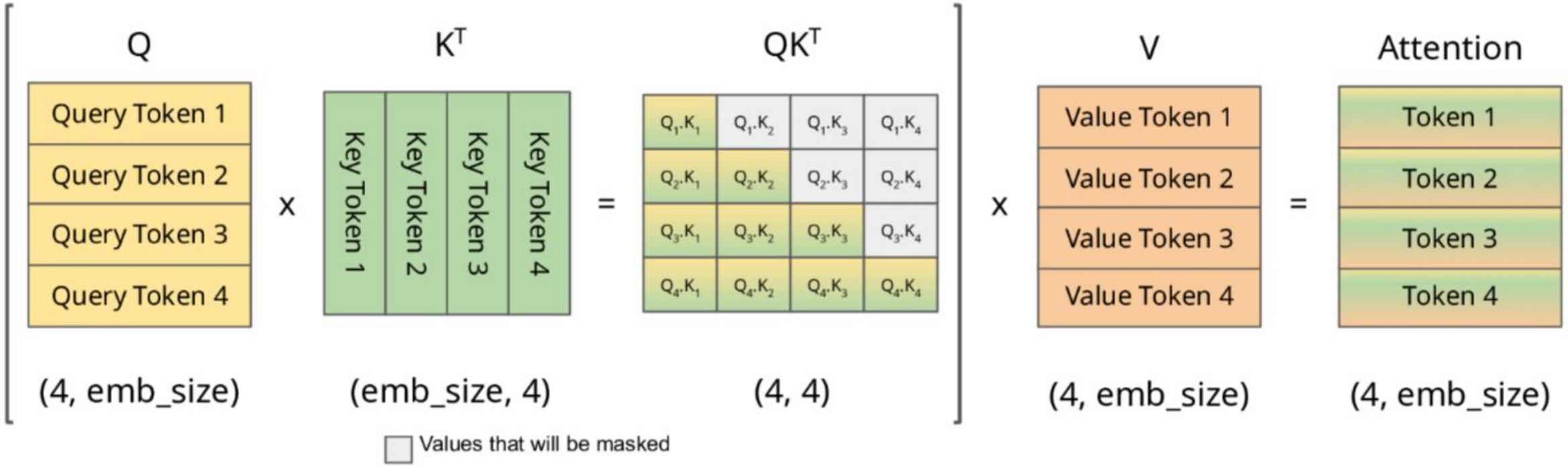
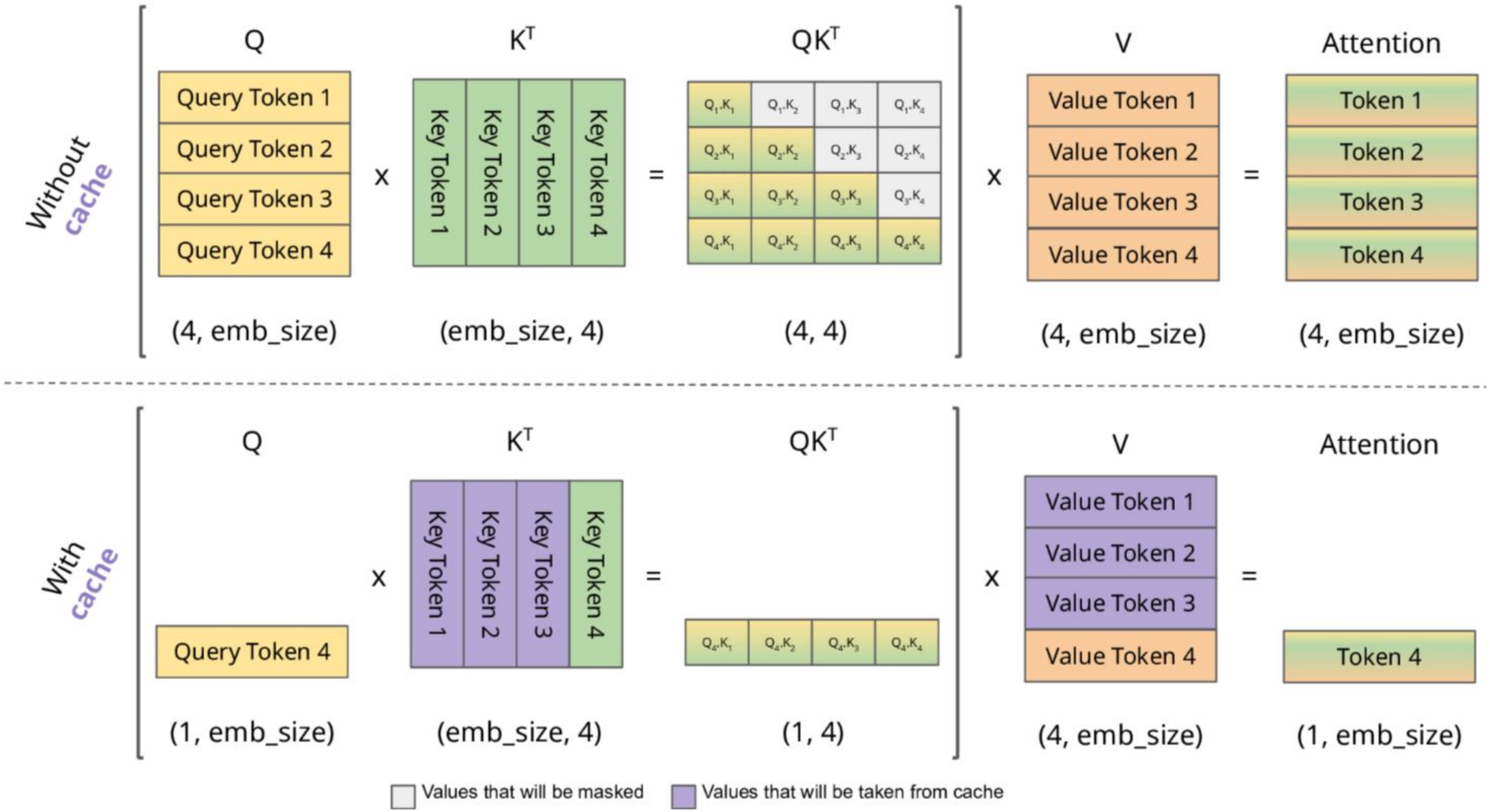

Q
K^T
QK^T
V
Attention
Query Token 1
Query Token 2
Query Token 3
Query Token 4
x
Key Token 1
Key Token 2
Key Token 3
Key Token 4
=
Q1.K1
Q1.K2
Q1.K3
Q1.K4
Q2.K1
Q2.K2
Q2.K3
Q2.K4
Q3.K1
Q3.K2
Q3.K3
Q3.K4
Q4.K1
Q4.K2
Q4.K3
Q4.K4
x
Value Token 1
Value Token 2
Value Token 3
Value Token 4
=
Token 1
Token 2
Token 3
Token 4
(4, emb_size)
(emb_size, 4)
(4, 4)
(4, emb_size)
(4, emb_size)
Values that will be masked

KV caching


Without cache
Q
Query Token 1
Query Token 2
Query Token 3
Query Token 4
(4, emb_size)
x
K^T
Key Token 1
Key Token 2
Key Token 3
Key Token 4
(emb_size, 4)
=
QK^T
(4, 4)
x
V
Value Token 1
Value Token 2
Value Token 3
Value Token 4
(4, emb_size)
=
Attention
Token 1
Token 2
Token 3
Token 4
(4, emb_size)
With cache
Q
Query Token 4
(1, emb_size)
x
K^T
Key Token 1
Key Token 2
Key Token 3
Key Token 4
(emb_size, 4)
=
QK^T
(1, 4)
x
V
Value Token 1
Value Token 2
Value Token 3
Value Token 4
(4, emb_size)
=
Attention
Token 4
(1, emb_size)
Values that will be masked
Values that will be taken from cache

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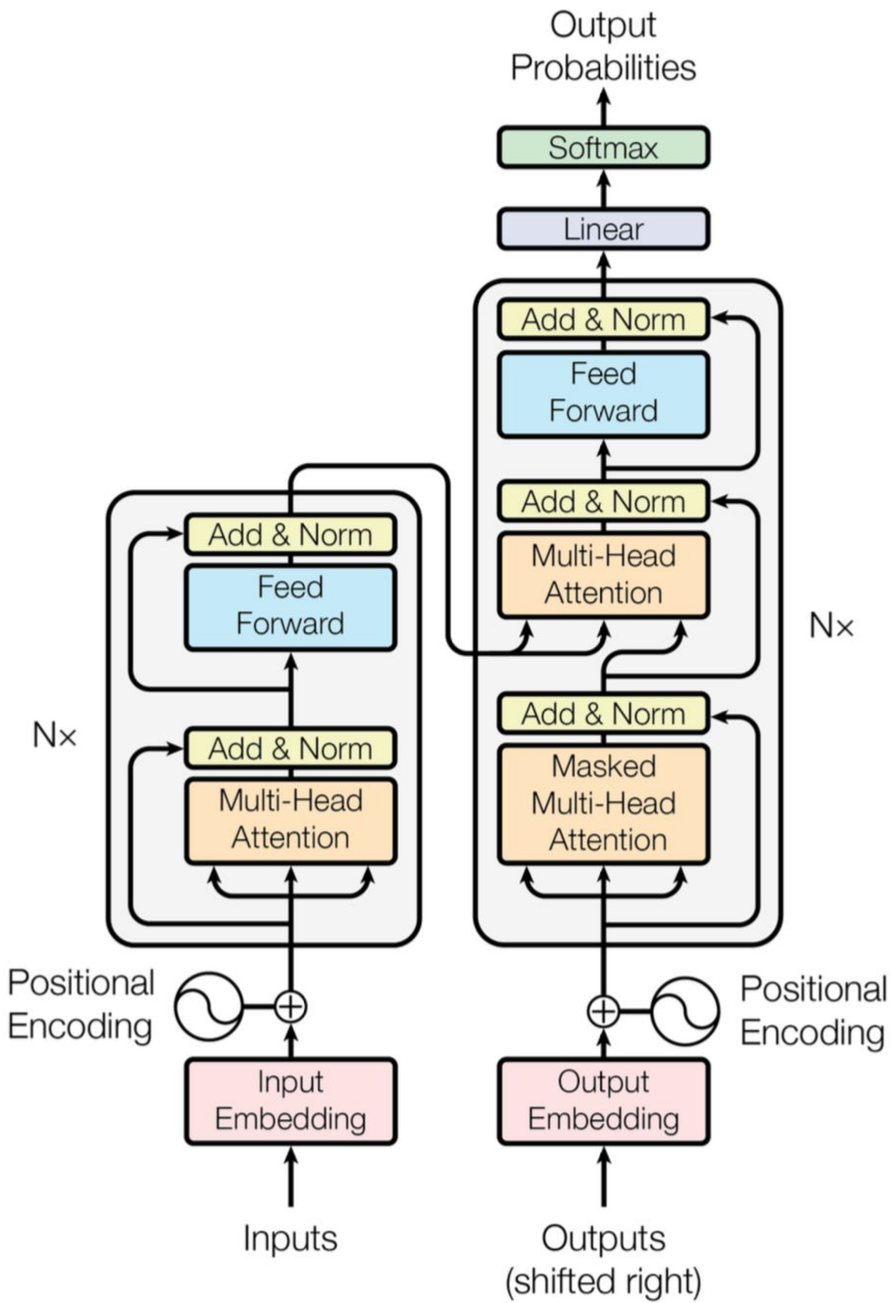
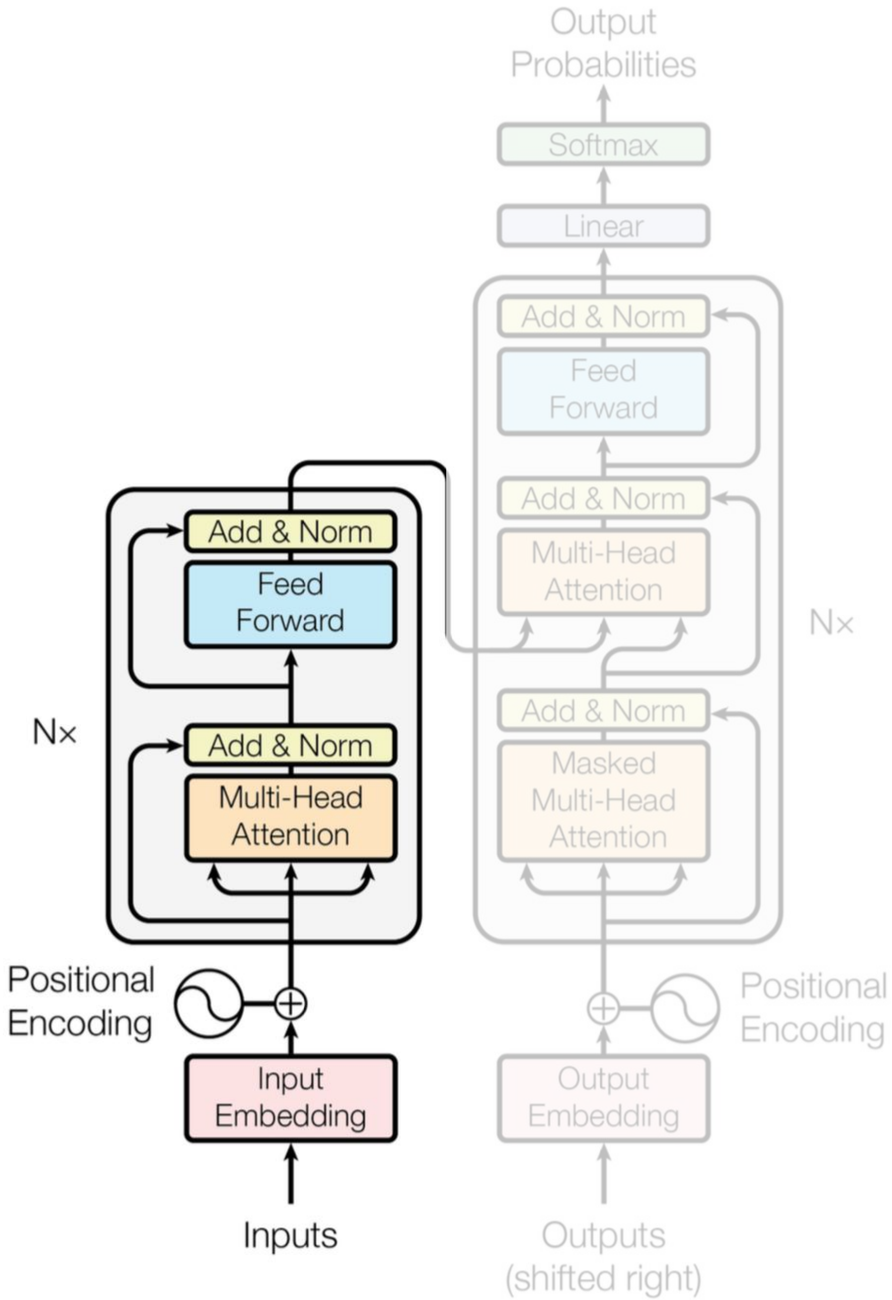
Transformer



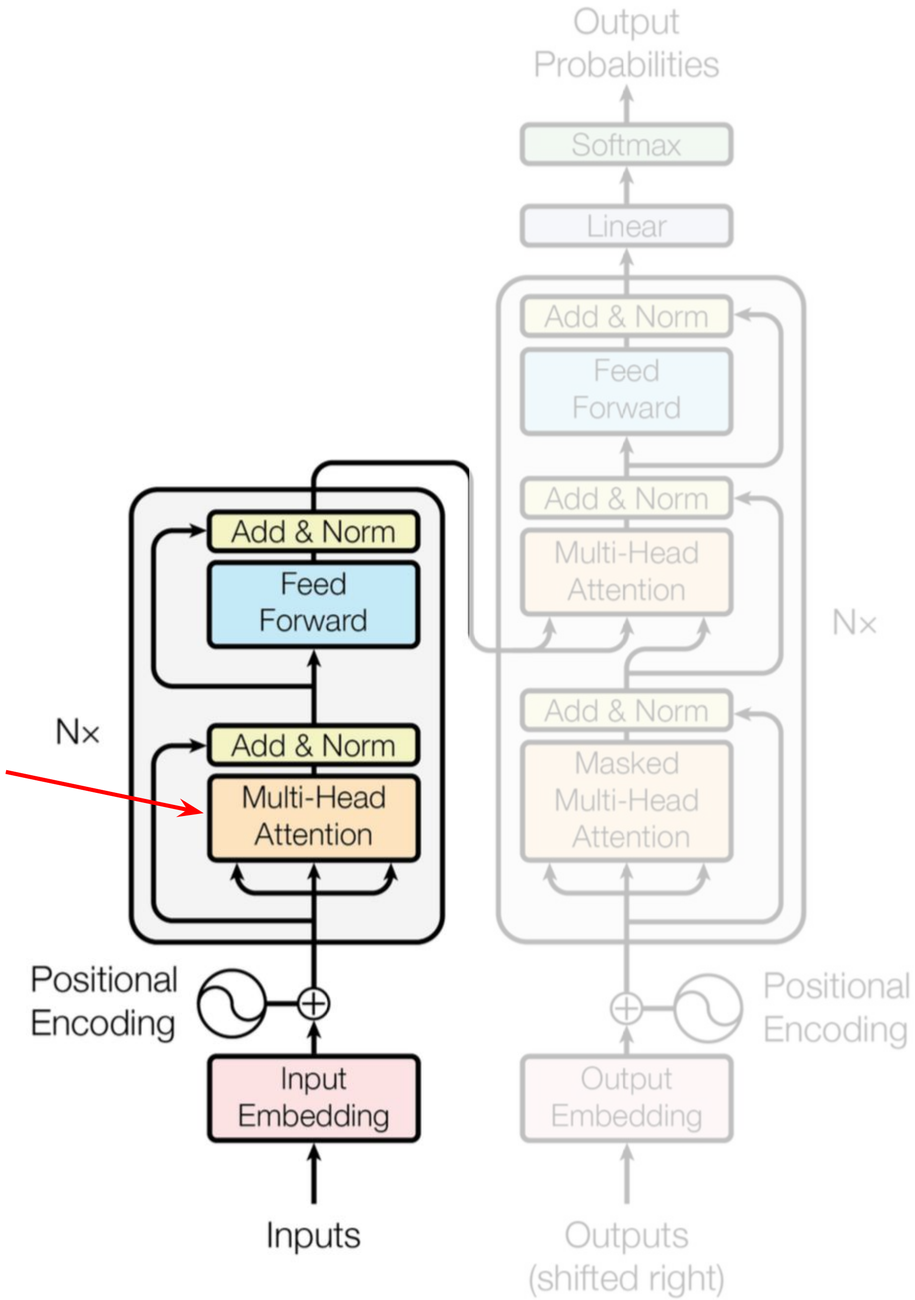
Transformer



Transformer

Encoder self-attention:
tokens look at each other

queries, keys, values
are computed from
encoder states



Transformer

Feed-forward network:
after taking information from other tokens, take a moment to think and process this information

Encoder self-attention:
tokens look at each other

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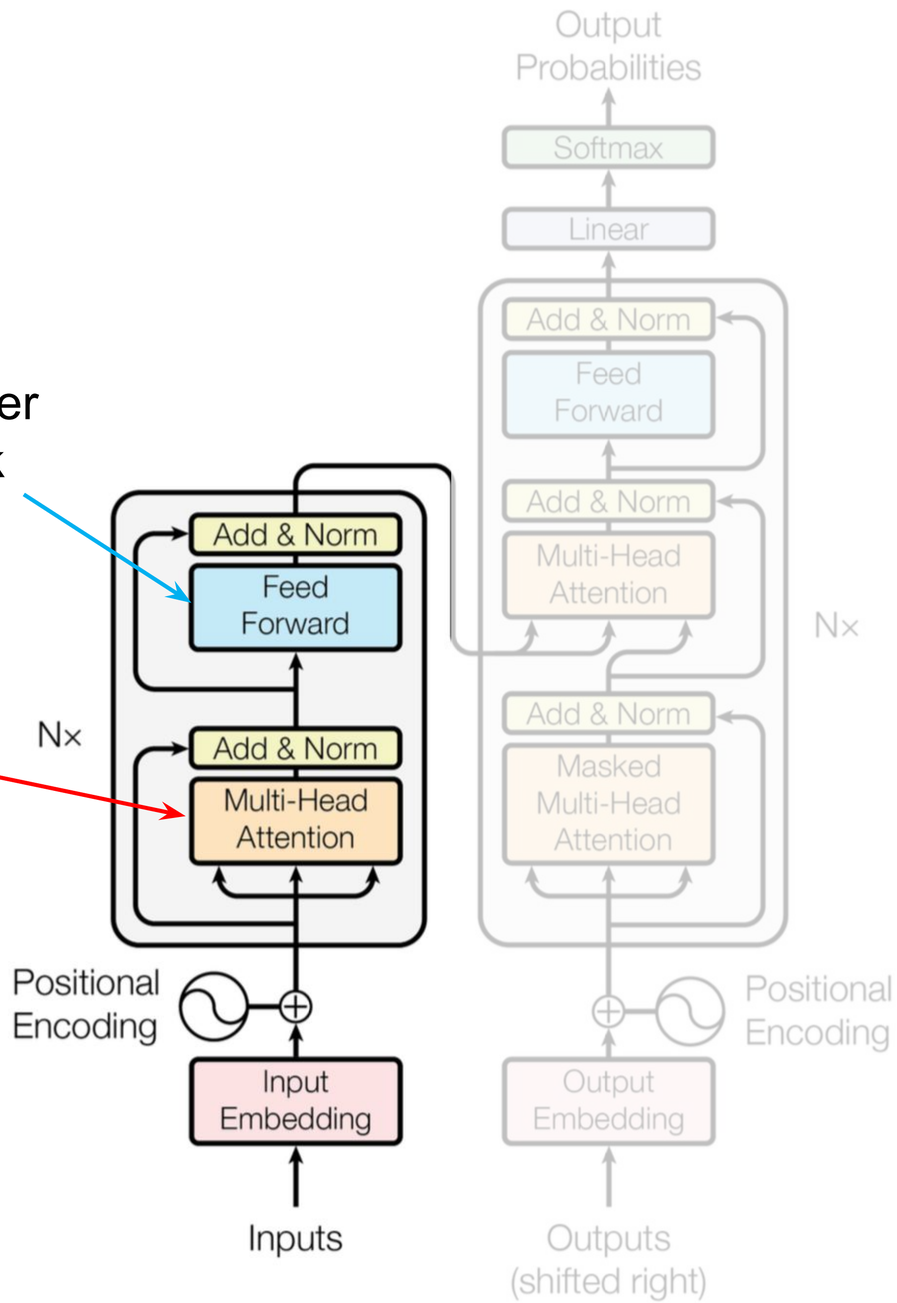

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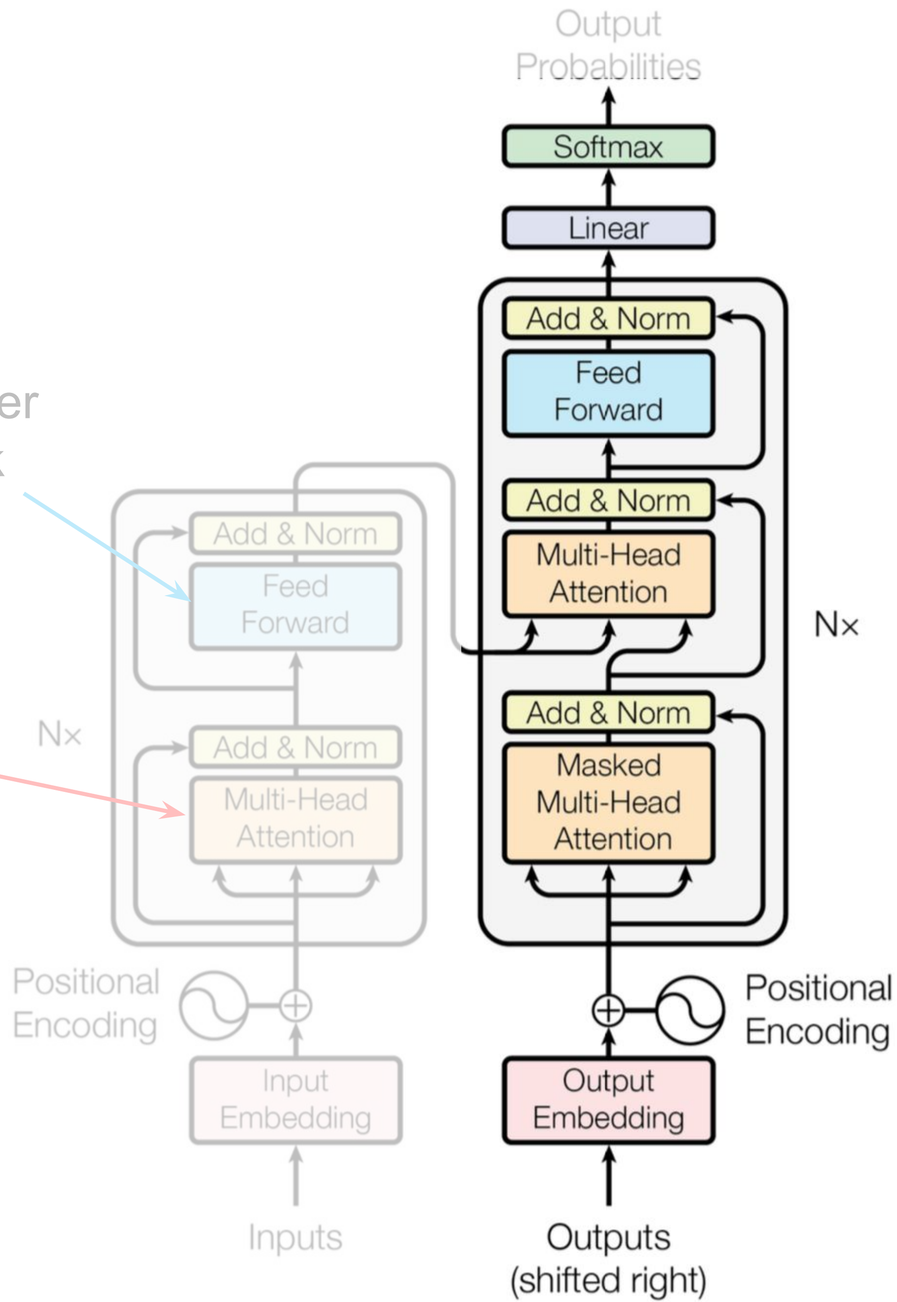

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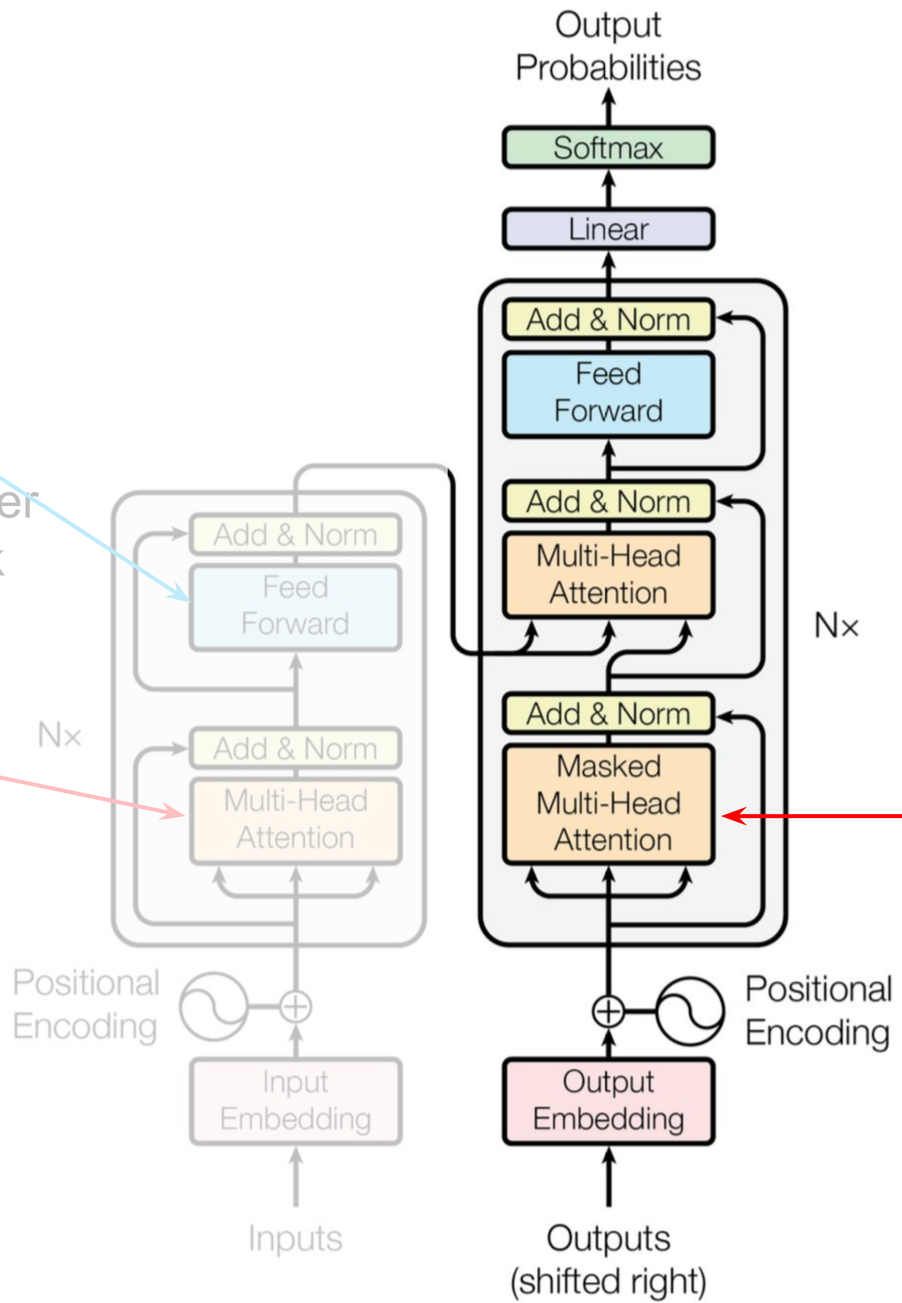

Decoder self-attention (masked):
tokens look at the previous tokens

queries, keys, values are computed from decoder states

Transformer

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Encoder self-attention:
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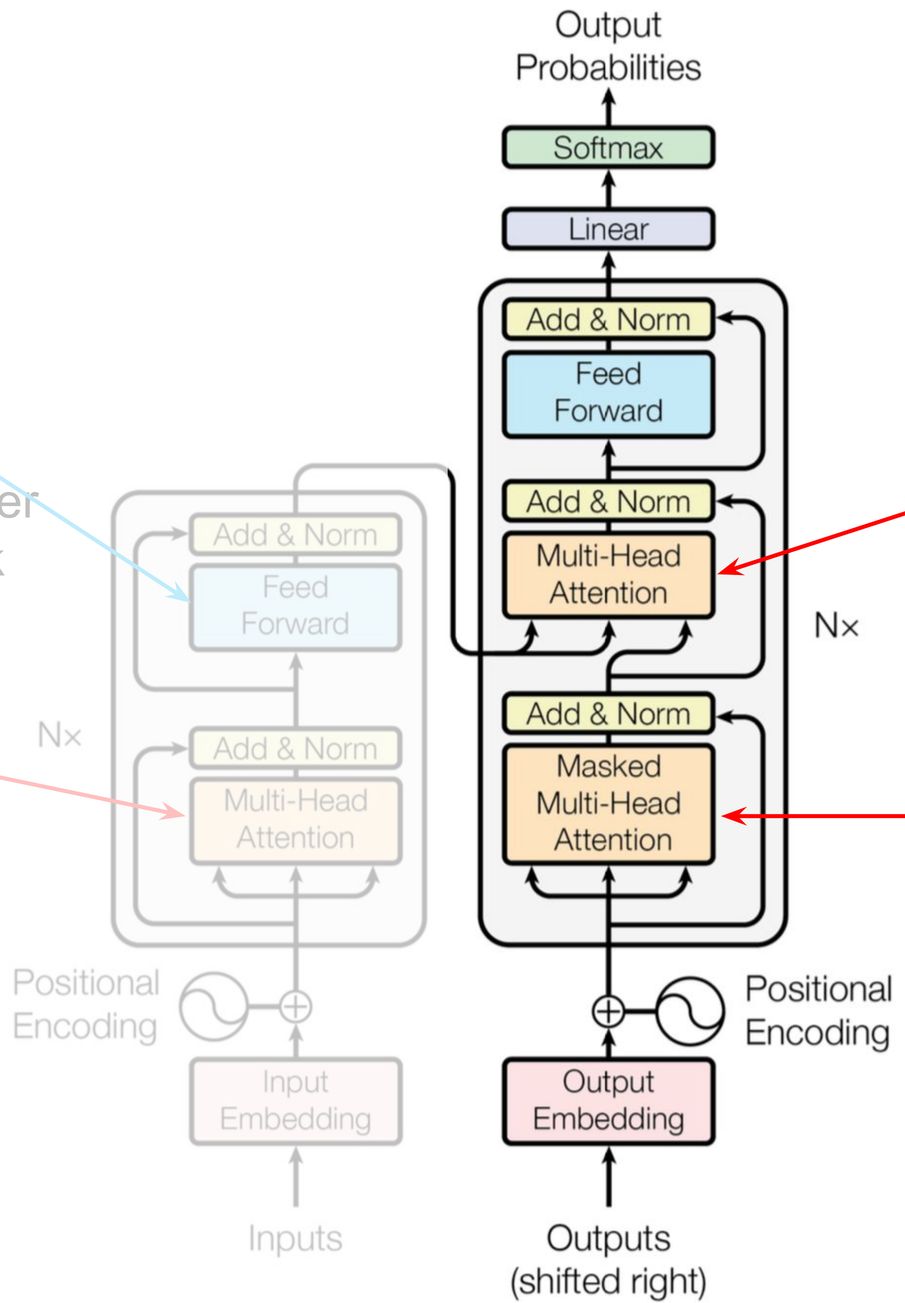

Decoder-encoder attention:
target token looks at the source

queries – from decoder states; keys and values from encoder states

Decoder self-attention (masked):
tokens look at the previous tokens

queries, keys, values are computed from decoder states

Transformer

Feed-forward network:
after taking information from other tokens, take a moment to think and process this information

↑

Encoder self-attention:
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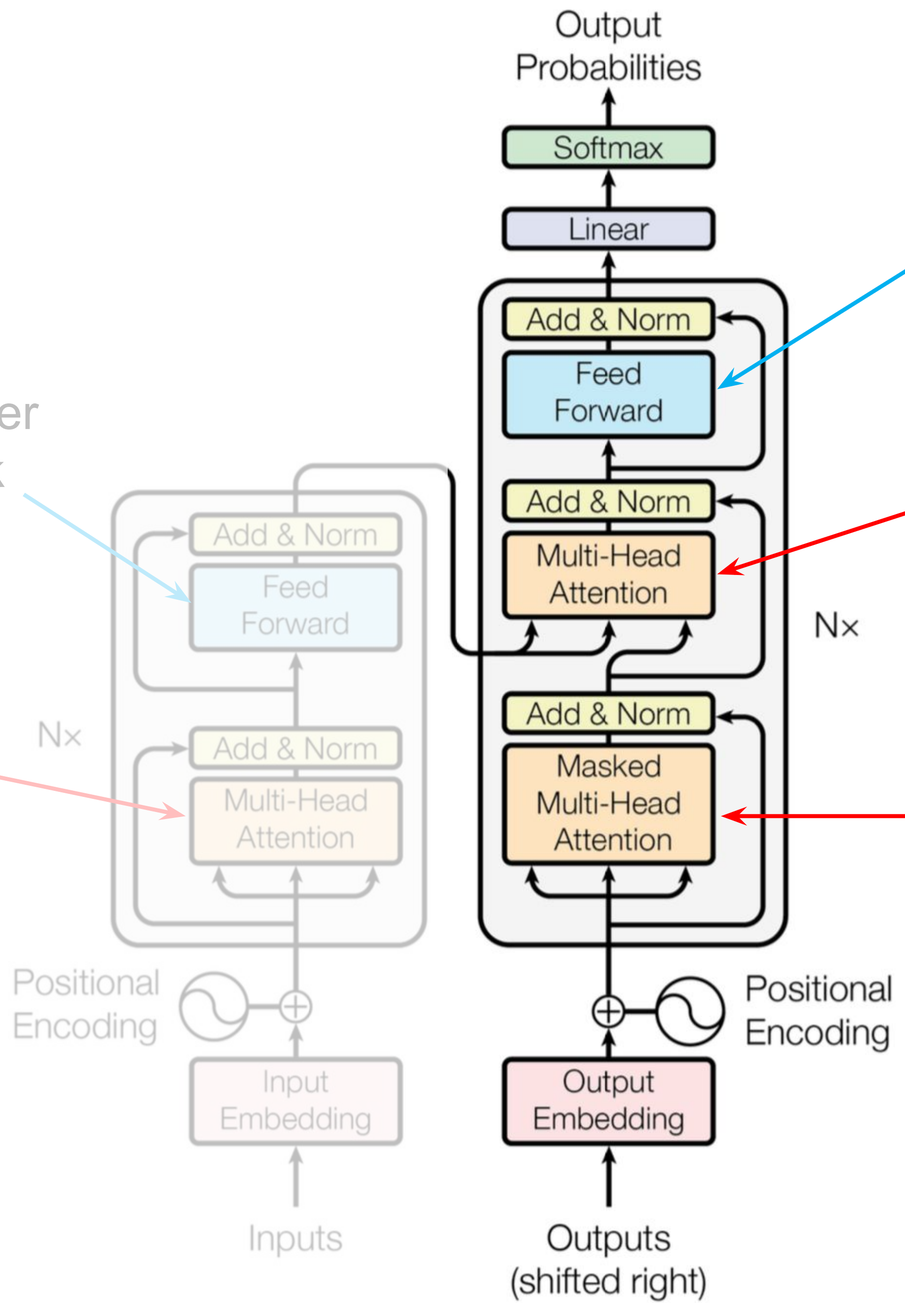

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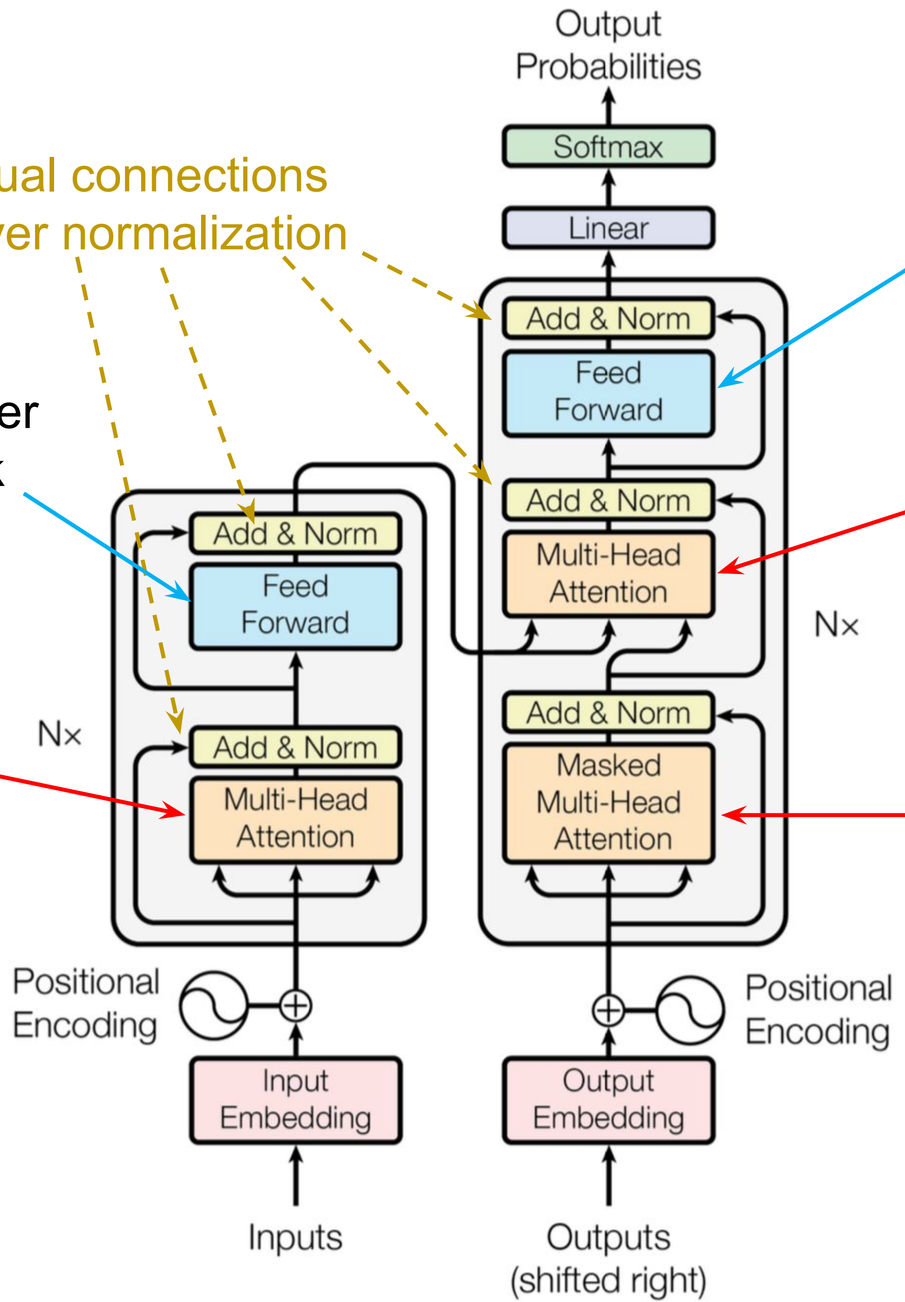

Residual connections and layer normalization

Feed-forward network:
after taking information from other tokens, take a moment to think and process this information

Encoder self-attention:
tokens look at each other

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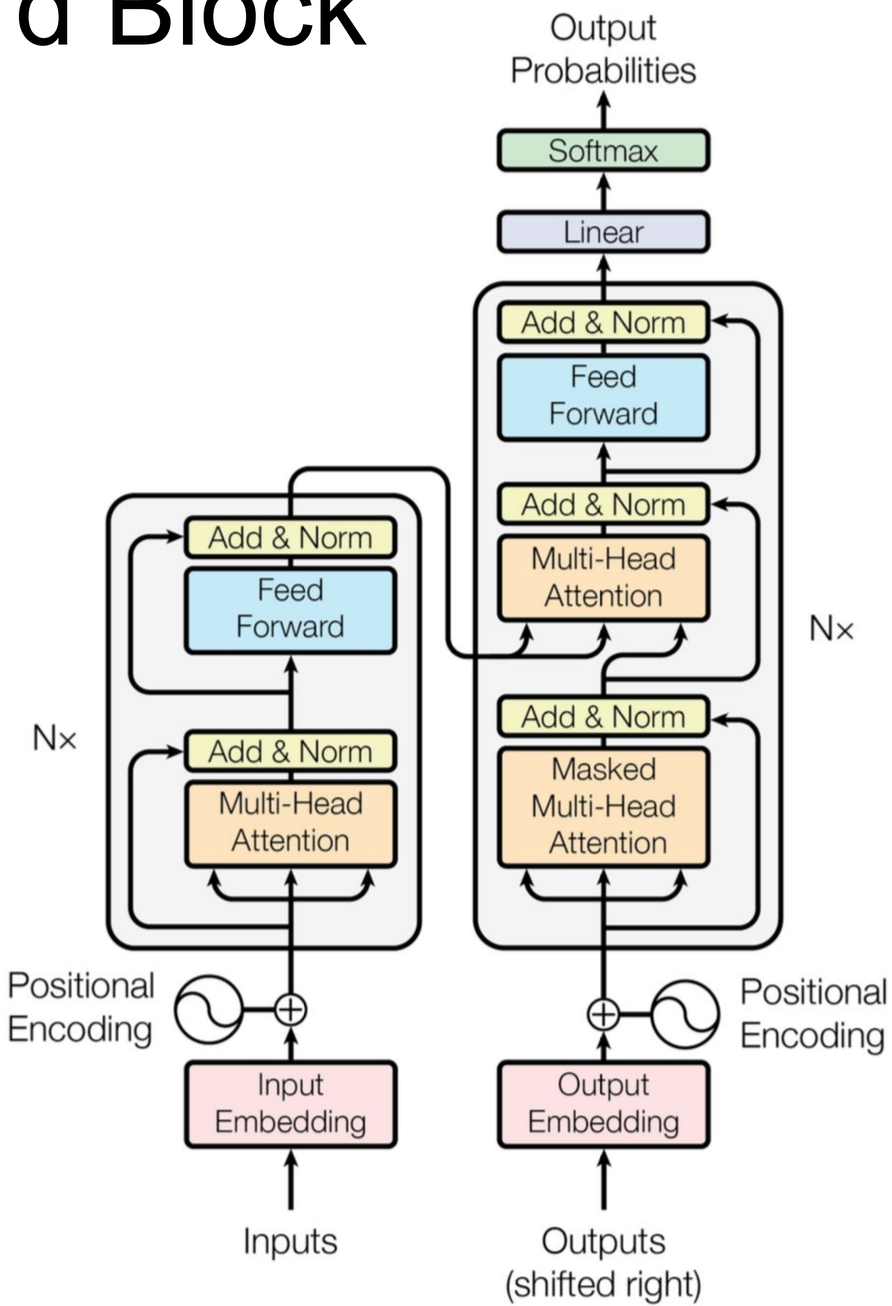
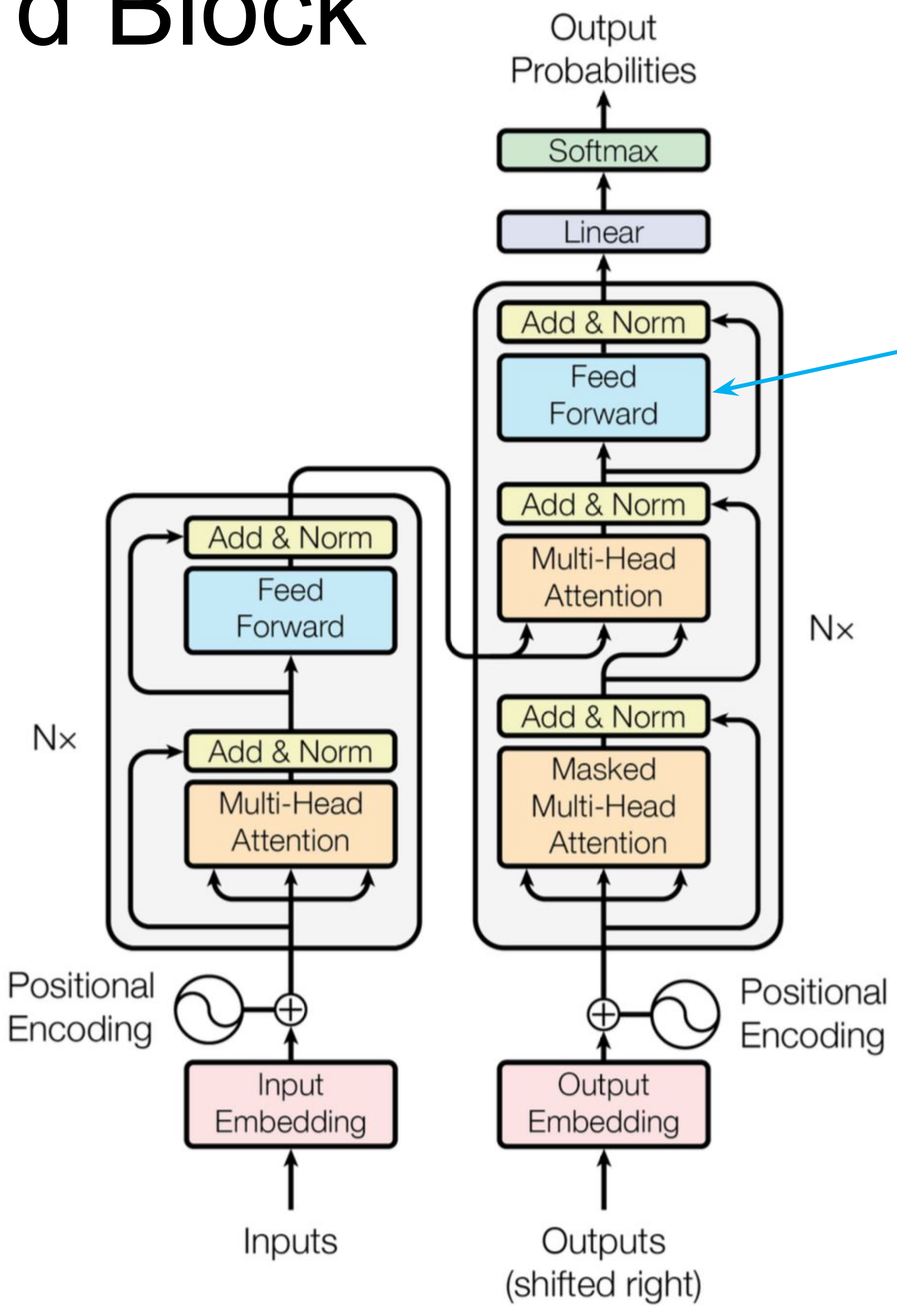
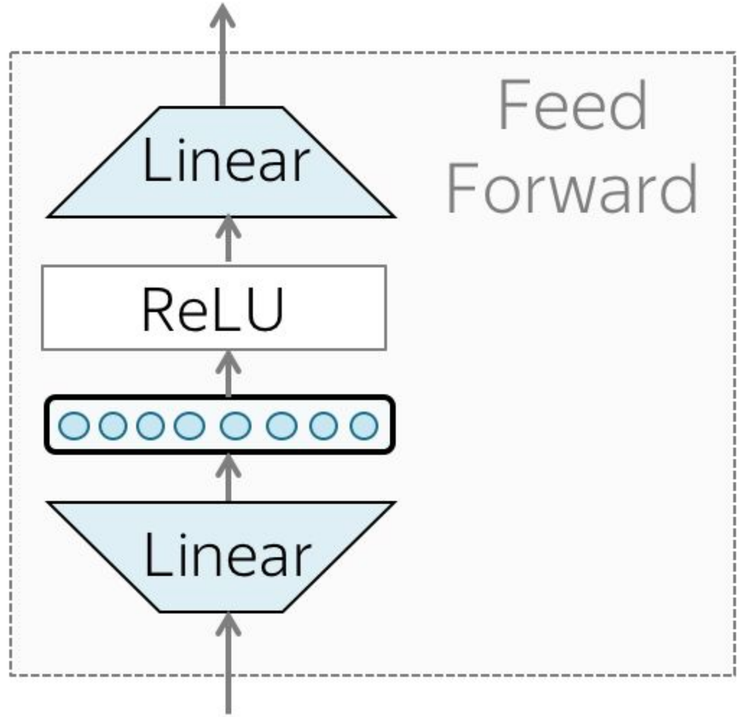
Feed Forward Block



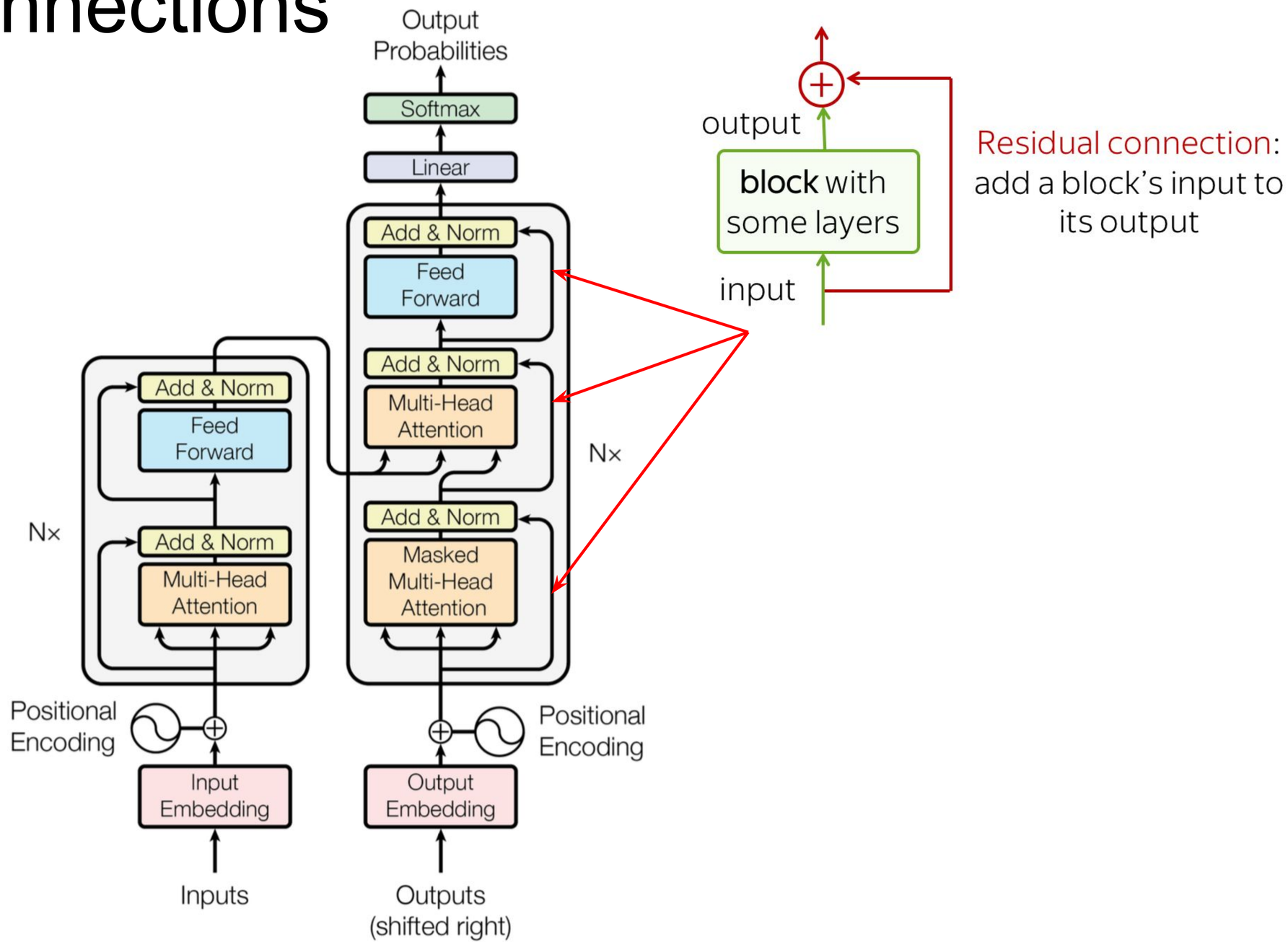
Feed Forward Block



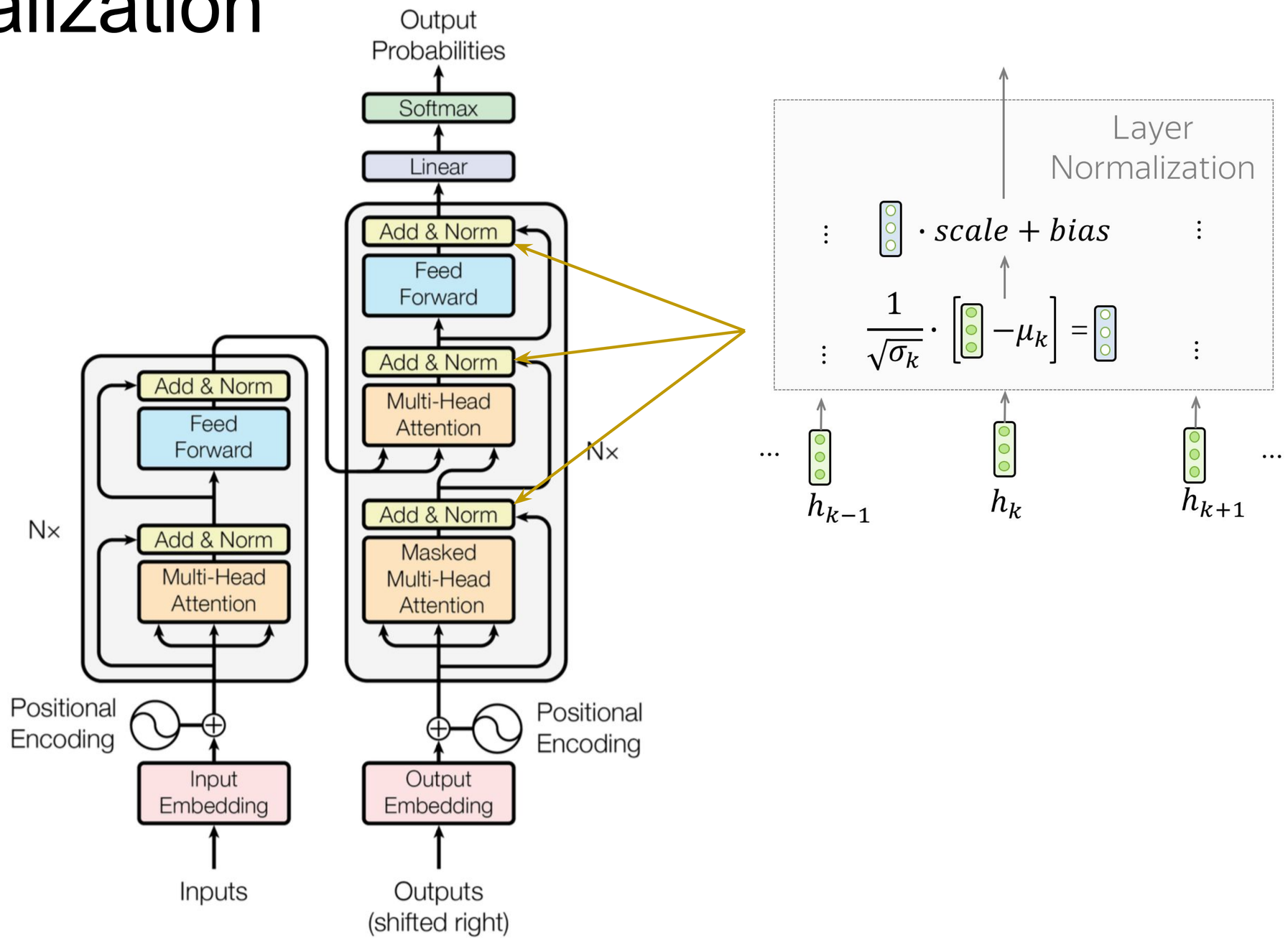


$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

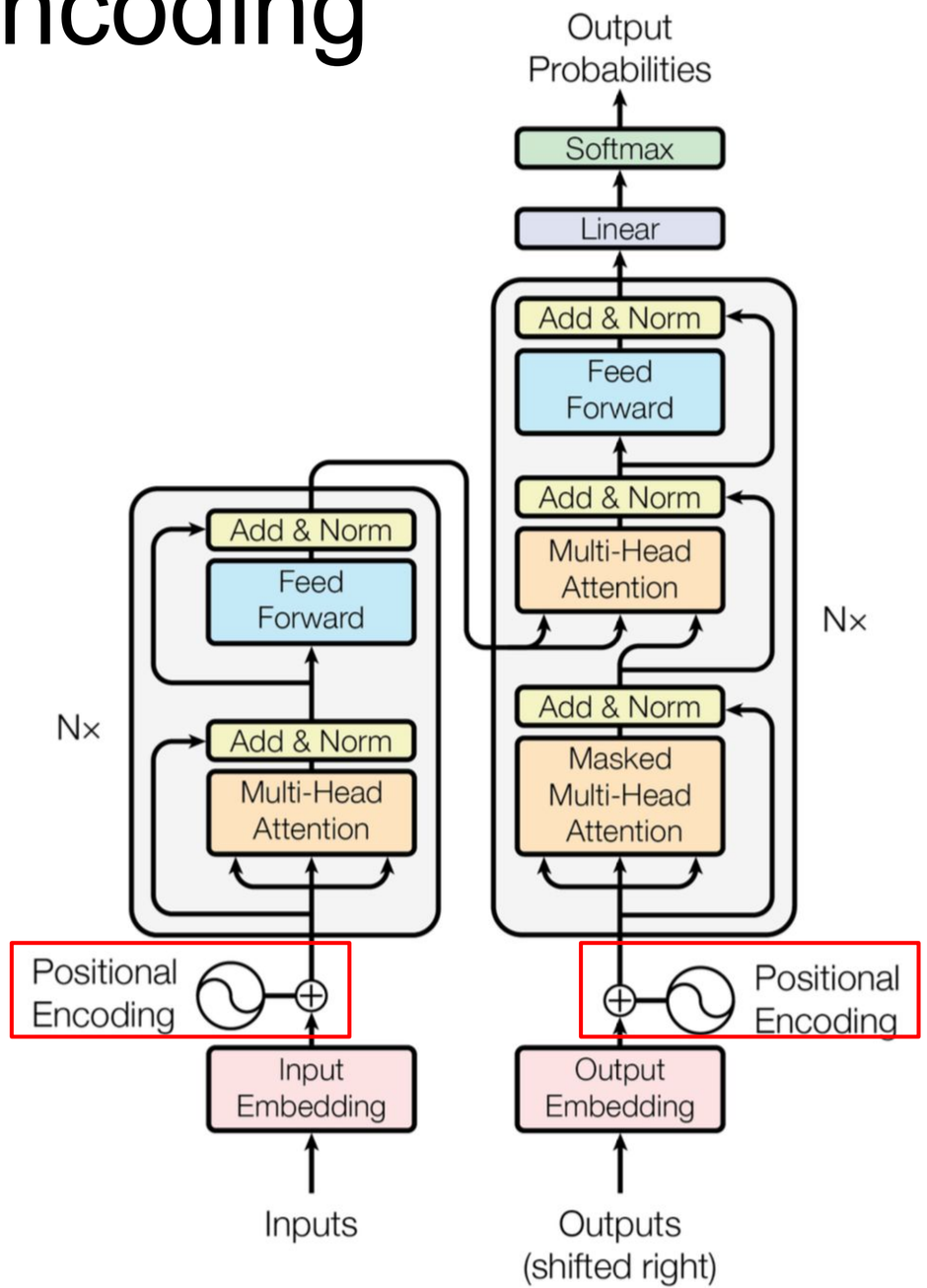
Residual Connections



Layer Normalization



Positional Encoding



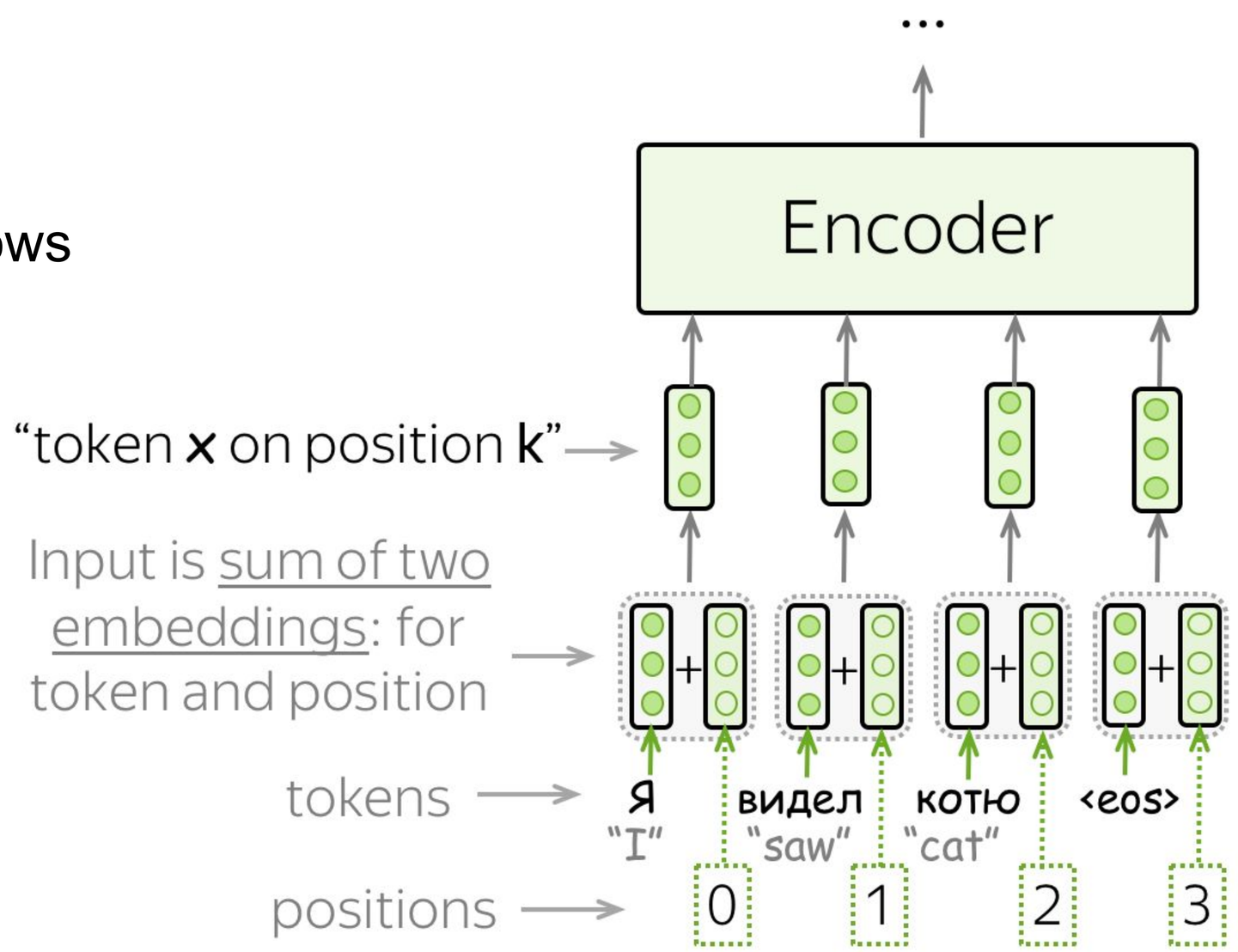
Positional Encoding

Problem:

- without recurrence or convolution, the model knows nothing about position

Solution:

- encode position explicitly and add



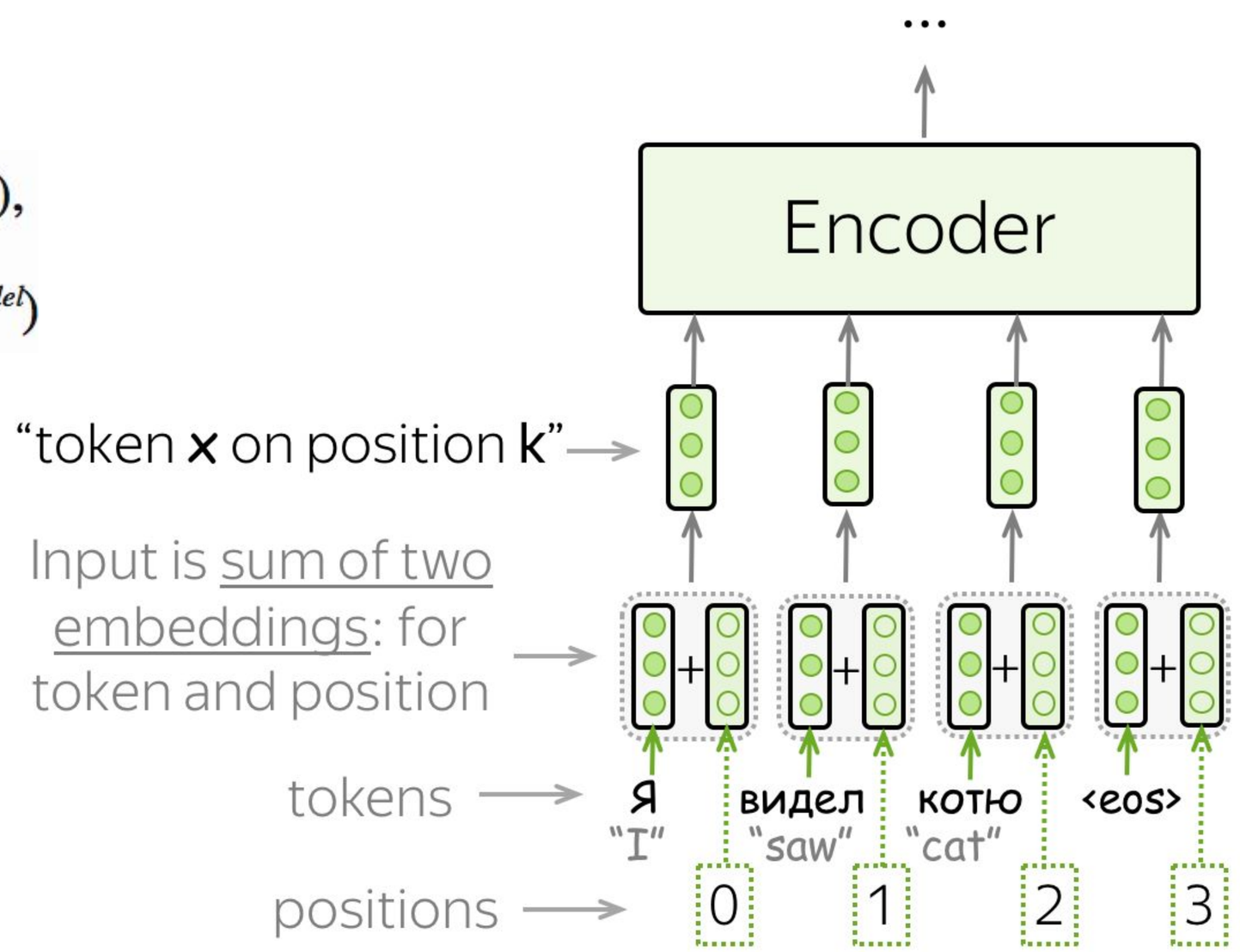
Positional Encoding

Fixed encodings:

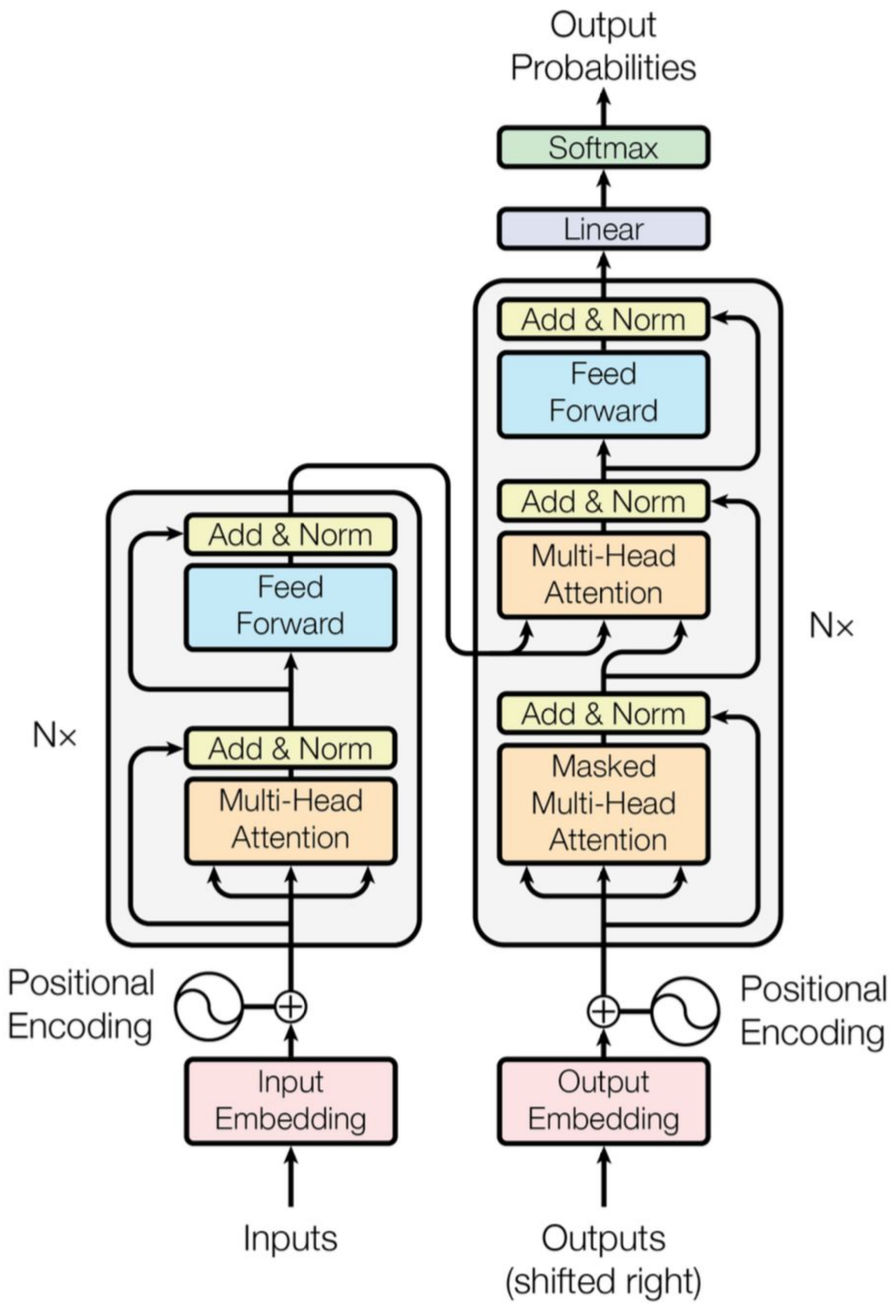
$$\mathrm{PE}_{pos,2i} = \sin(pos/10000^{2i/d_{model}}),$$

$$\mathrm{PE}_{pos,2i+1} = \cos(pos/10000^{2i/d_{model}})$$

pos – position, i - dimension



Transformer



Complexity of inference

n - length of input sequence, m - length of output sequence
d - size of embedding

RNN: $O(nd^2 + md^2)$

Transformer without caching: $O(n^2d + m^3d)$

Transformer with caching: $O(n^2d + m^2d)$

What is going to happen:

- Seq2seq Basics
- Attention
- Transformer
- Subword Segmentation: BPE
-  Analysis and Interpretability

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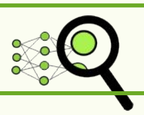
-  Analysis and Interpretability

Analysis Methods

Model-specific:

- Looking at model components
- …

Model-agnostic:

Analysis Methods

Model-specific:

- Looking at model components
- …

In the previous lectures:

- Convolutional filters of classifiers
- Neurons in RNN/CNN LMs

Model-agnostic:

Analysis Methods

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Today:

- Heads in Multi-Head Attention

Analysis Methods

Model-specific:

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- Look at the predictions: contrastive evaluation of specific phenomena

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- Probing: What do representations capture?

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Today:

- Probing: What do representations capture?

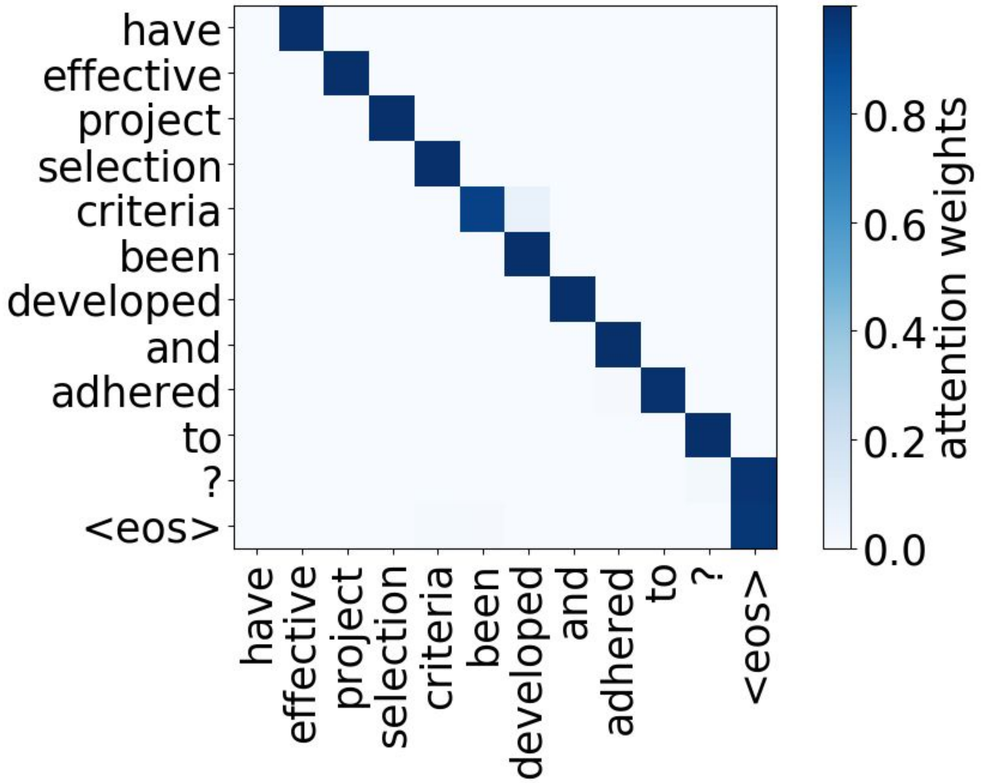
Multi-Head Attention: What are the Heads Doing?

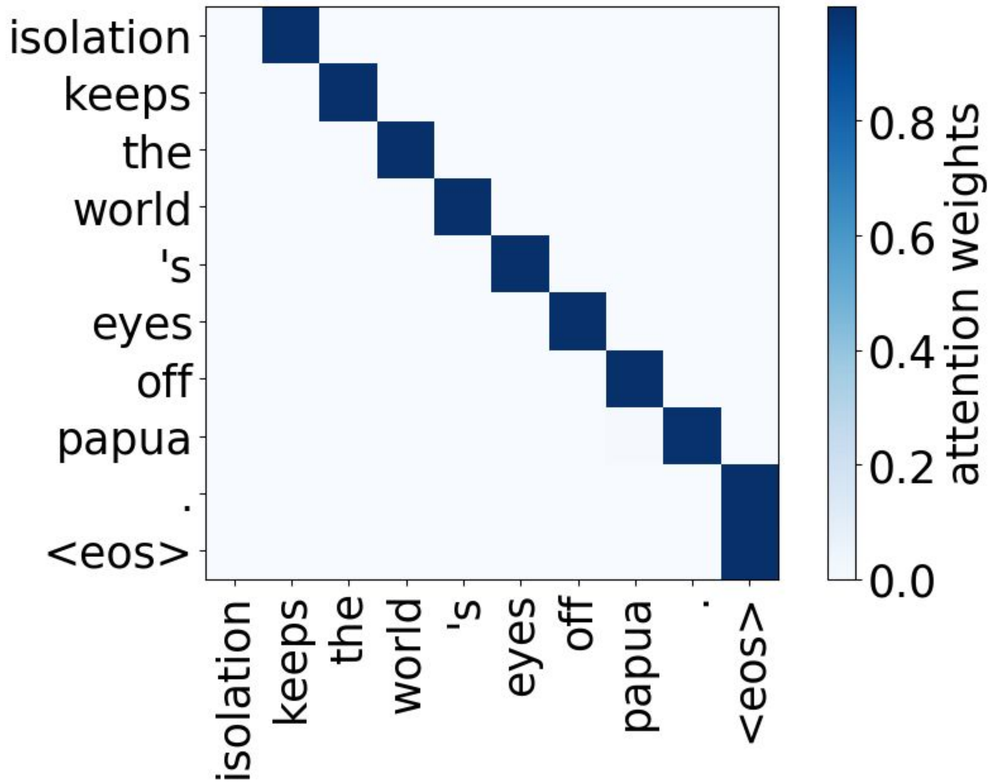
Heads which on average contribute more to generated translations (“important heads”) play interpretable roles:

- Positional
- Syntactic
- Attention to rare tokens

Paper: Analyzing Multi-Head Self-Attention:
Specialized Heads Do the Heavy Lifting, the Rest Can Be Pruned

Positional Heads: Attention to Neighbors





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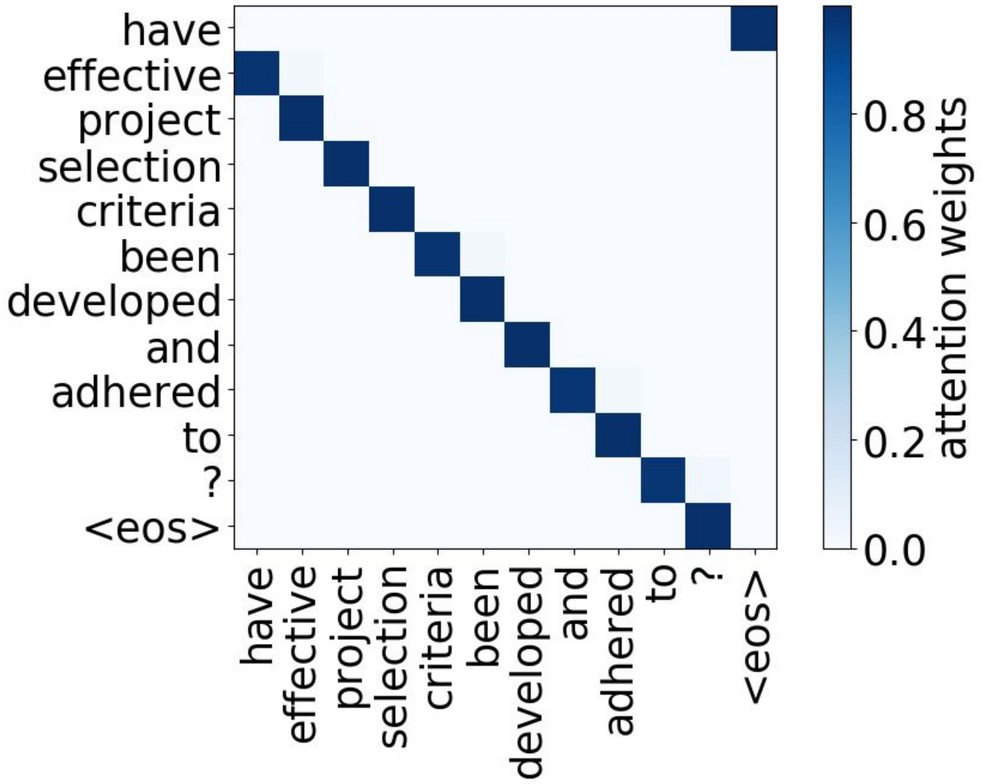
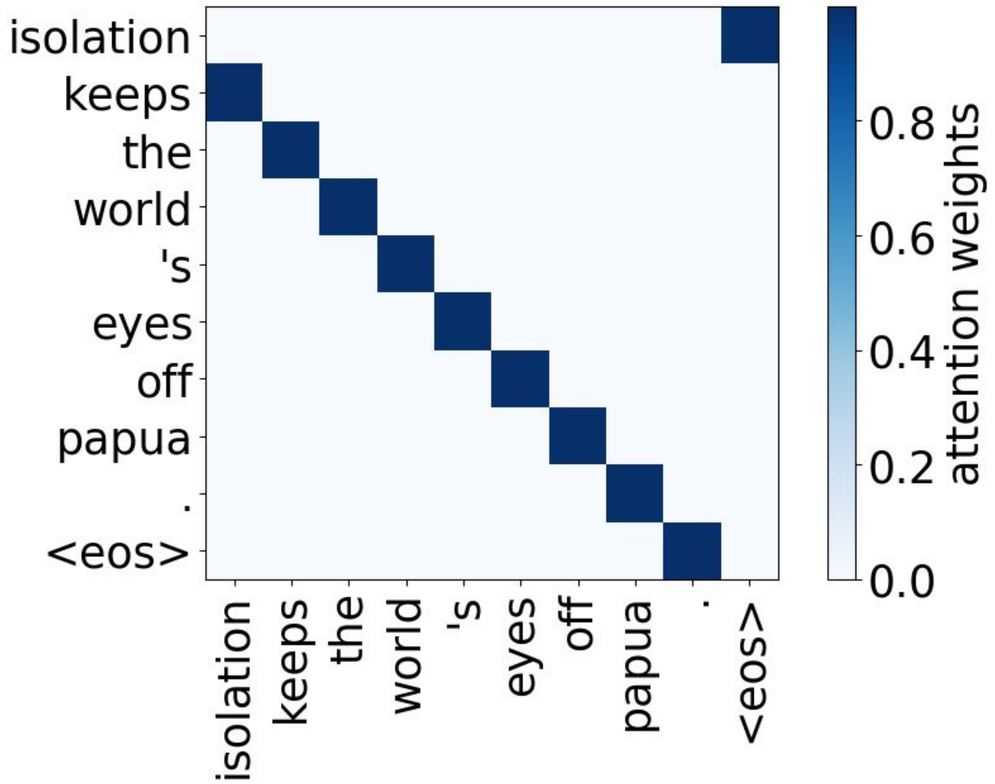
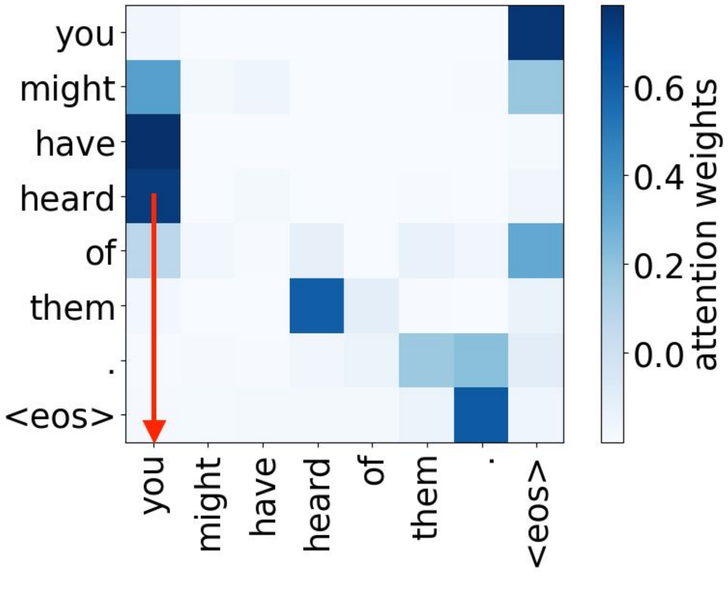


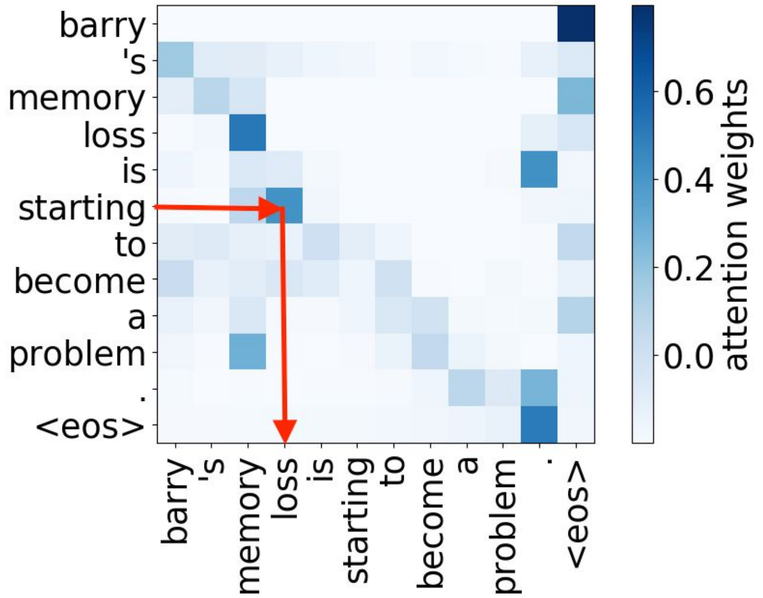


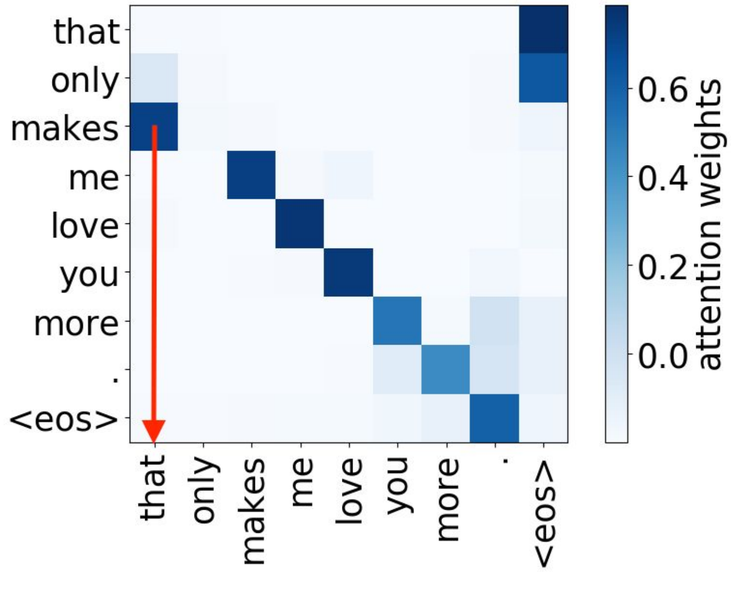
Paper: Analyzing Multi-Head Self-Attention: Specialized Heads Do the Heavy Lifting, the Rest Can Be

Syntactic Heads: Track Dependencies

- verb->subject







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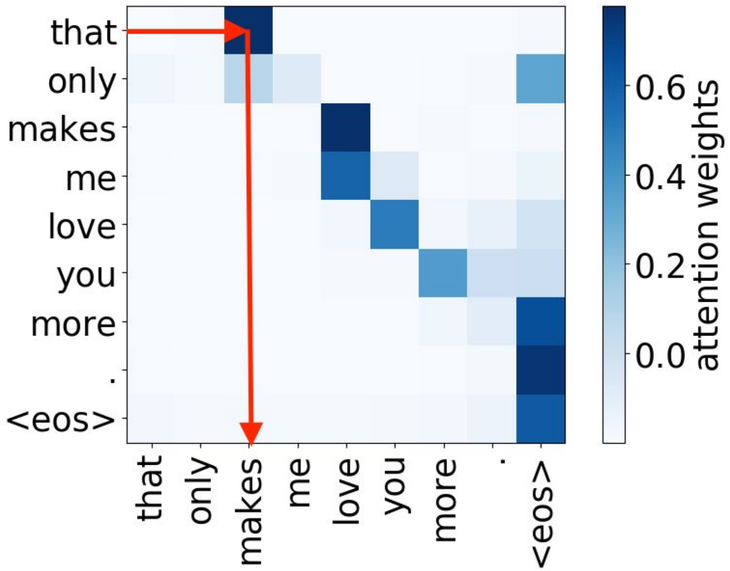
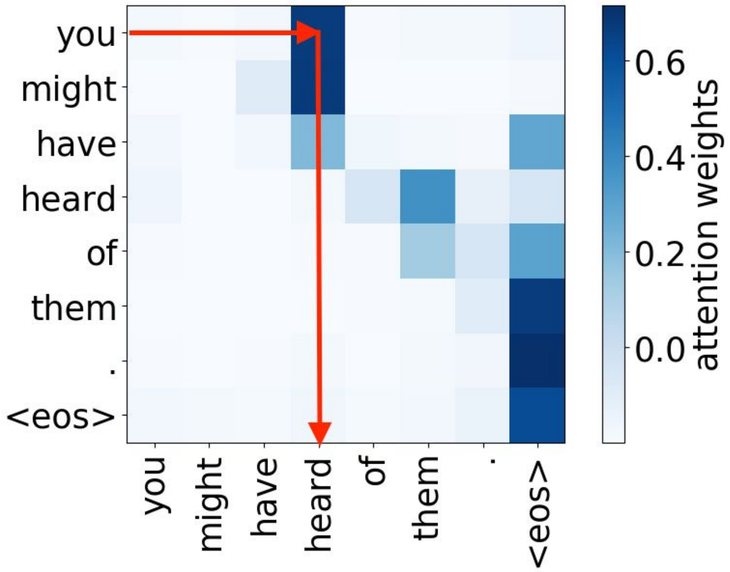
Syntactic Heads: Track Dependencies

- subject->verb

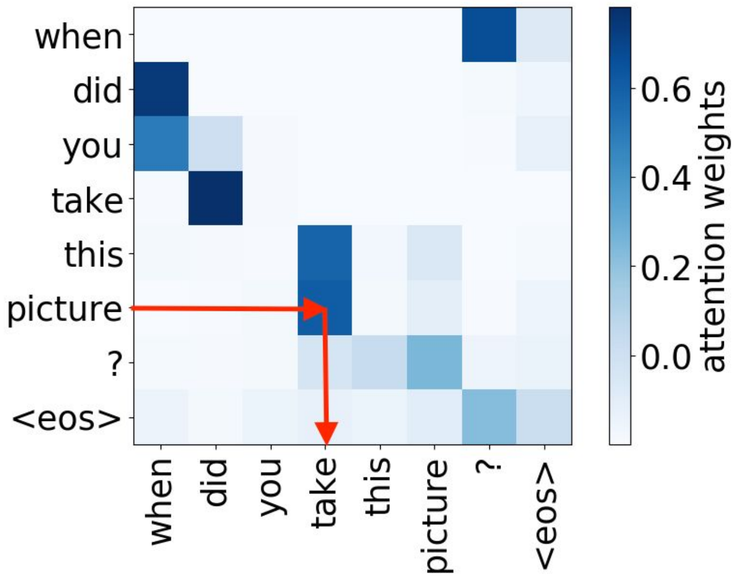


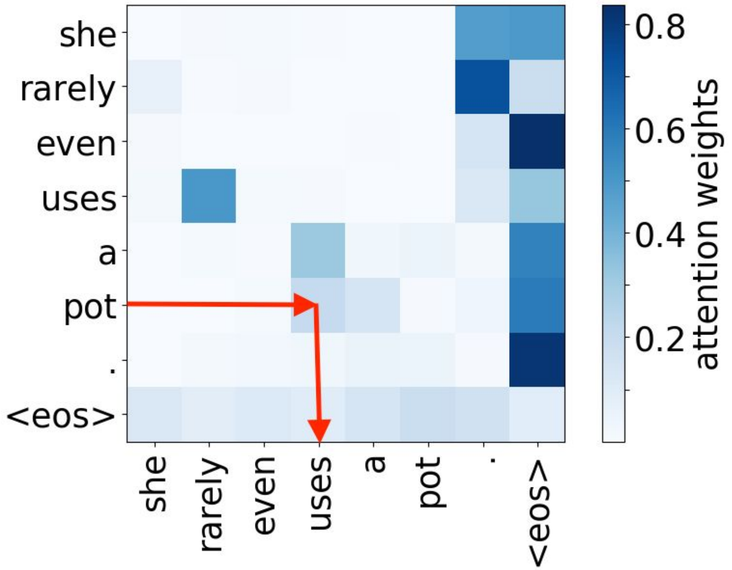


Paper: Analyzing Multi-Head Self-Attention: Specialized Heads Do the Heavy Lifting, the Rest Can Be Pruned

Syntactic Heads: Track Dependencies

- object->verb





Paper: Analyzing Multi-Head Self-Attention:
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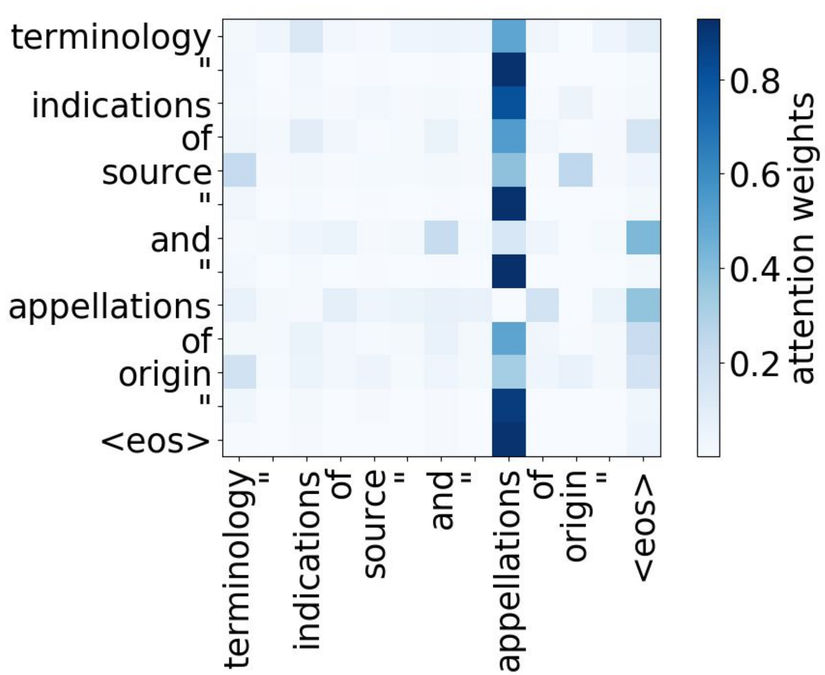
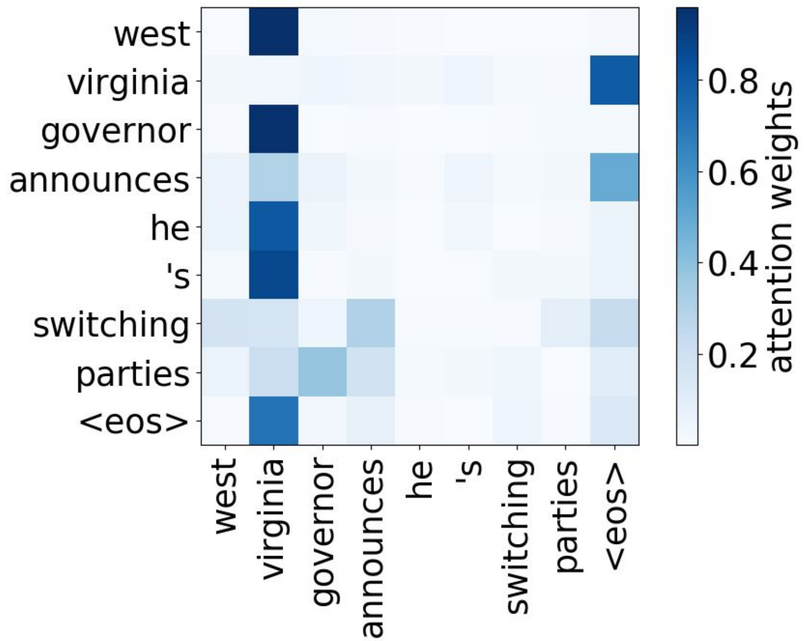
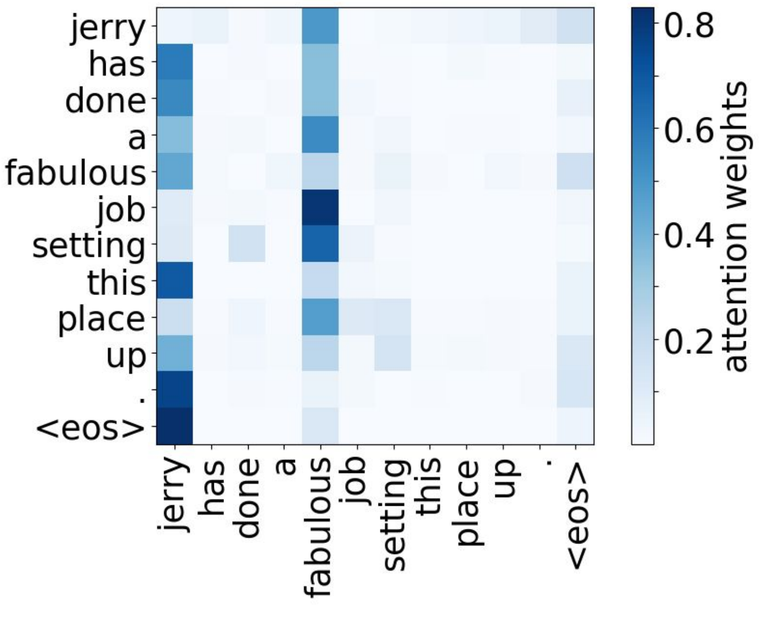
Attention to Rare Tokens







Paper: Analyzing Multi-Head Self-Attention:
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Analysis Methods

Model-specific:

- Looking at model components
- …

In the previous lectures:

- Convolutional filters of classifiers
- Neurons in RNN/CNN LMs

Today:

- Heads in Multi-Head Attention

Model-agnostic:

In the previous lecture:

- Look at the predictions: contrastive evaluation of specific phenomena

Today:

- Probing: What do representations capture?

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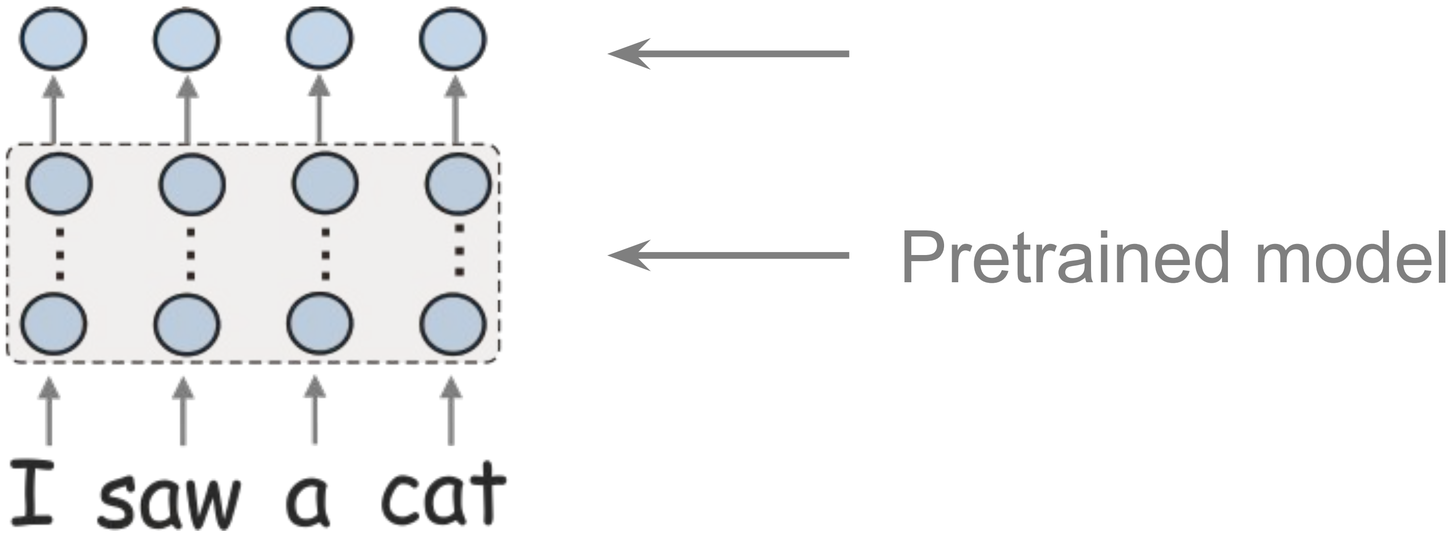

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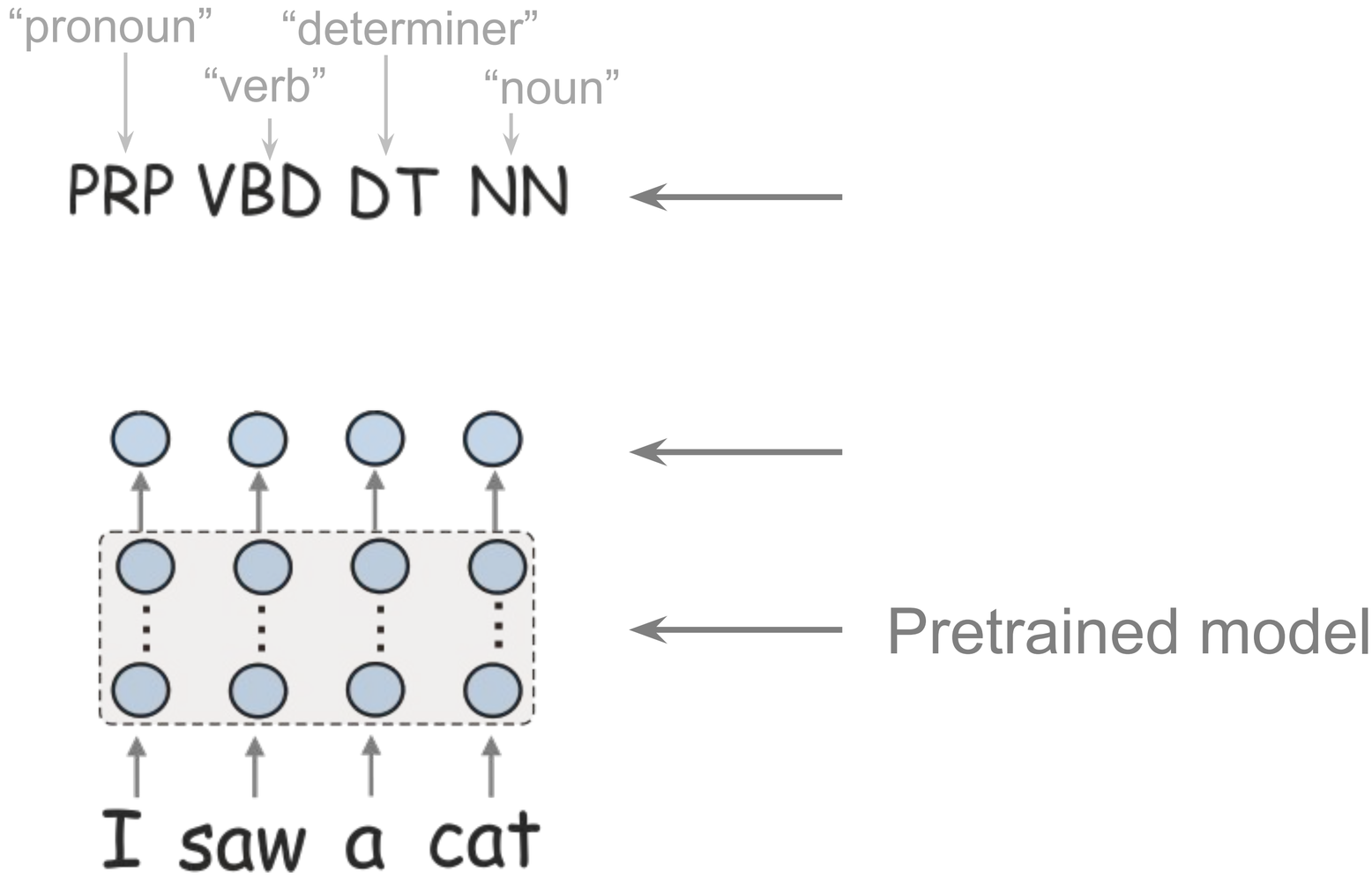

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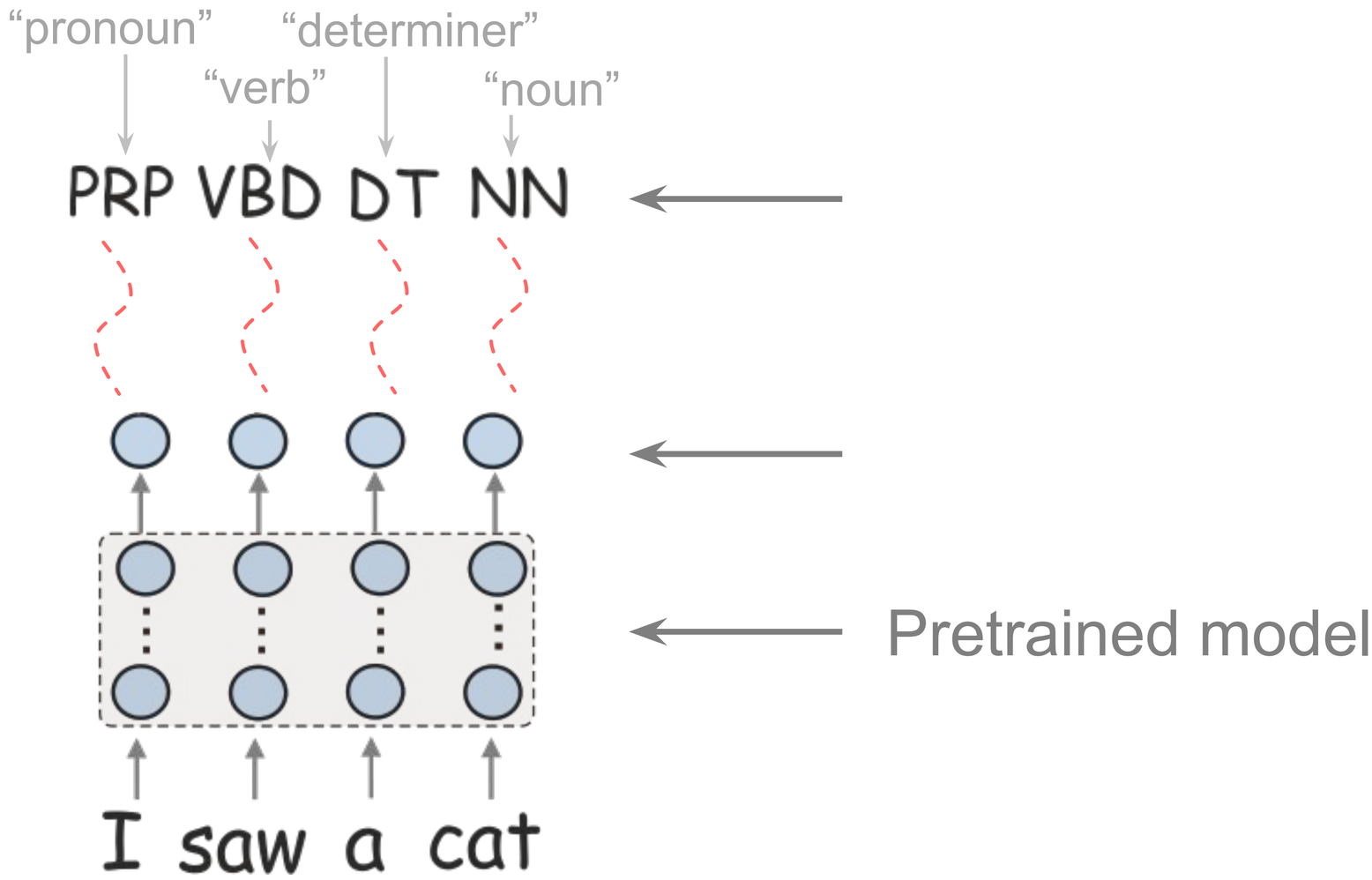

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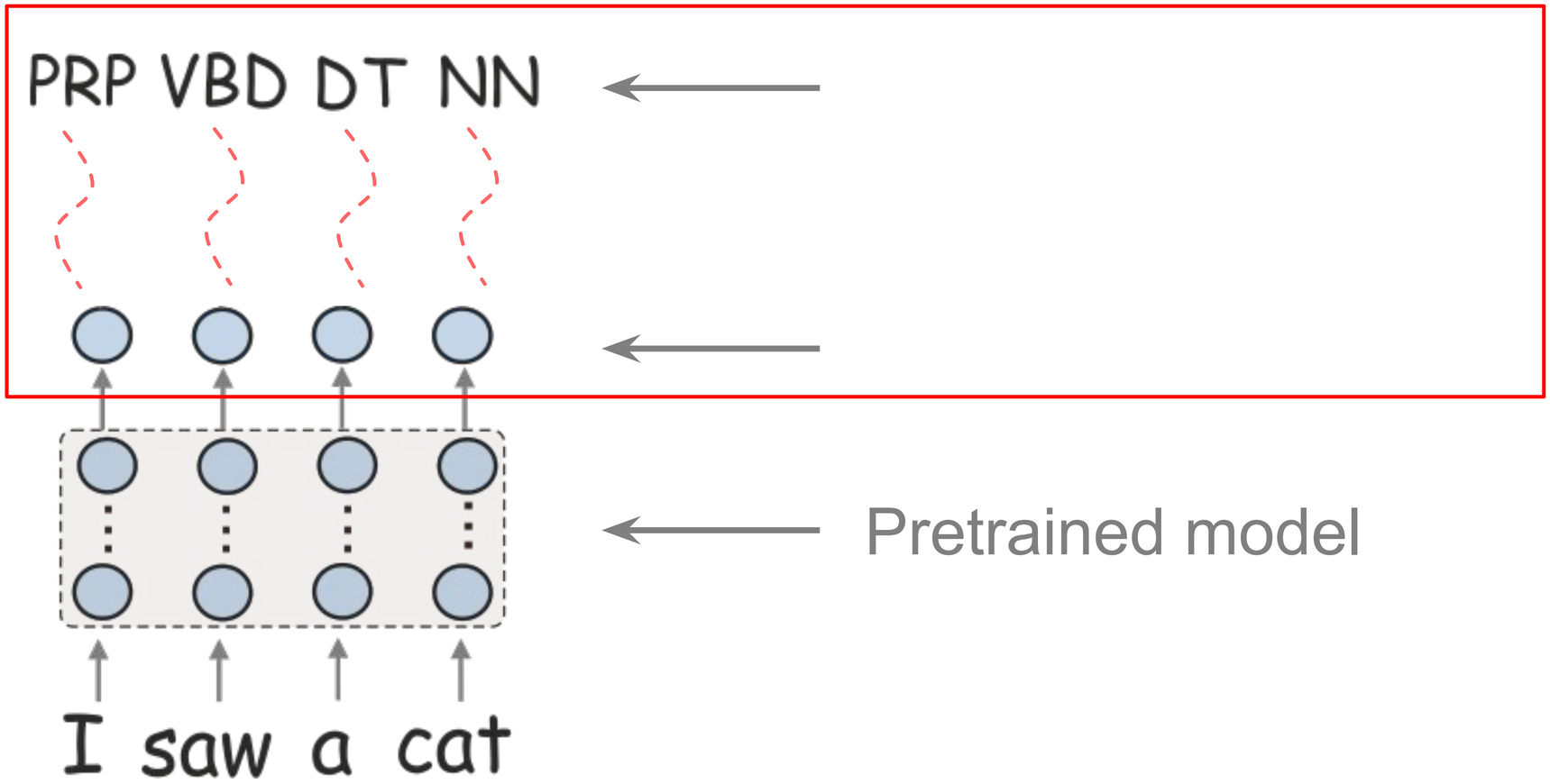
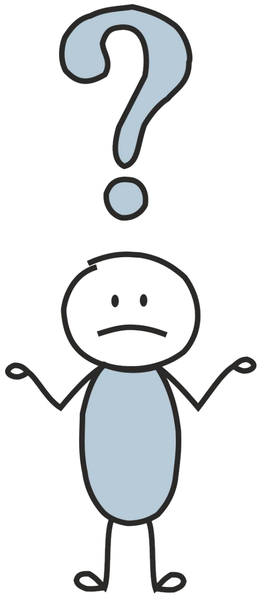
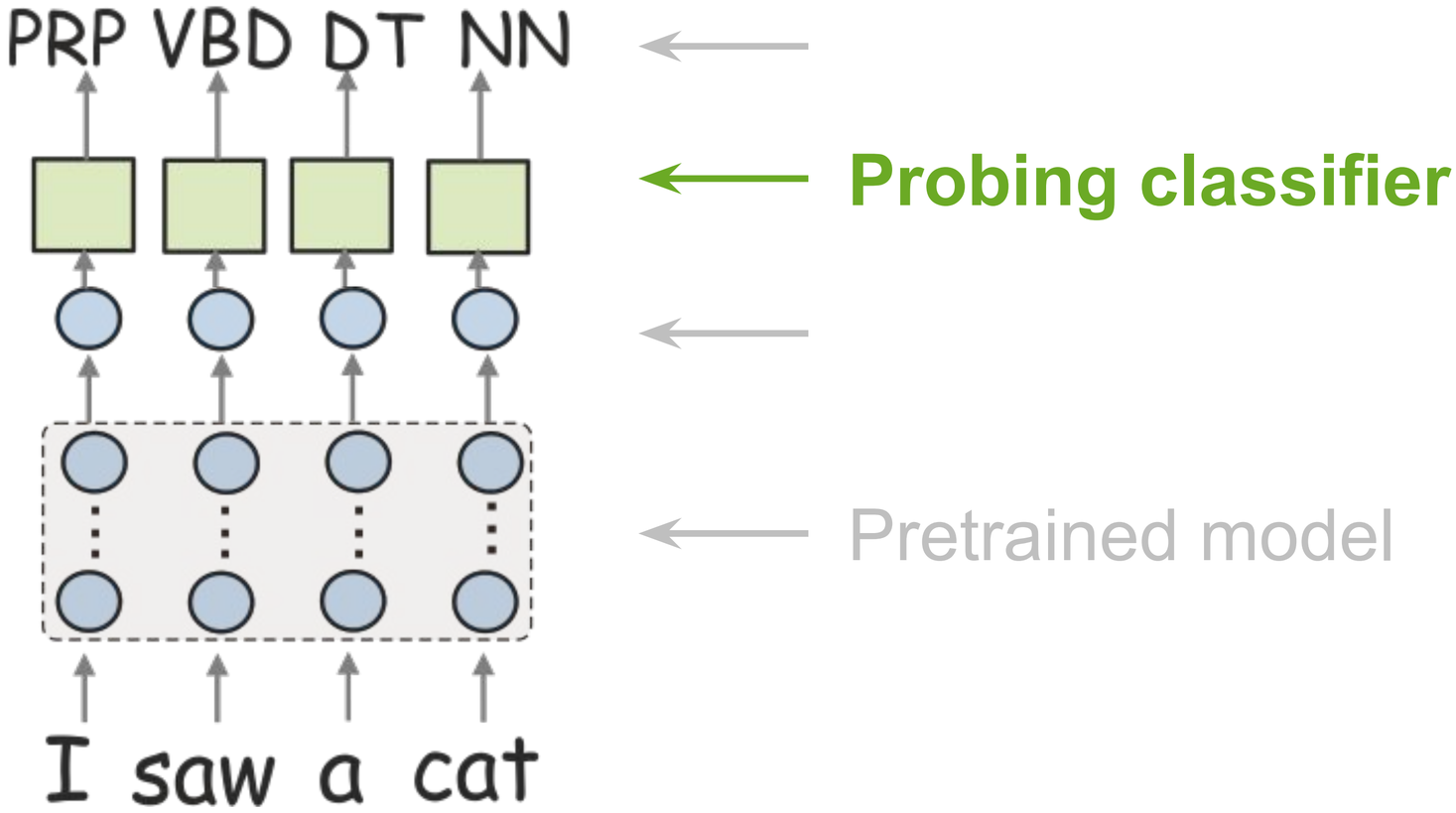


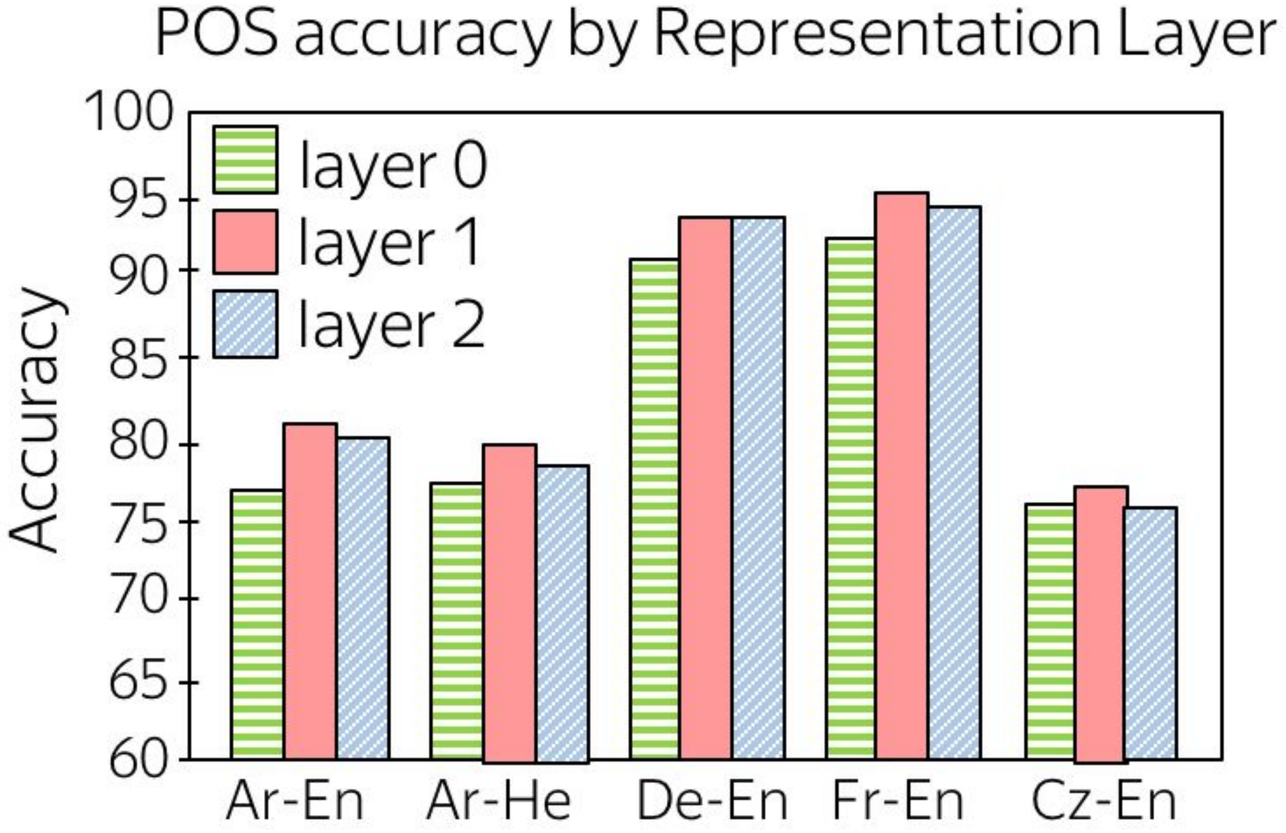


Standard Probing: train a classifier, use its accuracy



What Do NMT Models Learn About Morphology?

- Take NMT models for different language pairs
- Look at the encoder layers



Paper: What do Neural Machine Translation Models Learn about Morphology?

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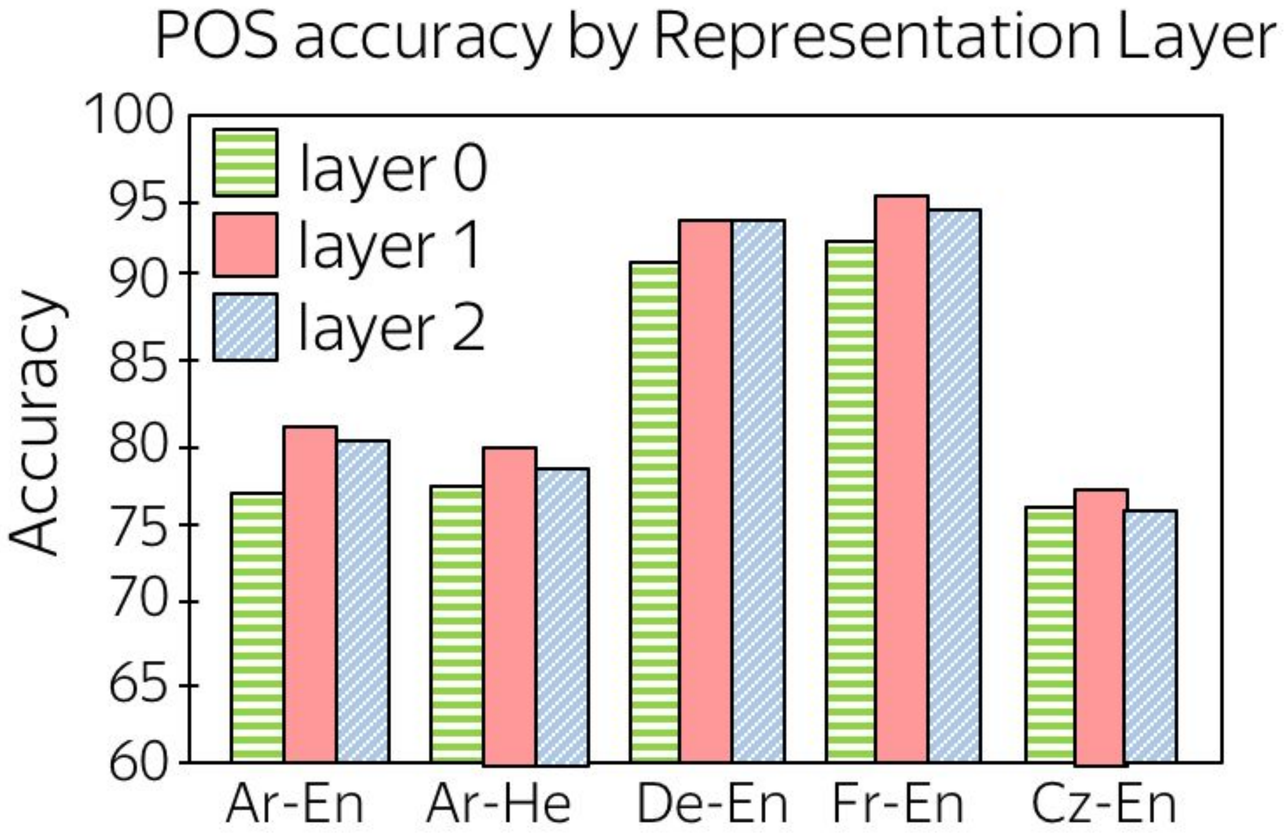
<u>Results:</u>

- Encoding helps: layer 0 (word embeddings) is the worst



Paper: <u>What do Neural Machine Translation Models Learn about Morphology?</u>

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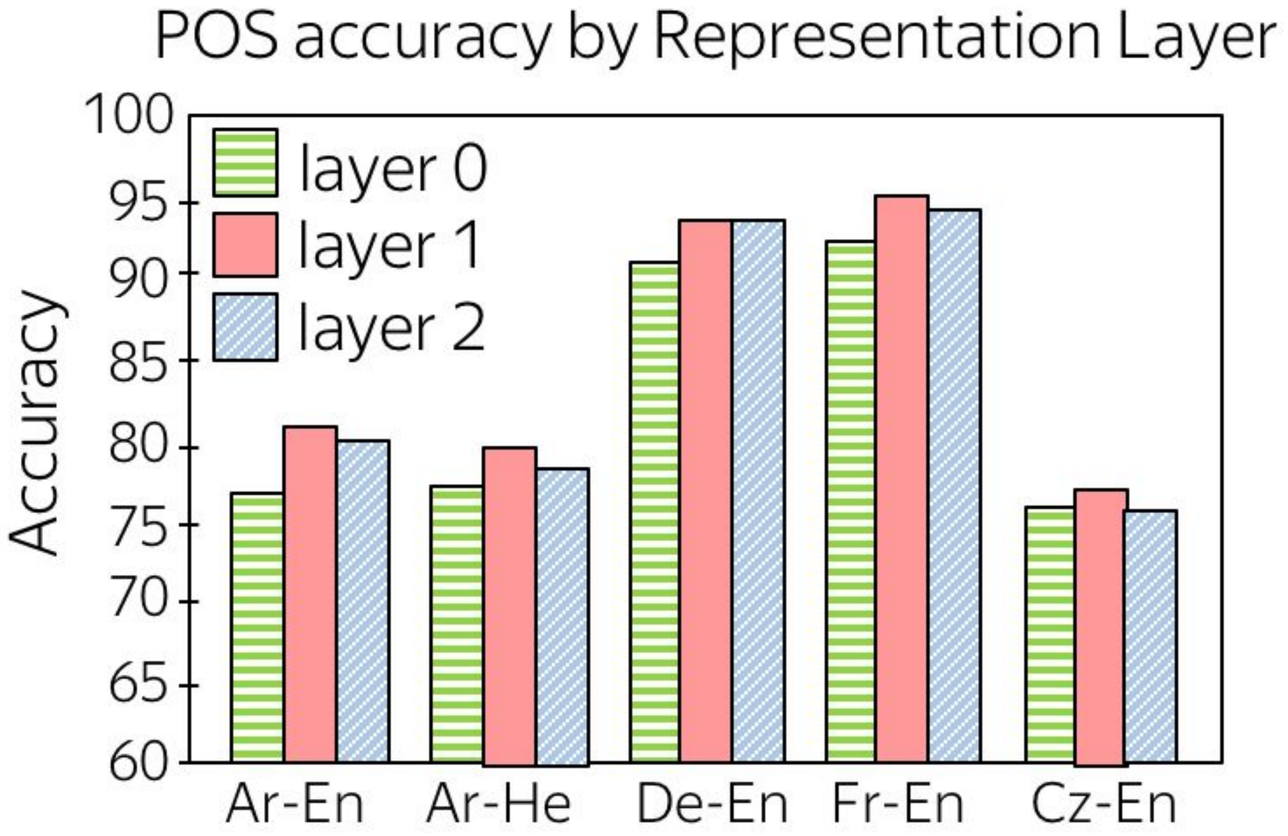
<u>Results:</u>

- Encoding helps: layer 0 (word embeddings) is the worst
- Layer 1 is better than layer 2



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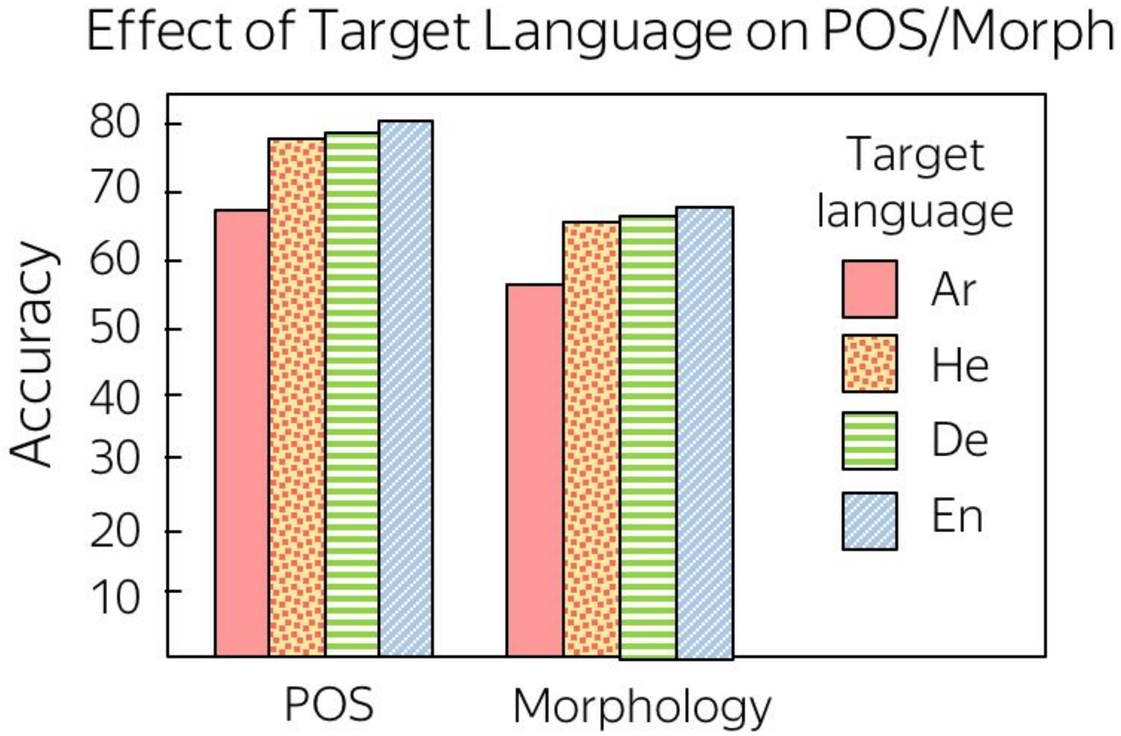

Paper: What do Neural Machine Translation Models Learn about Morphology?

What Do NMT Models Learn About Morphology?

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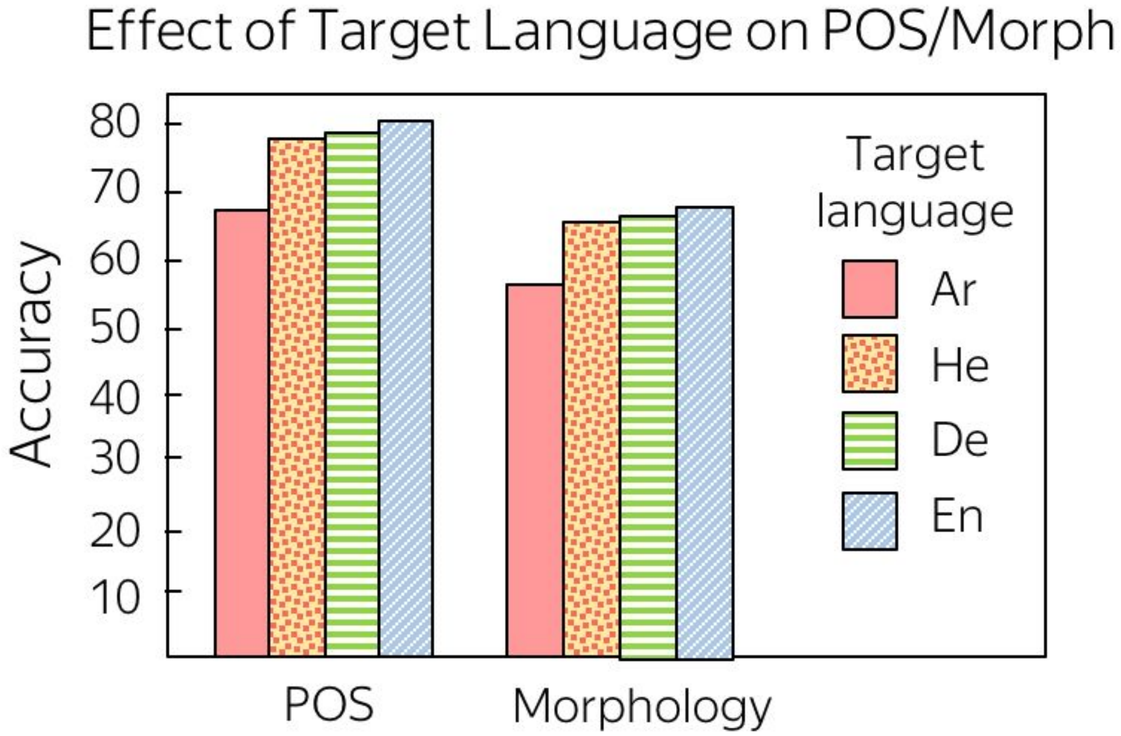
<u>Results:</u>

- Surprising: weaker target morphology leads to stronger encoder



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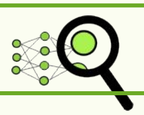
What is going to happen:

- Seq2seq Basics
- Attention
- Transformer
- Subword Segmentation: BPE
-  Analysis and Interpretability

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